Analog Electronics

Previous Year Questions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Analog Electronics (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
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- 5+ years of teaching experience

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- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Diode Circuits and applications

Topics

1. Diode Models with open and short circuit tests
2. Operating Point and Small Signal Analysis of a Diode
3. Zener Diode as a Voltage Regulator
4. Series and Shunt Clipper Circuits
5. Two-Level Clipper Circuits
6. Clamper Circuits and their examples
7. Peak Detector and Average Diode Current
8. Voltage Multiplier Circuits
9. Half Wave Rectifier and Centre Tapped Full Wave Rectifier
10. Full Wave Bridge Rectifier and Capacitive Filters

Unit Snapshot*

Stream: EC	**Questions:** 66	**MCQ:** 39	**MSQ:** 2	**NAT:** 25
1M: 29	**2M:** 30	**Descriptive:** 7	**Avg Weightage** ~3M	**Range:** 1M(2020)–5M(2011)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 2M

Consider the ideal diodes D1 and D2 as shown in the Figure with cut-in voltage $V_\gamma = 0$ Volt and $v_i(t)$ is in Volt. The maximum voltage (Volt) of the output $v_o(t)$ is _____. *(rounded off to two decimal places)*

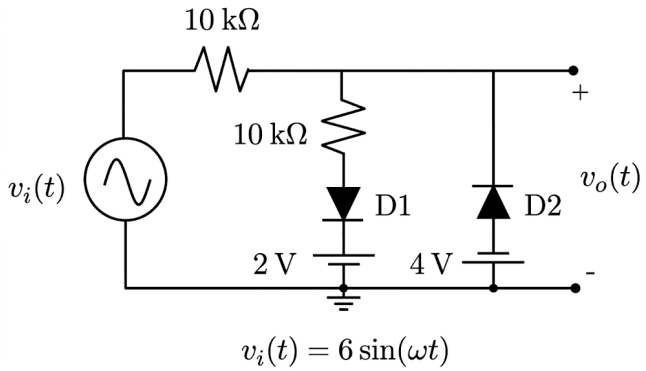


Fig. Q1.26.1

Q1.25.1 MSQ 2025 2M

All the diodes in the circuit given below are ideal. Which of the following plots is/are correct when V_I (in Volts) is swept from $-M$ to M ?

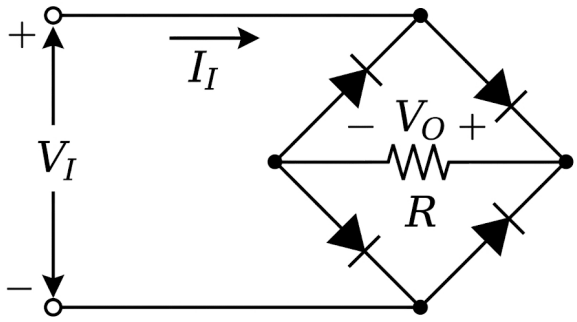
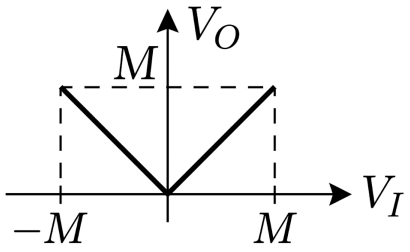
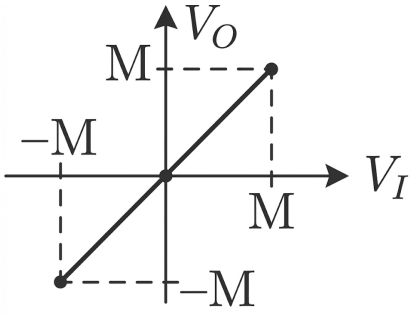


Fig. Q1.25.1

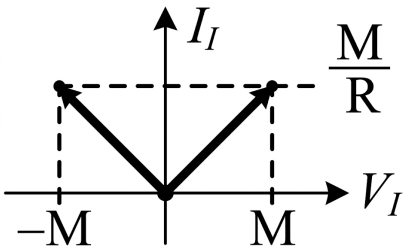
☐ **A.**



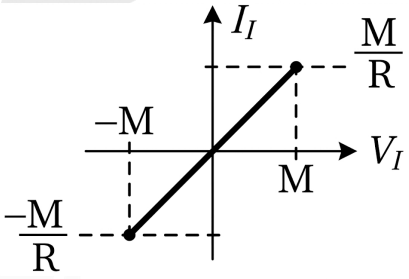
☐ **B.**



☐ **C.**



☐ **D.**



Q1.25.2 NAT 2025 2M

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $V_I = 10\sin 100\pi t$, where time t is in seconds.

The time duration (in ms, *rounded off to two decimal places*) for which the diode is forward biased during one period of the input is _____.

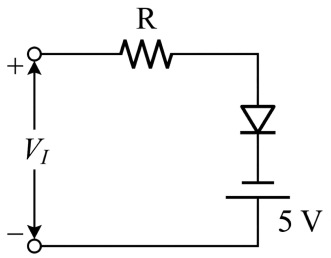


Fig. Q1.25.2

Q1.24.1 MCQ 2024 1M

In the circuit shown, the n : 1 step-down transformer and the diodes are ideal. The diodes have no voltage drop in forward biased condition. If the input voltage (in Volts) is $V_s(t) = 10 \sin \omega t$ and the average value of load voltage $V_L(t)$ (in Volts) is $2.5/\pi$, the value of n is _____.

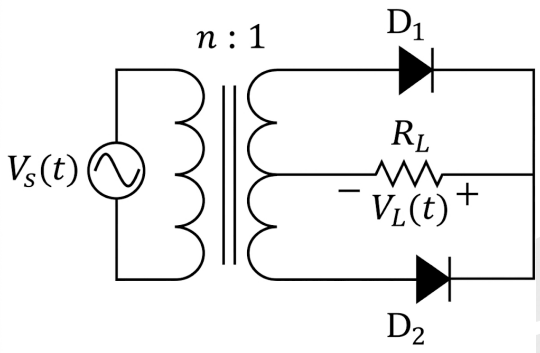


Fig. Q1.24.1

A. 4 **B.** 8 **C.** 12 **D.** 16

Q1.24.2 NAT 2024 1M

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is then closed at $t = 0$. The total energy dissipated in the ideal Zener diode ($V_Z = 5$ V) after the switch is closed (in mJ, *rounded off to three decimal places*) is _____.

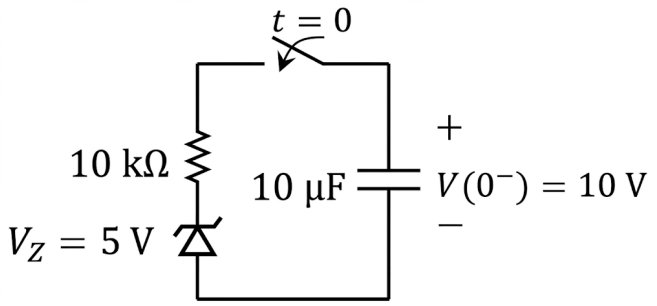


Fig. Q1.24.2

Q1.23.1 MCQ 2023 2M

In the circuit shown below, D_1 and D_2 are silicon diodes with cut-in voltage of 0.7 V. V_{IN} and V_{OUT} are input and output voltages in volts. The transfer characteristic is

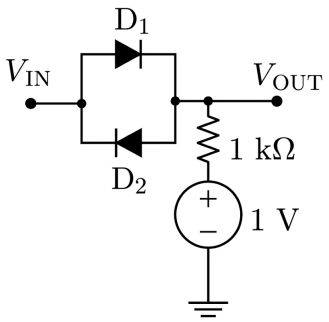
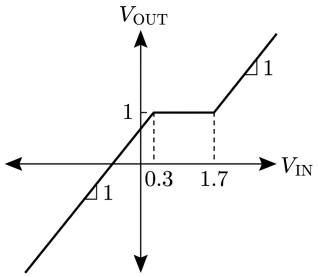
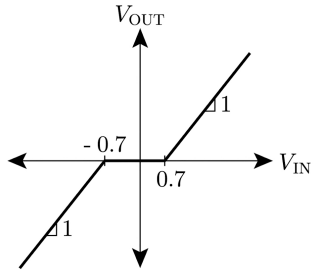


Fig. Q1.23.1

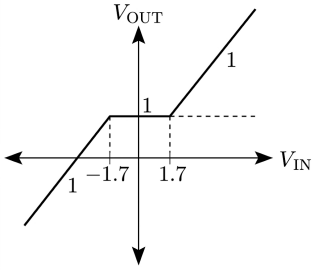
A.



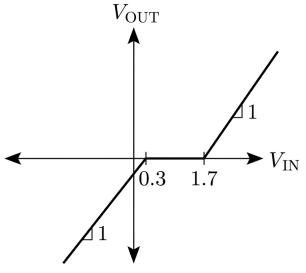
B.



C.



D.



Q1.22.1 NAT 2022 2M

A circuit and the characteristics of the diode (D) in it are shown. The ratio of the minimum to the maximum small signal voltage gain $\frac{\partial V_{out}}{\partial V_{in}}$ is _____ (*rounded off to two decimal places*).

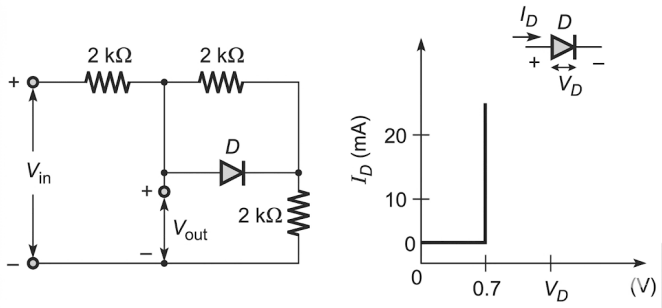


Fig. Q1.22.1

Q1.21.1 NAT 2021 2M

An asymmetrical periodic pulse train v_{in} of 10 V amplitude with on-time T_{ON} = 1 ms and off-time T_{OFF} = 1 μs is applied to the circuit shown in the figure. The diode D_1 is ideal.

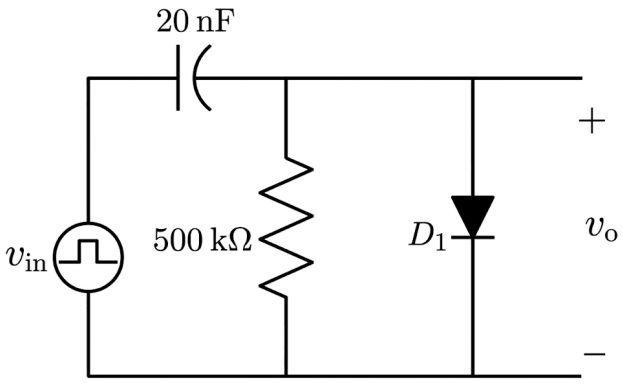


Fig. Q1.21.1

The difference between the maximum voltage and minimum voltage of the output waveform v_o (*in integer*) is _____ V.

Q1.20.1 NAT 2020 1M

In the circuit shown below, all the components are ideal and the input voltage is sinusoidal. The magnitude of the steady-state output V_o (*rounded off to two decimal places*) is _____ V.

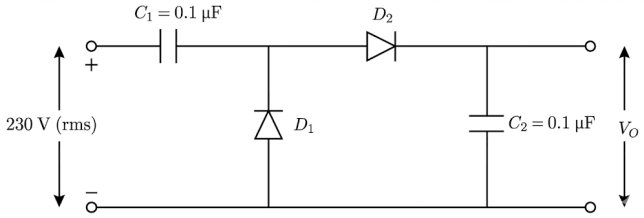


Fig. Q1.20.1

Q1.19.1 NAT 2019 1M

In the circuit shown, V_s is a square wave of period T with maximum and minimum values of 8 V and −10 V, respectively. Assume that the diode is ideal and $R_1 = R_2 = 50\,\Omega$. The average value of V_L is _____ volts *(rounded off to 1 decimal place)*.

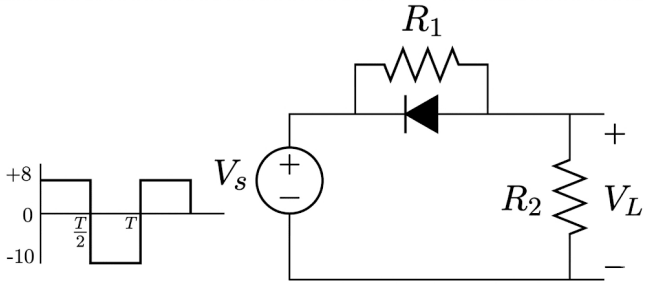


Fig. Q1.19.1

Q1.19.2 MCQ 2019 2M

In the circuit shown, the breakdown voltage and the maximum current of the Zener diode are 20 V and 60 mA, respectively. The values of R_1 and R_L are 200 Ω and 1 kΩ, respectively. What is the range of V_i that will maintain the Zener diode in the 'on' state?

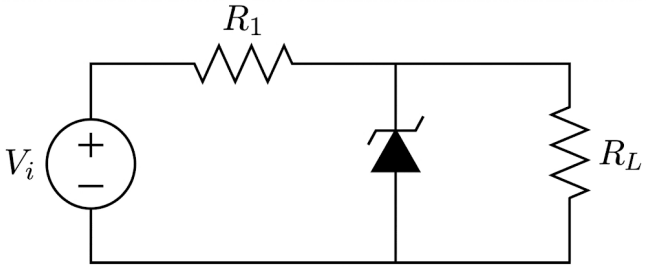


Fig. Q1.19.2

A. 22 V to 34 V
B. 24 V to 36 V
C. 18 V to 24 V
D. 20 V to 28 V

Q1.18.1 MCQ 2018 2M

The circuit shown in the figure is used to provide regulated voltage (5 V) across the 1 kΩ resistor. Assume that the Zener diode has a constant reverse breakdown voltage for a current range, starting from a minimum required Zener current, $I_{Z\,\min} = 2$ mA to its maximum allowable current. The input voltage V_I may vary by 5% from its nominal value of 6 V. The resistance of the diode in the breakdown region is negligible.

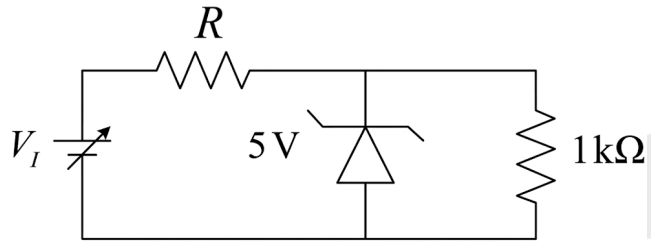


Fig. Q1.18.1

The value of R and the minimum required power dissipation rating of the diode, respectively, are

A. 186 Ω and 10 mW **B.** 100 Ω and 40 mW
C. 100 Ω and 10 mW **D.** 186 Ω and 40 mW

Q1.18.2 NAT 2018 1M

A dc current of 26 μA flows through the circuit shown. The diode in the circuit is forward biased and it has an ideality factor of one. At the quiescent point, the diode has a junction capacitance of 0.5 nF . Its neutral region resistances can be neglected. Assume that the room temperature thermal equivalent voltage is 26 mV.

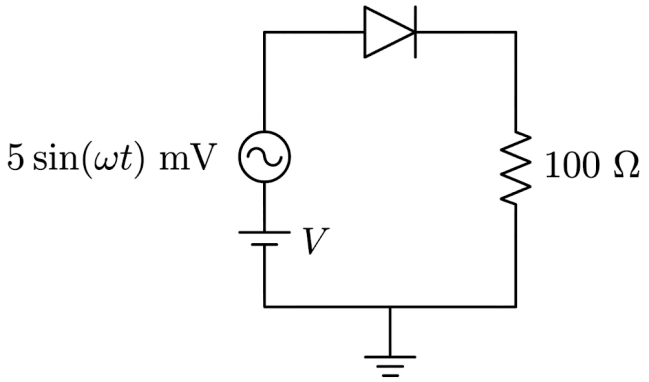


Fig. Q1.18.2

For $\omega = 2 \times 10^6$ rad/s , the amplitude of the small-signal component of diode current (in μA, correct to one decimal place) is _____.

Q1.17.1 NAT 2017 1M

The output V_0 of the diode circuit shown in the figure is connected to an averaging DC voltmeter. The reading on the DC voltmeter in Volts, neglecting the voltage drop across the diode, is _____.

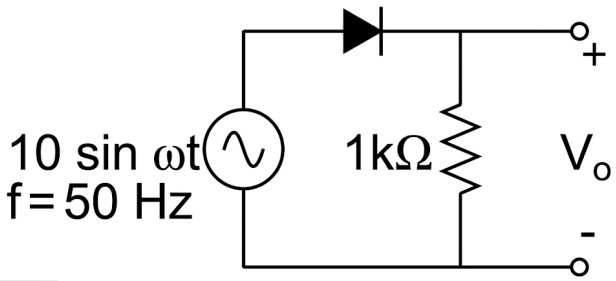


Fig. Q1.17.1

Q1.17.2 MCQ 2017 1M

In the figure, D1 is a real silicon pn junction diode with a drop of 0.7 V under forward bias condition and D2 is a Zener diode with breakdown voltage of −6.8 V. The input $V_{\text{in}}(t)$ is a periodic square wave of period T, whose one period is shown in the figure.

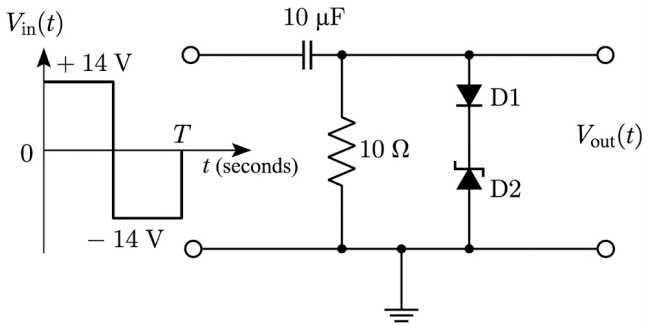


Fig. Q1.17.2

Assuming $10\tau \ll$ T, where τ is the time constant of the circuit, the maximum and minimum values of the output waveform are respectively,

A. 7.5 V and −20.5 V **B.** 6.1 V and −21.9 V
C. 7.5 V and −21.2 V **D.** 6.1 V and −22.6 V

Q1.16.1 NAT 2016 2M

An AC voltage source $V = 10\sin(t)$ volts is applied to the following network. Assume that $R_1 = 3\,k\Omega$, $R_2 = 6\,k\Omega$ and $R_3 = 9\,k\Omega$, and that the diode is ideal.

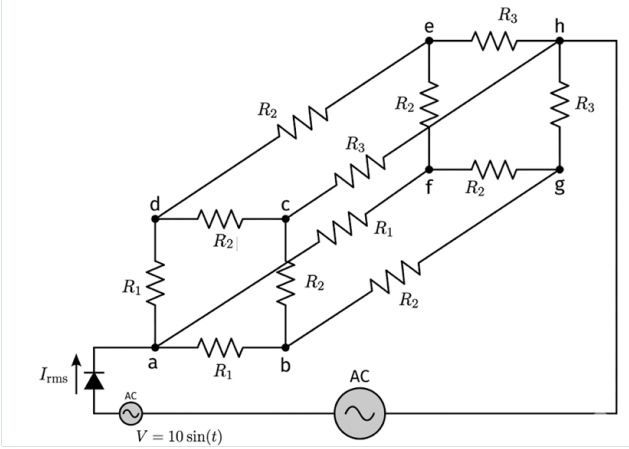


Fig. Q1.16.1

RMS current I_{rms} (in mA) through the diode is _____

Q1.16.2 NAT 2016 1M

Assume that the diode in the figure has $V_{on} = 0.7$ V, but is otherwise ideal.

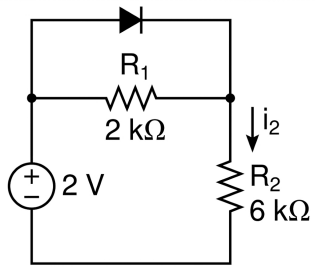


Fig. Q1.16.2

The magnitude of the current i_2 (in mA) is equal to _____

Q1.16.3 NAT 2016 2M

The figure shows a half-wave rectifier with a 475 μF filter capacitor. The load draws a constant current $I_O = 1$ A from the rectifier. The figure also shows the input voltage V_i, the output voltage V_C and the peak-to-peak voltage ripple u on V_C. The input voltage V_i is a triangle-wave with an amplitude of 10 V and a period of 1 ms.

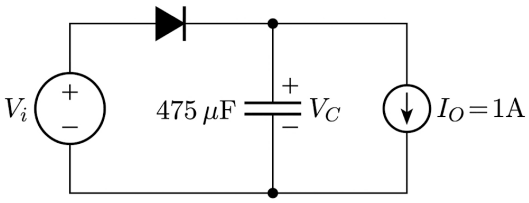


Fig. Q1.16.3

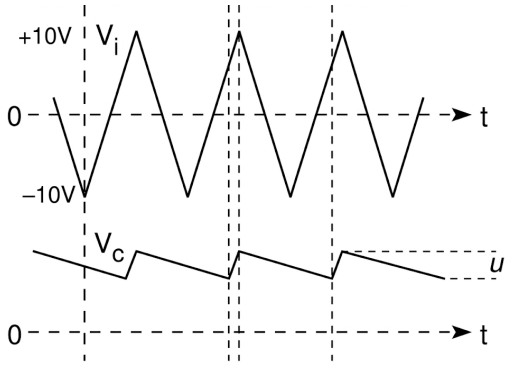


Fig. Q1.16.3

The value of the ripple u (in volts) is _____

Q1.16.4 NAT 2016 1M

The diodes D1 and D2 in the figure are ideal and the capacitors are identical. The product RC is very large compared to the time period of the ac voltage. Assuming that the diodes do not breakdown in the reverse bias, the output voltage V_o (in volt) at the steady state is _____

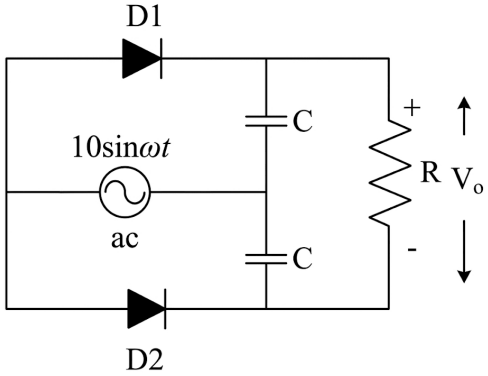


Fig. Q1.16.4

Q1.16.5 MCQ 2016 2M

The I-V characteristics of the zener diodes D1 and D2 are shown in Figure I. These diodes are used in the circuit given in Figure II. If the supply voltage is varied from 0 to 100 V, then breakdown occurs in

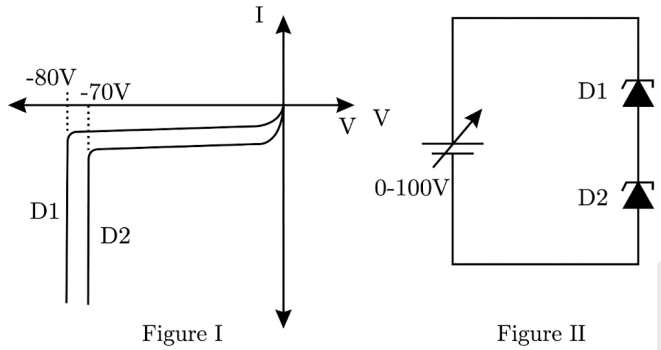


Fig. Q1.16.5

A. D1 only
B. D2 only
C. both D1and D2
D. none of D1 and D2

Q1.16.6 NAT 2016 2M

The switch S in the circuit shown has been closed for a long time. It is opened at time $t = 0$ and remains open after that. Assume that the diode has zero reverse current and zero forward voltage drop.

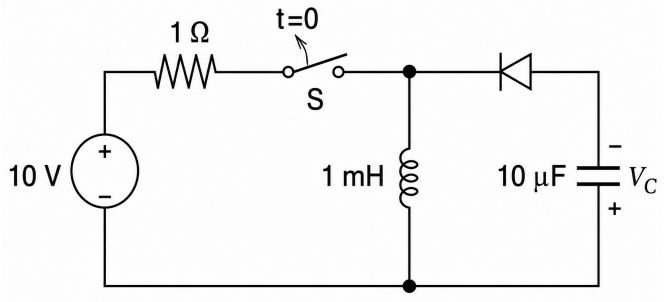


Fig. Q1.16.6

The steady state magnitude of the capacitor voltage V_C (in volts) is _____

Q1.15.1 MCQ 2015 1M

For the circuit with ideal diodes shown in the figure, the shape of the output (v_{out}) for the given sine wave input (v_{in}) will be

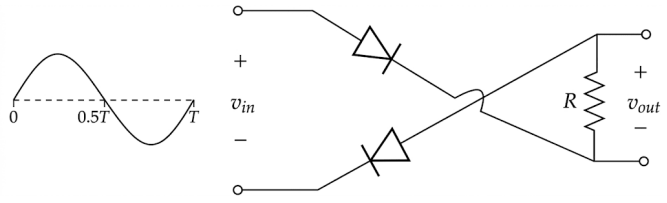
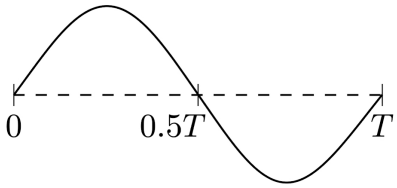
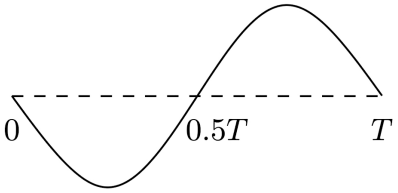
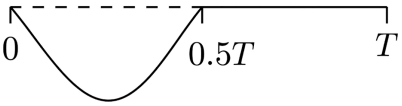
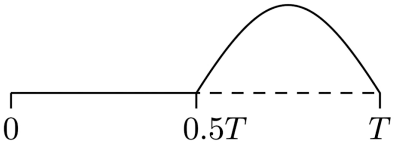


Fig. Q1.15.1

A.


B.


C.


D.


Q1.15.2 NAT 2015 1M

In the circuit shown below, the Zener diode is ideal and the Zener voltage is 6 V. The output voltage V_o (in volts) is _____.

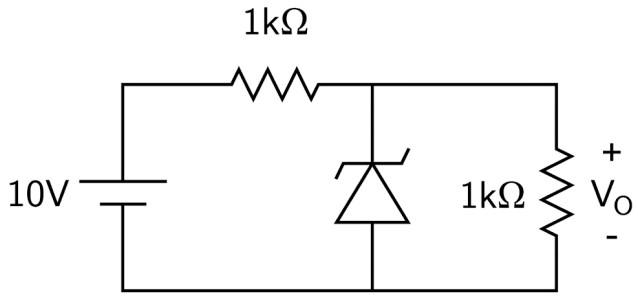


Fig. Q1.15.2

Q1.15.3 MCQ 2015 1M

If the circuit shown has to function as a clamping circuit, then which one of the following conditions should be satisfied for the sinusoidal signal of period T ?

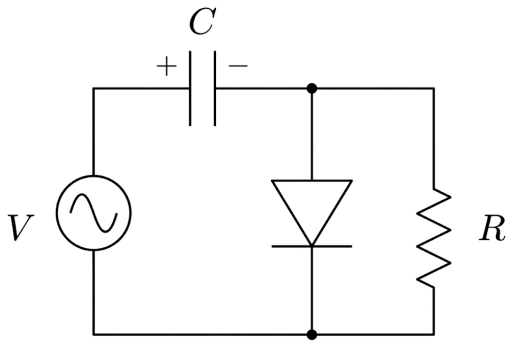


Fig. Q1.15.3

A. $RC \ll T$ **B.** $RC = 0.35T$ **C.** $RC \approx T$ **D.** $RC \gg T$

Q1.15.4 NAT 2015 2M

The diode in the circuit given below has $V_{ON} = 0.7$ V but is ideal otherwise. The current (in mA) in the 4 kΩ resistor is _____.

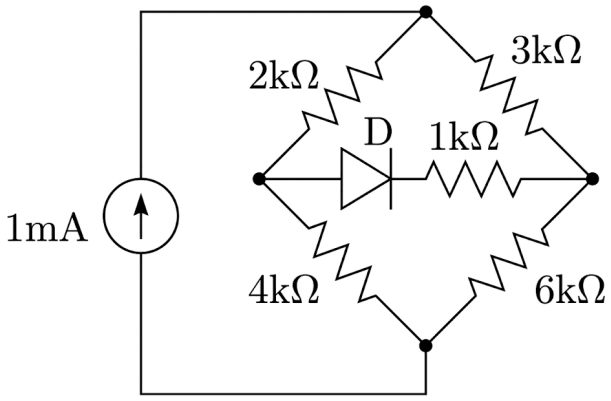


Fig. Q1.15.4

Q1.15.5 NAT 2015 2M

In the circuit shown, assume that the diodes D_1 and D_2 are ideal. The average value of voltage V_{ab} (in Volts), across terminals 'a' and 'b' is _____.

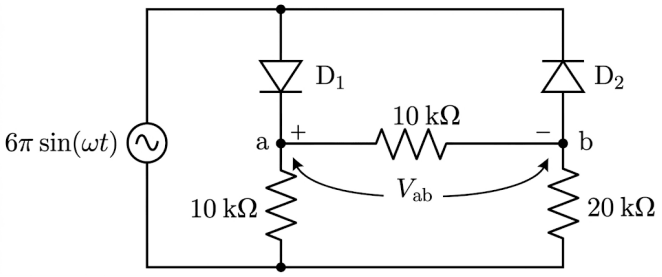


Fig. Q1.15.5

Q1.15.6 NAT 2015 1M

In the circuit shown, assume that diodes D_1 and D_2 are ideal. In the steady state condition, the average voltage V_{ab} (in Volts) across the 0.5 μF capacitor is _____.

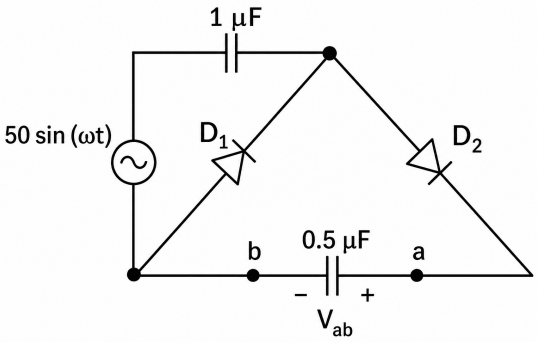


Fig. Q1.15.6

Q1.14.1 MCQ 2014 2M

In the figure, assume that the forward voltage drops of the PN diode D_1 and Schottky diode D_2 are 0.7 V and 0.3 V, respectively. If ON denotes conducting state of the diode and OFF denotes non-conducting state of the diode, then in the circuit,

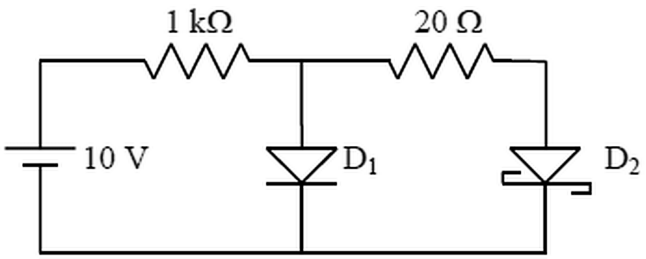


Fig. Q1.14.1

A. both D_1 and D_2 are ON

B. D_1 is ON and D_2 is OFF

C. both D_1 and D_2 are OFF

D. D_1 is OFF and D_2 is ON

Q1.14.2 MCQ 2014 2M

The diode in the circuit shown has $V_{on} = 0.7$ Volts but is ideal otherwise. If $V_i = 5\sin(\omega t)$ Volts, the minimum and maximum values of V_o (in Volts) are, respectively,

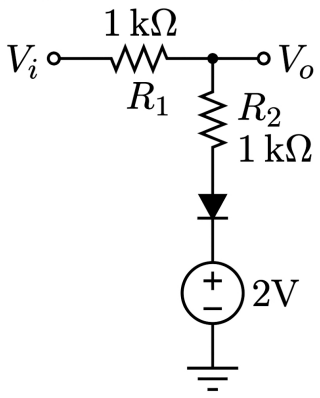


Fig. Q1.14.2

A. −5 and 2.7
B. 2.7 and 5
C. −5 and 3.85
D. 1.3 and 5

Q1.14.3 NAT 2014 1M

The figure shows a half-wave rectifier. The diode D is ideal. The average steady-state current (in Amperes) through the diode is approximately _____.

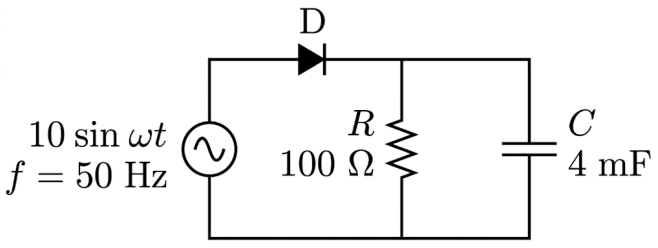


Fig. Q1.14.3

Q1.14.4 MCQ 2014 1M

Two silicon diodes, with a forward voltage drop of 0.7 V, are used in the circuit shown in the figure. The range of input voltage V_i for which the output voltage $V_o = V_i$, is

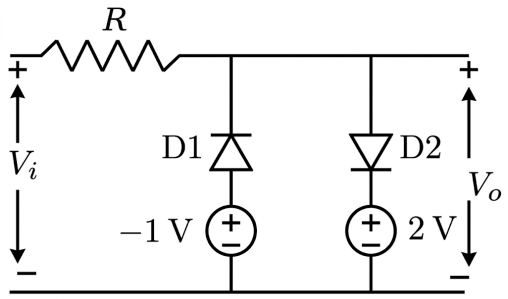


Fig. Q1.14.4

A. $-0.3\,\text{V} < V_i < 1.3\,\text{V}$

B. $-0.3\,\text{V} < V_i < 2\,\text{V}$

C. $-1.0\,\text{V} < V_i < 2.0\,\text{V}$

D. $-1.7\,\text{V} < V_i < 2.7\,\text{V}$

Q1.13.1 MCQ 2013 2M

In the circuit shown below, the knee current of the ideal Zener diode is 10 mA. To maintain 5 V across R_L, the minimum value of R_L in Ω and the minimum power rating of the Zener diode in mW, respectively, are

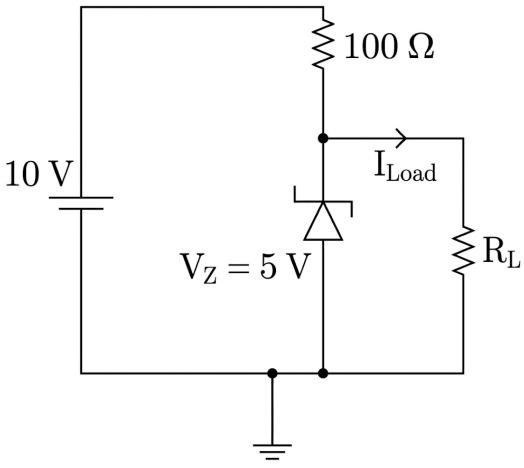


Fig. Q1.13.1

A. 125 and 125
B. 125 and 250
C. 250 and 125
D. 250 and 250

Q1.13.2 MCQ 2013 2M

A voltage 1000 sin ωt Volts is applied across YZ. Assuming ideal diodes, the voltage measured across WX in Volts, is

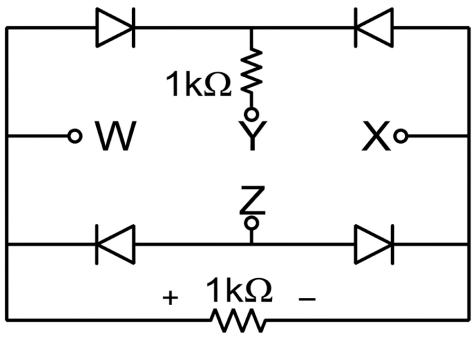


Fig. Q1.13.2

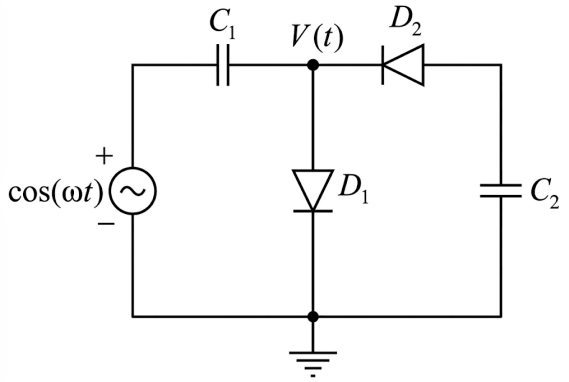


Fig. Q1.12.2

A. $\sin \omega t$ **B.** $(\sin \omega t + |\sin \omega t|)/2$
C. $(\sin \omega t - |\sin \omega t|)/2$ **D.** 0 for all t

Q1.12.1 MCQ 2012 1M

The i-v characteristics of the diode in the circuit given below are

$$i = \begin{cases} \dfrac{v - 0.7}{500}\ \text{A}, & v \geq 0.7\ \text{V} \\ 0\ \text{A}, & v < 0.7\ \text{V} \end{cases}$$

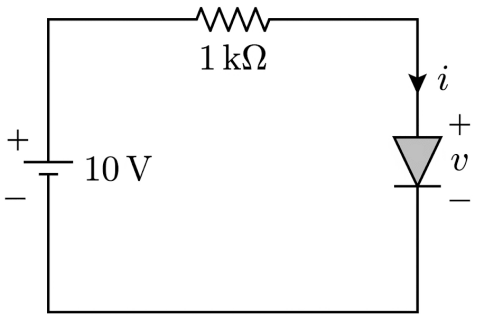


Fig. Q1.12.1

The current in the circuit is

A. 10 mA **B.** 9.3 mA **C.** 6.67 mA **D.** 6.2 mA

Q1.12.2 MCQ 2012 1M

The diodes and capacitors in the circuit shown are ideal. The voltage $V(t)$ across the diode D_1 is

A. $\cos(\omega t) - 1$ **B.** $\sin(\omega t)$
C. $1 - \cos(\omega t)$ **D.** $1 - \sin(\omega t)$

Q1.11.1 MCQ 2011 1M

A Zener diode, when used in voltage stabilization circuits, is biased in

A. reverse bias region below the breakdown voltage

B. reverse breakdown region

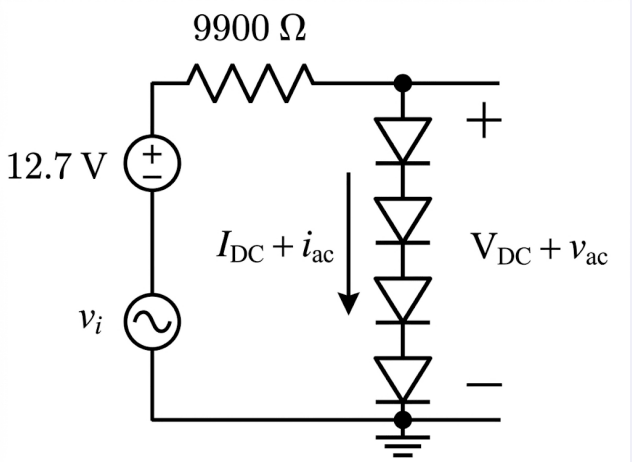
C. forward bias region

D. forward bias constant current mode

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q1.11.2 and Q1.11.3:
In the circuit shown below, assume that the voltage drop across a forward biased diode is 0.7 V. The thermal voltage V_t = kT/q = 25mV. The small signal input $v_i = V_p \cos(\omega t)$ where V_p = 100 mV.



Q1.11.2 MCQ 2011 2M

The bias current I_{DC} through the diodes is

A. 1 mA **B.** 1.28 mA **C.** 1.5 mA **D.** 2 mA

Q1.11.3 MCQ 2011 2M

The ac output voltage v_{ac} is

A. $0.25\cos(\omega t)$ mV **B.** $1\cos(\omega t)$ mV
C. $2\cos(\omega t)$ mV **D.** $22\cos(\omega t)$ mV

Q1.09.1 MCQ 2009 2M

In the circuit below, the diode is ideal. The voltage V is given by

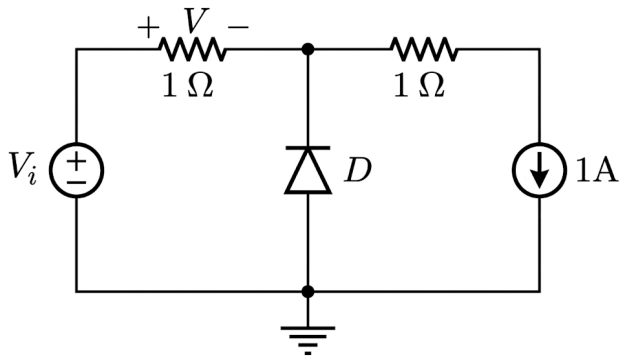


Fig. Q1.09.1

A. $\min(V_i, 1)$ **B.** $\max(V_i, 1)$
C. $\min(-V_i, 1)$ **D.** $\max(-V_i, 1)$

Q1.08.1 MCQ 2008 1M

In the following limiter circuit, an input voltage $V_i = 10\sin 100\pi t$ V is applied. Assume that the diode drop is 0.7 V when it is forward biased. The Zener breakdown voltage is 6.8 V.

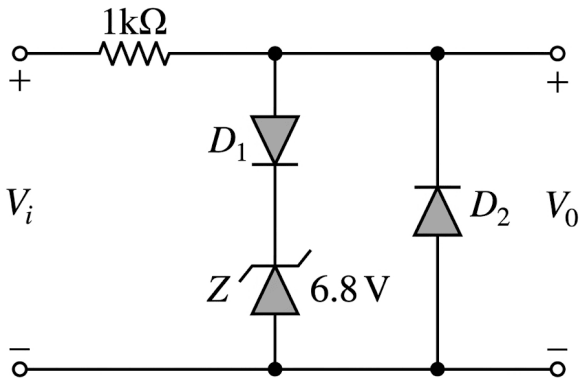


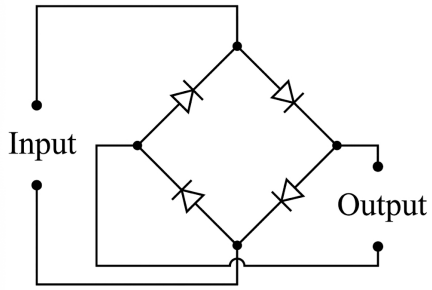
Fig. Q1.08.1

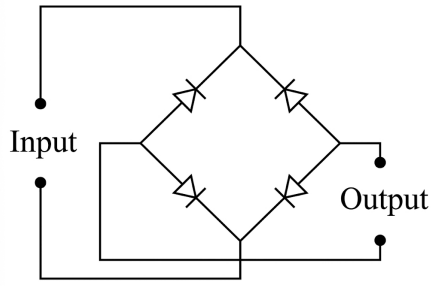
The maximum and minimum values of the output voltage respectively are

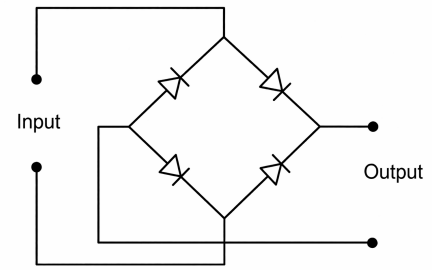
A. 6.1 V, −0.7 V **B.** 0.7 V, −7.5 V
C. 7.5 V, −0.7 V **D.** 7.5 V, −7.5 V

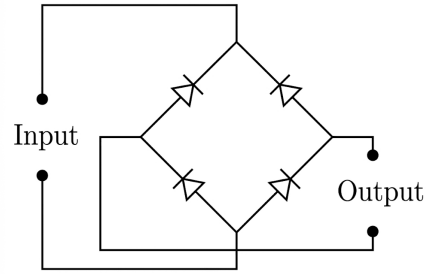
Q1.07.1 MCQ 2007 1M

The correct full wave rectifier circuit is

A.


B.


C.


D.


Q1.07.2 MCQ 2007 2M

For the Zener diode shown in the figure, the Zener voltage at knee is 7 V, the knee current is negligible and the Zener dynamic resistance is 10 Ω. If the input voltage (V_i) range is from 10 to 16 V, the output voltage (V_0) ranges from

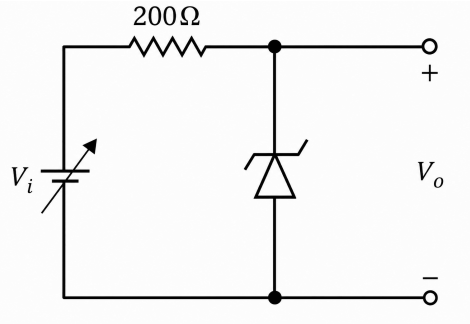


Fig. Q1.07.2

A. 7.00 to 7.29 V **B.** 7.14 to 7.29 V
C. 7.14 to 7.43 V **D.** 7.29 to 7.43 V

Q1.06.1 MCQ 2006 2M

For the circuit shown below, assume that the Zener diode is ideal with a breakdown voltage of 6 volts. The waveform observed across R is

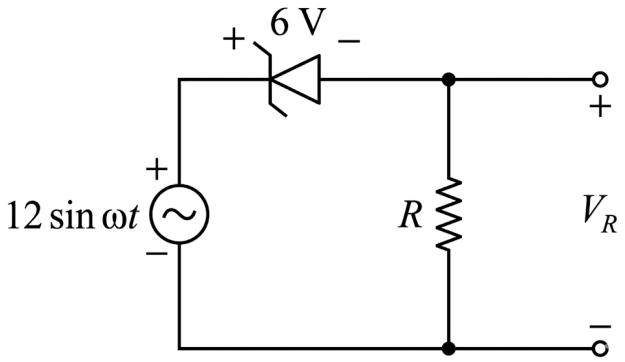
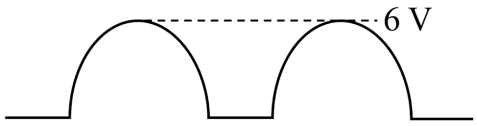
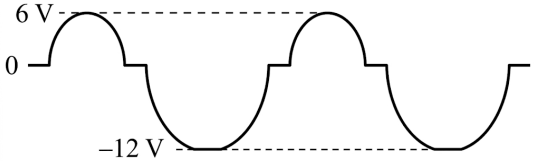


Fig. Q1.06.1

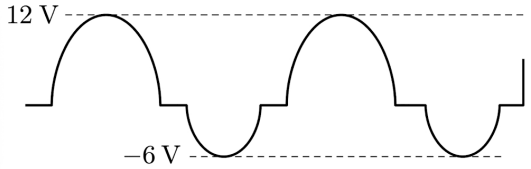
A.



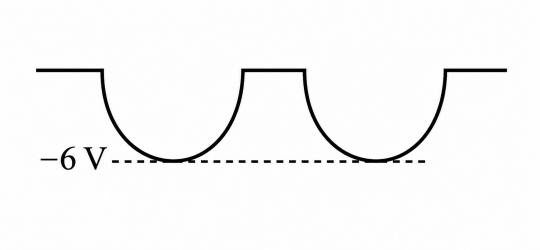
B.



C.



D.



Q1.05.1 MCQ 2005 2M

The Zener diode in the regulator circuit shown in figure has a Zener voltage of 5.8 Volts and a Zener knee current of 0.5 mA. The maximum load current drawn from this circuit ensuring proper functioning over the input voltage range between 20 and 30 Volts, is

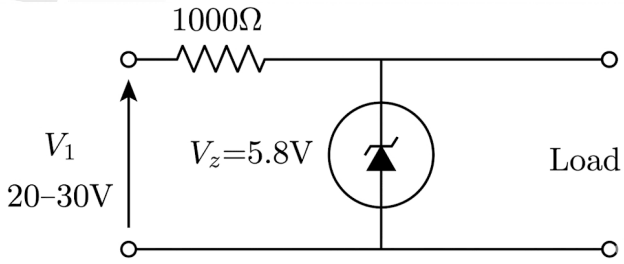


Fig. Q1.05.1

A. 23.7 mA **B.** 14.2 mA **C.** 13.7 mA **D.** 24.2 mA

Q1.04.1 MCQ 2004 2M

In a full-wave rectifier using two ideal diodes, V_{dc} and V_m are the dc and peak values of the voltage respectively across a resistive load. If PIV is the peak inverse voltage of the diode, then the appropriate relationships for this rectifier are

A. $V_{dc} = \frac{V_m}{\pi}$, PIV = $2V_m$

B. $V_{dc} = 2\frac{V_m}{\pi}$, PIV = $2V_m$

C. $V_{dc} = 2\frac{V_m}{\pi}$, PIV = V_m

D. $V_{dc} = \frac{V_m}{\pi}$, PIV = V_m

Q1.04.2 MCQ 2004 2M

In the voltage regulator shown in the figure, the load current can vary from 100 mA to 500 mA. Assuming that the Zener diode is ideal (i.e., the Zener knee current is negligibly small and Zener resistance is zero in the breakdown region), the value of R is

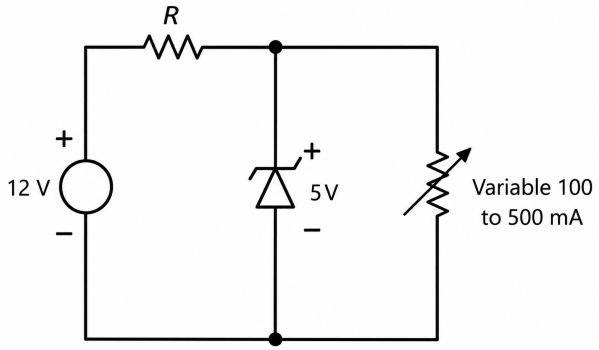


Fig. Q1.04.2

A. 7 Ω **B.** 70 Ω **C.** $\frac{70}{3}$ Ω **D.** 14 Ω

Q1.03.1 MCQ 2003 1M

The circuit shown in the figure is best described as a

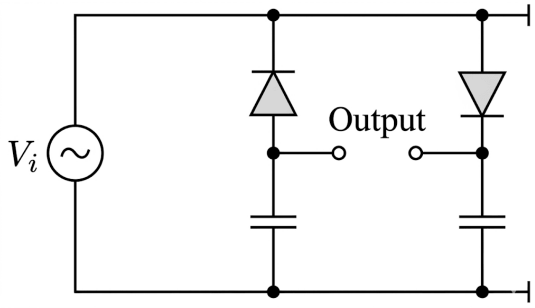


Fig. Q1.03.1

A. bridge rectifier **B.** ring modulator
C. frequency discriminatory **D.** voltage doubler

Q1.02.1 MCQ 2002 1M

In the given figure, a silicon is carrying a constant current of 1 mA. When the temperature of the diode is 20°C, V_D is found to be 700 mV. If the temperature rises to 40°C, V_D becomes approximately equal to

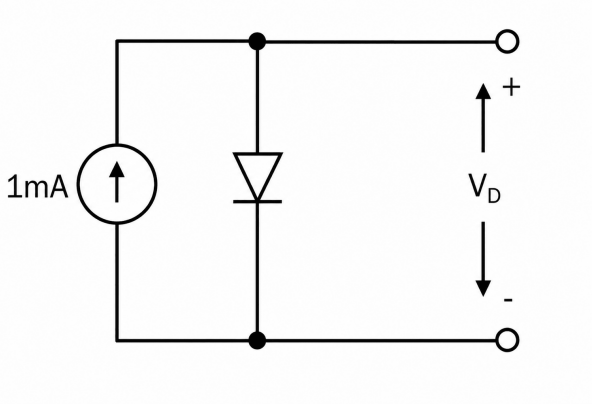


Fig. Q1.02.1

A. 740 mV **B.** 660 mV **C.** 680 mV **D.** 700 mV

Q1.02.2 MCQ 2002 2M

A zener diode regulator in figure is to be designed to meet the specifications: I_L = 10 mA, V_0 = 10 V and V_{in} varies from 30 V to 50 V. The zener diode has V_z = 10 V and I_{zk} (knee current) = 1 mA. For satisfactory operation

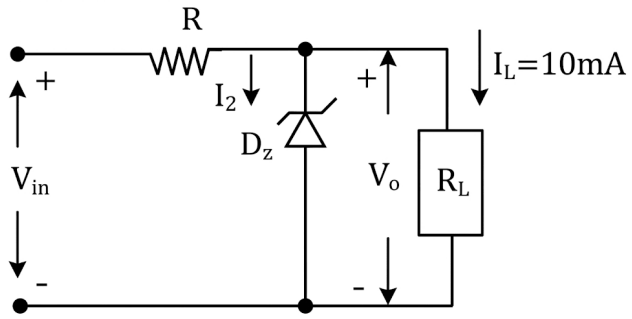


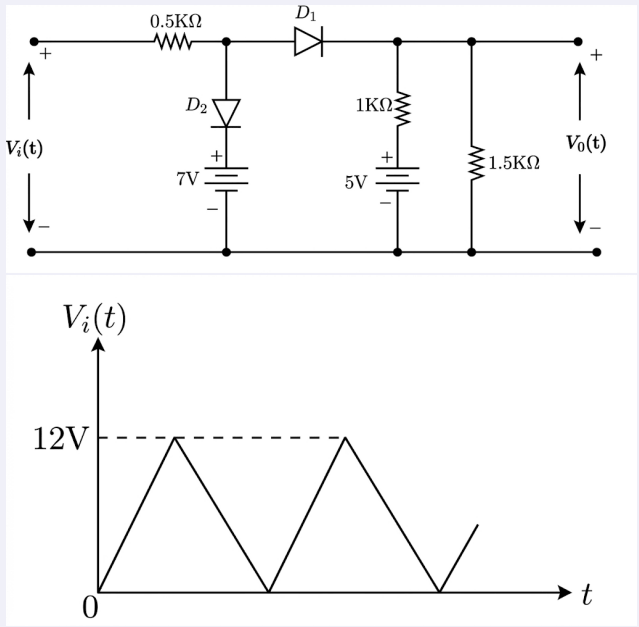
Fig. Q1.02.2

A. $R \leq 1800\,\Omega$ **B.** $2000\,\Omega \leq R \leq 2200\,\Omega$
C. $3700\,\Omega \leq R \leq 4000\,\Omega$ **D.** $R > 4000\,\Omega$

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions Q1.02.3 and Q1.02.4 :
A triangular voltage waveform $V_i(t)$ is applied at the input to the circuit of figure. Assume the diodes to be ideal.



Q1.02.3 NAT 2002

Determine the output $V_0(t)$.

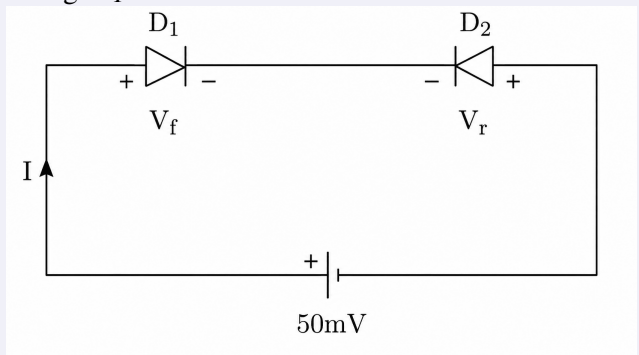
Q1.02.4 NAT 2002

Neatly sketch the output waveform superimposed on the input $V_i(t)$ and label the key points.

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions Q1.01.1 and Q1.01.2 :
For the circuit shown in figure, D_1 and D_2 are identical diodes with ideality factor of unity. The thermal voltage V_T = 25 mV.



Q1.01.1 NAT 2001

Calculate V_f and V_r.

Q1.01.2 NAT 2001

For the shared circuit, the two identical diodes are oppositely oriented in series across 50 mV, with ideality factor 1 and thermal voltage 25 mV. If their reverse saturation current I_s is 1 pA, compute the current I (in pA, rounded off to three decimal places): _____.

Q1.00.1 MCQ 2000 2M

For the circuit in the figure, the voltage v_0 is

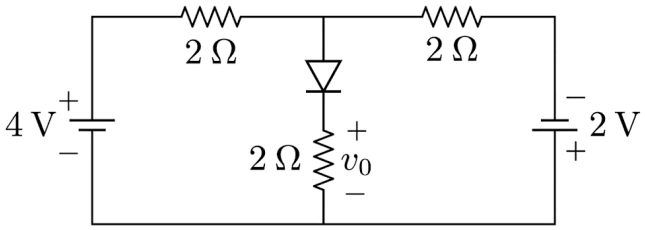


Fig. Q1.00.1

A. 2 V **B.** 1 V **C.** −1 V **D.** None of these

Q1.99.1 MCQ 1999 1M

A dc power supply has a no-load voltage of 30 V, and a full-load voltage of 25 V at a full-load current of 1 A. Its output resistance and load regulation, respectively, are

A. 5 Ω and 20% **B.** 25 Ω and 20%
C. 5 Ω and 16.7% **D.** 25 Ω and 16.7%

Q1.98.1 MCQ 1998 1M

For full wave rectification, a four diode bridge rectifier is claimed to have the following advantages over a two diode circuit.
1. Less expensive transformer
2. Smaller size transformer
3. Suitability for higher voltage application

Of these,

A. Only (1) and (2) are true

B. Only (1) and (3) are true

C. Only (2) and (3) are true

D. (1), (2), as well as (3) are true

Q1.98.2 MCQ 1998 1M

For small signal a.c. operation, a practical forward biased diode can be modeled as

A. a resistance and a capacitance in series.

B. an ideal diode and resistance in parallel.

C. a resistance and an ideal diode in series.

D. a resistance.

Q1.97.1 MCQ 1997 1M

In the circuit of in the figure is the current i_D through the ideal diode (zero cut in voltage and forward resistance) equals

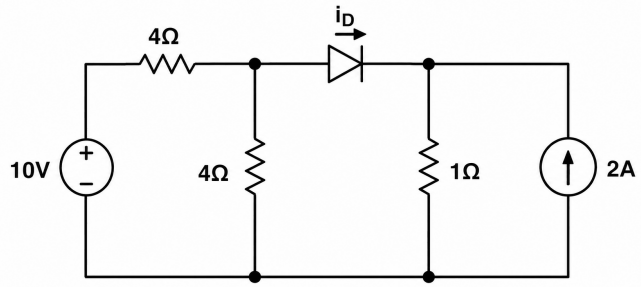


Fig. Q1.97.1

A. 0 A **B.** 4 A **C.** 1 A **D.** None of the above

Q1.96.1 MCQ 1996 1M

In the circuit of the given figure, assume that the diodes are ideal and the meter is an average indicating ammeter. The ammeter will read

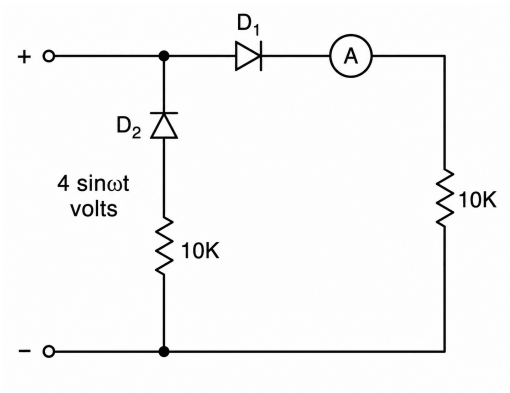


Fig. Q1.96.1

A. $0.4\sqrt{2}$ mA **B.** 0.4 mA **C.** $\frac{0.8}{\pi}$ mA **D.** $\frac{0.4}{\pi}$ mA

Q1.96.2 MCQ 1996 2M

A zener diode in the circuit shown in the figure is has a knee current of 5 mA, and a maximum allowed power dissipation of 300 mW. What are the minimum and maximum load currents that can be drawn safely from the circuit, keeping the output voltage V_o constant at 6 V ?

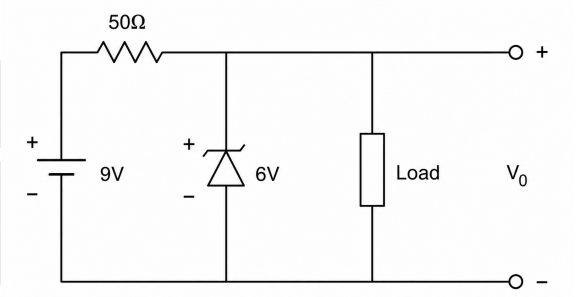


Fig. Q1.96.2

A. 0 mA, 180 mA **B.** 5 mA, 110 mA
C. 10 mA, 55 mA **D.** 60 mA, 180 mA

Q1.93.1 MCQ 1993 1M

The wave shape of V_0 in figure is

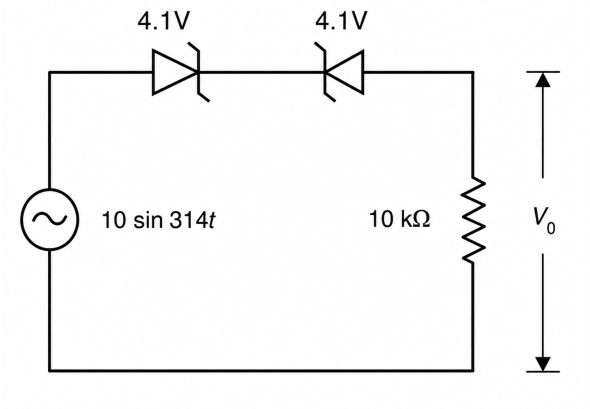
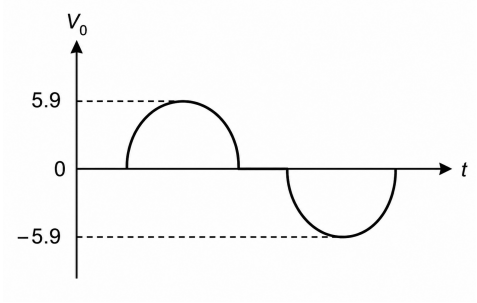
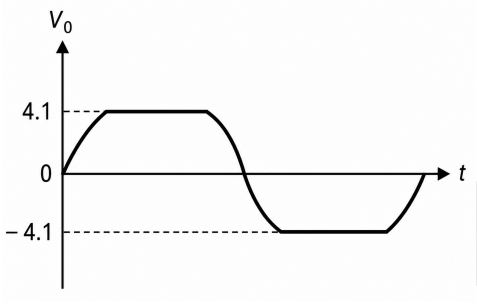


Fig. Q1.93.1

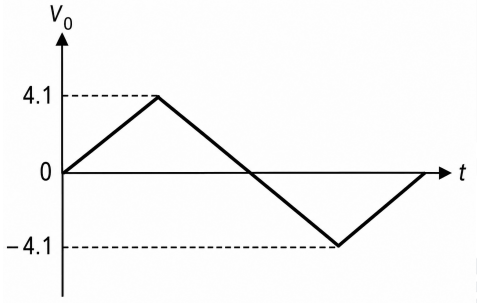
A.



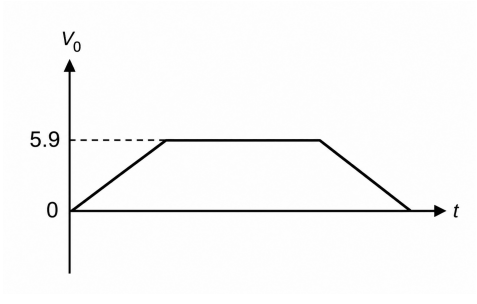
B.



C.



D.



Q1.93.2 MCQ 1993 1M

In figure, the ideal moving iron voltmeter M will read

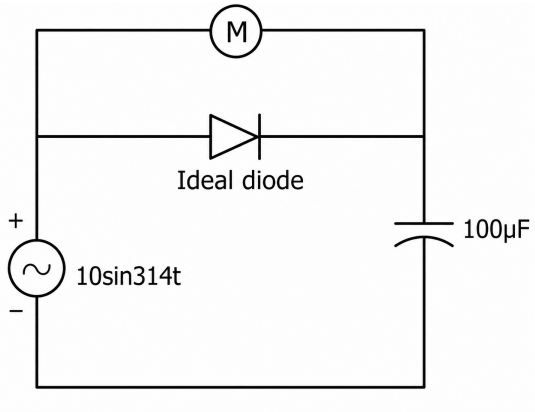


Fig. Q1.93.2

A. 7.07 V **B.** 12.24 V **C.** 14.14 V **D.** 20.0 V

Q1.92.1 MCQ 1992 2M

The 6 V Zener diode shown in figure has zero zener resistance and a knee current of 5 mA. The minimum value of R so that the voltage across it does not fall below 6 V is:

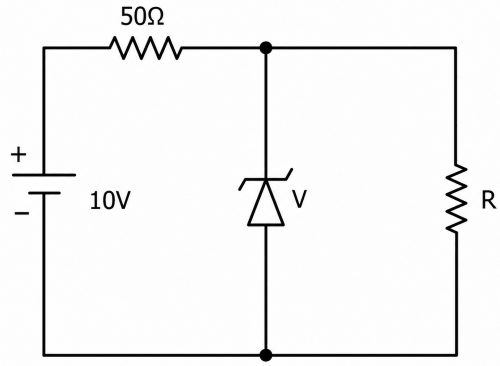


Fig. Q1.92.1

A. 1.2 kΩ **B.** 80 Ω **C.** 50 Ω **D.** 0 Ω

Q1.91.1 NAT 1991

In figure the input V_i is a 100 Hz triangular wave having a peak to peak amplitude of 2 volts and an average value of zero volts. Given that the diode is ideal, the average value of the output V_0 is _____.

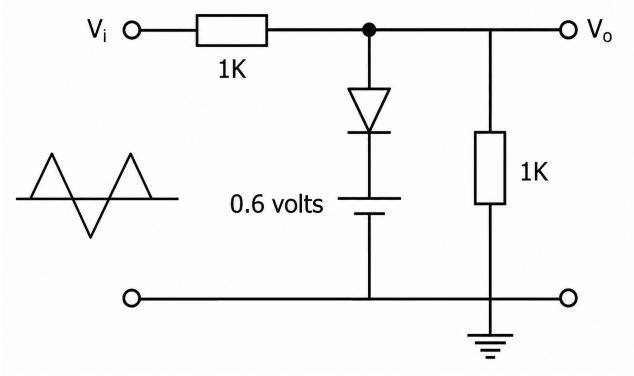


Fig. Q1.91.1

Q1.90.1 MSQ 1990

For the same a.c. voltage and load impedance, which of the following statements about rectifier are correct?

- **A.** The average load current in a full wave rectifier is twice than in a half wave rectifier
- **B.** The average load current in a full wave rectifier is n times that in a half wave rectifier
- **C.** Half wave rectifier will have a bigger sized transformer compared to full wave rectifier
- **D.** Half wave rectifier will have a small sized transformer compared to full wave rectifier

Q1.90.2 NAT 1990

Two ideal and identical (ideality factor $\eta = 1$) junction diodes are connected in series as shown in figure. Show that

$$\exp(eV_1/KT) + \exp(-eV_2/KT) = 2$$

where V_1 and V_2 are the voltage drops across the diodes D_1 and D_2.

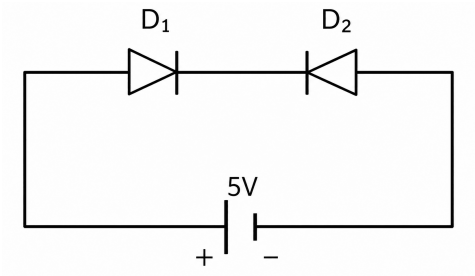


Fig. Q1.90.2

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q1.26.1 4	Q1.25.1 A,D	Q1.25.2 13.33
Q1.24.1 A	Q1.24.2 0.25	Q1.23.1 A
Q1.22.1 0.75	Q1.21.1 10	Q1.20.1 650.54
Q1.19.1 -3	Q1.19.2 B	Q1.18.1 B
Q1.18.2 6.4	Q1.17.1 3.18	Q1.17.2 A
Q1.16.1 1	Q1.16.2 0.25	Q1.16.3 1.9 to 2.2
Q1.16.4 0	Q1.16.5 A	Q1.16.6 100
Q1.15.1 C	Q1.15.2 5	Q1.15.3 D
Q1.15.4 0.6	Q1.15.5 5	Q1.15.6 100
Q1.14.1 D	Q1.14.2 C	Q1.14.3 0.1
Q1.14.4 D	Q1.13.1 B	Q1.13.2 D
Q1.12.1 D	Q1.12.2 A	Q1.11.1 B
Q1.11.2 A	Q1.11.3 B	Q1.09.1 A
Q1.08.1 C	Q1.07.1 C	Q1.07.2 C
Q1.06.1 B	Q1.05.1 C	Q1.04.1 B
Q1.04.2 D	Q1.03.1 D	Q1.02.1 B
Q1.02.2 A	Q1.02.3 *	Q1.02.4 *
Q1.01.1 *	Q1.01.2 0.762	Q1.00.1 D
Q1.99.1 A	Q1.98.1 D	Q1.98.2 D
Q1.97.1 C	Q1.96.1 D	Q1.96.2 C
Q1.93.1 A	Q1.93.2 B	Q1.92.1 B
Q1.91.1 0	Q1.90.1 A,D	Q1.90.2 *

UNIT

II

MOSFETs

Topics

1. Introduction to MOS - Depletion and Enhancement Types, MOS Physics
2. Biasing a MOSFET - Transconductance, Output Impedance and Small Signal Analysis
3. Small Signal Model of MOSFET and Port Impedance
4. Common Source Amplifier - With and Without Degeneration, Different Biasing
5. Miller's Theorem and GmRout Method for Gain
6. Common Gate, Common Drain and PMOS Amplifiers
7. Cascode Current Source, Amplifiers and Current Mirror Circuits
8. Multistage Amplifiers and Pass Transistor Logic
9. Frequency Response of MOSFET and Common Gate Amplifiers
10. Introduction to Differential Amplifiers, Current Source, Voltage Source as Tail
11. Large Signal Analysis of Differential Amplifiers, Concept of Half Circuit
12. Building a Two-Stage Op Amp
13. Types and Effects of Feedback
14. Feedback Topologies - Voltage-Voltage, Voltage-Current, Current-Voltage, Current-Current
15. MOS Amplifiers with Body Effect

Unit Snapshot*

Stream: EC	**Questions:** 74	**MCQ:** 49	**MSQ:** 2	**NAT:** 23
1M: 28	**2M:** 44	**Descriptive:** 2	**Avg Weightage** ~4M	**Range:** 2M(2026)–8M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 NAT 2026 2M

An n-channel MOSFET is connected as shown in the Figure. Assume $V_{TH} = 1\,\text{V}$, $V_{DD} = 5\,\text{V}$, and $\mu C_{ox}\left(\frac{W}{L}\right) = 2\,\text{mA V}^{-2}$ and neglect channel length modulation effects.

The gate voltage (V_G) of the n-channel MOSFET (in Volt) is _____. *(rounded off to two decimal places)*

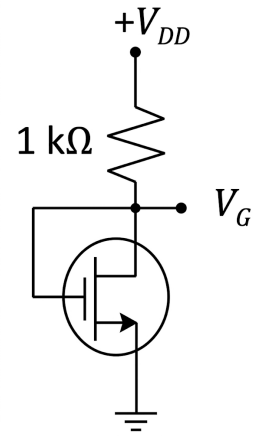


Fig. Q2.26.1

Q2.25.1 MSQ 2025 1M

Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers having finite load resistors?

- ☐ **A.** The common-gate amplifier has an infinite input resistance
- ☐ **B.** The common-source amplifier has an infinite input resistance
- ☐ **C.** The input and output voltages of the common-source amplifier are in phase
- ☐ **D.** The input and output voltages of the common-drain amplifier are in phase

Q2.25.2 MCQ 2025 2M

The identical MOSFETs M_1 and M_2 in the circuit given below are ideal and biased in the saturation region. M_1 and M_2 have a transconductance g_m of 5 mS.

The input signals (in Volts) are:

$V_1 = 2.5 + 0.01 \sin \omega t$

$V_2 = 2.5 - 0.01 \sin \omega t$

The output signal V_3 (in Volts) is _____.

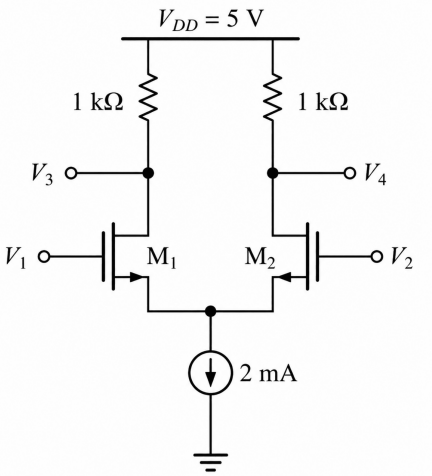


Fig. Q2.25.2

A. $3 + 0.05 \sin \omega t$ **B.** $3 - 0.1 \sin \omega t$

C. $4 + 0.1 \sin \omega t$ **D.** $4 - 0.05 \sin \omega t$

Q2.24.1 MCQ 2024 1M

In the circuit below, assume that the long channel NMOS transistor is biased in saturation. The small signal transconductance of the transistor is g_m. Neglect body effect, channel length modulation and intrinsic device capacitances. The small signal input impedance $Z_{in}(j\omega)$ is _____.

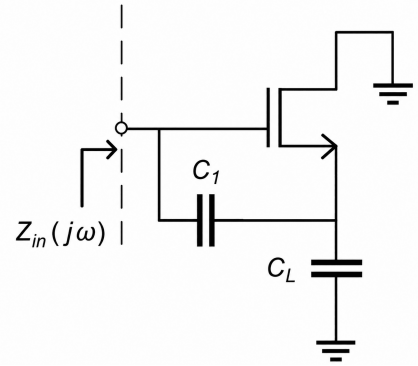


Fig. Q2.24.1

A. $\dfrac{-g_m}{C_1 C_L \omega^2} + \dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$

B. $\dfrac{g_m}{C_1 C_L \omega^2} + \dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$

C. $\dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$

D. $\dfrac{-g_m}{C_1 C_L \omega^2} + \dfrac{1}{j\omega C_1 + j\omega C_L}$

Q2.24.2 MCQ 2024 1M

For the closed loop amplifier circuit shown below, the magnitude of open loop low frequency small signal voltage gain is 40. All the transistors are biased in saturation. The current source I_{SS} is ideal. Neglect body effect, channel length modulation and intrinsic device capacitances. The closed loop low frequency small signal voltage gain $\frac{v_{out}}{v_{in}}$ (rounded off to three decimal places) is _____.

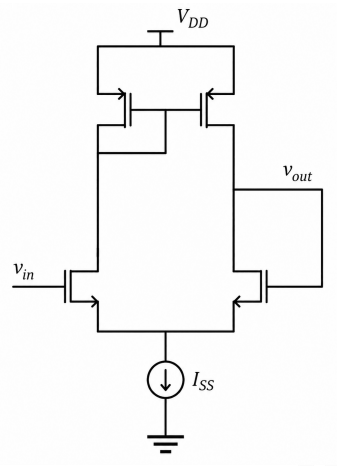


Fig. Q2.24.2

A. 0.976 **B.** 1.000 **C.** 1.025 **D.** 0.488

Q2.24.3 MCQ 2024 2M

In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are g_{m1} and g_{m2} respectively. Neglect body effect, channel length modulation and intrinsic device capacitances.

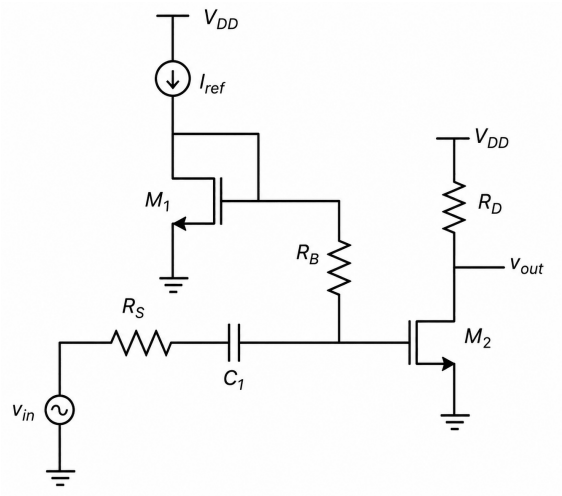


Fig. Q2.24.3

Assuming that capacitor C_1 is a short circuit for AC analysis, the exact magnitude of small signal voltage gain $\left|\frac{v_{out}}{v_{in}}\right|$ is _____.

A. $g_{m2}R_D$

B. $\dfrac{g_{m2}R_D\left(R_B+\dfrac{1}{g_{m1}}\right)}{R_B+\dfrac{1}{g_{m1}}+R_S}$

C. $\dfrac{g_{m2}R_D\left(R_B+\dfrac{1}{g_{m1}}+R_S\right)}{R_B+\dfrac{1}{g_{m1}}}$

D. $\dfrac{g_{m2}R_D\left(\dfrac{1}{g_{m1}}\right)}{\dfrac{1}{g_{m1}}+R_S}$

Q2.23.1 MCQ 2023 1M

In the circuit shown below, V_1 and V_2 are bias voltages. Based on input and output impedances, the circuit behaves as a

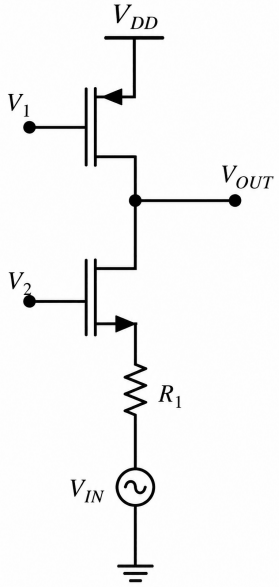


Fig. Q2.23.1

A. voltage controlled voltage source.

B. voltage controlled current source.

C. current controlled voltage source.

D. current controlled current source.

Q2.23.2 MCQ 2023 1M

A cascade of common-source amplifiers in a unity gain feedback configuration oscillates when

A. the loop gain is less than 1 and the phase shift is less than 180°.

B. the loop gain is greater than 1 and the phase shift is less than 180°.

C. the loop gain is less than 1 and the phase shift is greater than 180°.

D. the loop gain is greater than 1 and the phase shift is greater than 180°.

Q2.23.3 NAT 2023 2M

In the circuit below, the voltage V_L is _____ V (*rounded off to two decimal places*).

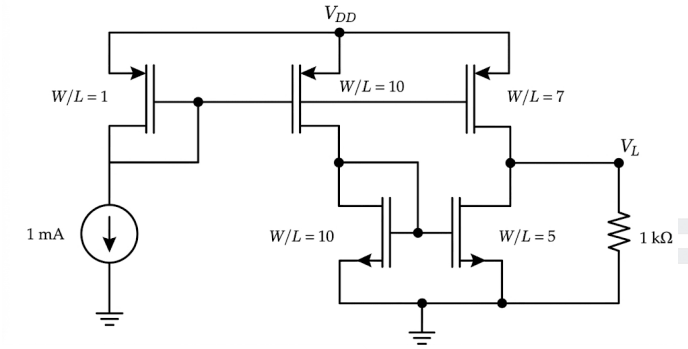


Fig. Q2.23.3

Q2.22.1 MCQ 2022 1M

Consider the CMOS circuit shown in the figure (substrates are connected to their respective sources). The gate width (W) to gate length (L) ratios $\left(\frac{W}{L}\right)$ of the transistors are as shown. Both the transistors have the same gate oxide capacitance per unit area. For the pMOSFET, the threshold voltage is -1 V and the mobility of holes is $40\frac{\text{cm}^2}{\text{V.s}}$. For the nMOSFET, the threshold voltage is 1 V and the mobility of electrons is $300\frac{\text{cm}^2}{\text{V.s}}$. The steady state output voltage V_o is _____.

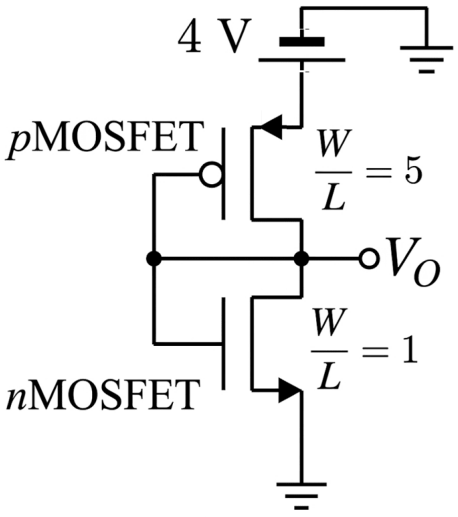


Fig. Q2.22.1

A. equal to 0 V
B. more than 2 V
C. less than 2 V
D. equal to 2 V

Q2.22.2 NAT 2022 2M

Consider the circuit shown with an ideal long channel nMOSFET (enhancement-mode, substrate is connected to the source). The transistor is appropriately biased in the saturation region with V_{GG} and V_{DD} such that it acts as a linear amplifier. v_i is the small-signal ac input voltage. v_A and v_B represent the small-signal voltages at the nodes A and B, respectively. The value of $\frac{v_A}{v_B}$ is _____ (*rounded off to one decimal place*).

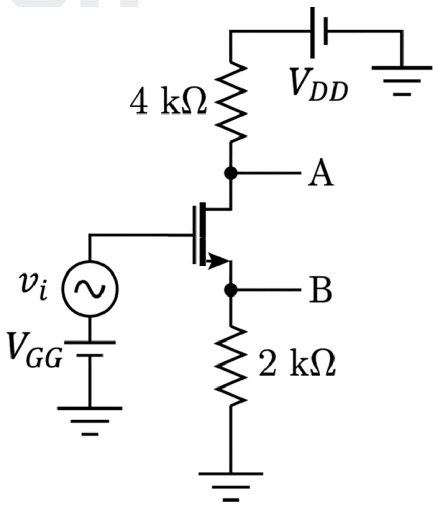


Fig. Q2.22.2

Q2.22.3 MCQ 2022 2M

Consider an ideal long channel *n*MOSFET (enhancement-mode) with gate length 10 μm and width 100 μm. The product of electron mobility (μ_n) and oxide capacitance per unit area (C_{OX}) is $\mu_n C_{OX} = 1$ mA/V^2. The threshold voltage of the transistor is 1 V. For a gate-to-source voltage $V_{GS} = [2 - \sin(2t)]$ V and drain-to-source voltage $V_{DS} = 1$ V (substrate connected to the source), the maximum value of the drain-to-source current is _____.

A. 40 mA **B.** 20 mA **C.** 15 mA **D.** 5 mA

Q2.21.1 MCQ 2021 1M

In the circuit shown in the figure, the transistors M_1 and M_2 are operating in saturation. The channel length modulation coefficients of both the transistors are non-zero. The transconductance of the MOSFETs M_1 and M_2 are g_{m1} and g_{m2}, respectively, and the internal resistance of the MOSFETs M_1 and M_2 are r_{o1} and r_{o2}, respectively.

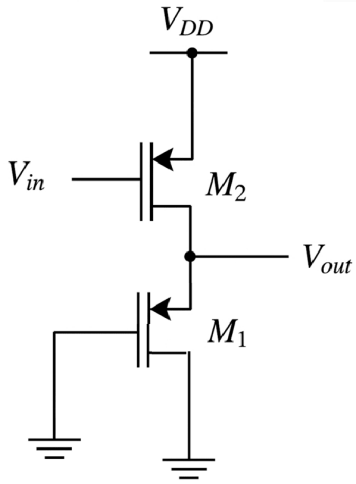


Fig. Q2.21.1

Ignoring the body effect, the ac small signal voltage gain ($\partial V_{out}/\partial V_{in}$) of the circuit is

A. $-g_{m2}(r_{o1}||r_{o2})$ **B.** $-g_{m2}\left(\frac{1}{g_{m1}}||r_{o2}\right)$

C. $-g_{m1}\left(\frac{1}{g_{m2}}||r_{o1}||r_{o2}\right)$ **D.** $-g_{m2}\left(\frac{1}{g_{m1}}||r_{o1}||r_{o2}\right)$

Q2.21.2 NAT 2021 2M

For the transistor M_1 in the circuit shown in the figure, $\mu_n C_{ox} = 100\ \mu$A/V^2 and $(W/L) = 10$, where μ_n is the mobility of electron, C_{ox} is the oxide capacitance per unit area, W is the width and L is the length.

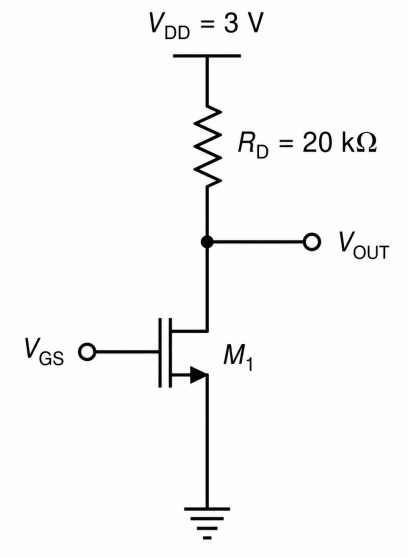


Fig. Q2.21.2

The channel length modulation coefficient is ignored. If the gate-to-source voltage V_{GS} is 1 V to keep the transistor at the edge of saturation, then the threshold voltage of the transistor (*rounded off to one decimal place*) is _____ V.

Q2.20.1 MCQ 2020 2M

Using the incremental low frequency small-signal model of the MOS device, the Norton equivalent resistance of the following circuit is

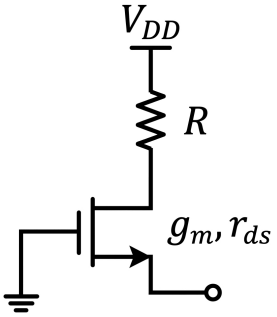


Fig. Q2.20.1

A. $r_{ds} + R + g_m r_{ds} R$ **B.** $\frac{r_{ds} + R}{1 + g_m r_{ds}}$

C. $r_{ds} + \frac{1}{g_m} + R$ **D.** $r_{ds} + R$

Q2.20.2 MCQ 2020 2M

An enhancement MOSFET of threshold voltage 3 V is being used in the sample and hold circuit given below. Assume that the substrate of the MOS device is connected to −10 V. If the input voltage V_I lies between ±10 V, the minimum and the maximum values of V_G required for proper sampling and holding respectively, are

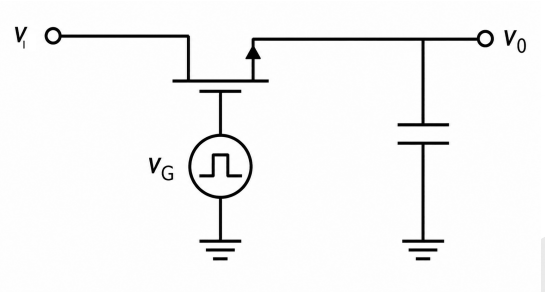


Fig. Q2.20.2

A. 3 V and −3 V.
B. 10 V and −10 V.
C. 13 V and −7 V.
D. 10 V and −13 V.

Q2.19.1 MCQ 2019 2M

In the circuits shown, the threshold voltage of each nMOS transistor is 0.6 V. Ignoring the effect of channel length modulation and body bias, the values of V_{out1} and V_{out2}, respectively, in volts, are

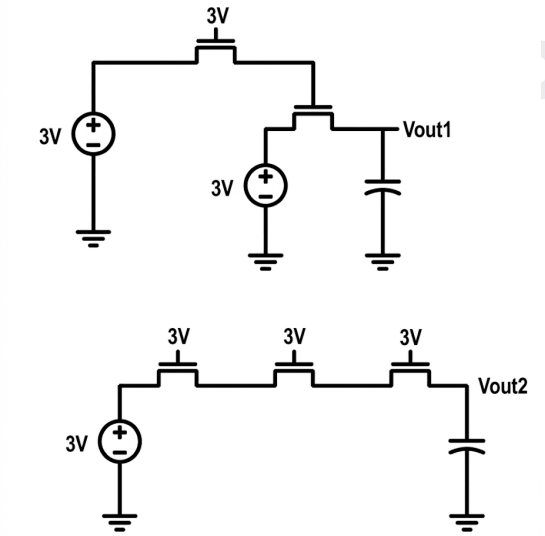


Fig. Q2.19.1

A. 1.8 and 1.2
B. 2.4 and 2.4
C. 1.8 and 2.4
D. 2.4 and 1.2

Q2.19.2 NAT 2019 2M

In the circuit shown, the threshold voltages of the pMOS ($|V_{tp}|$) and nMOS (V_{tn}) transistors are both equal to 1 V. All the transistors have the same output resistance r_{ds} of 6 MΩ. The other parameters are listed below:

$\mu_n C_{ox} = 60\,\mu A/V^2; (W/L)_{nMOS} = 5$
$\mu_p C_{ox} = 30\,\mu A/V^2; (W/L)_{pMOS} = 10$

μ_n and μ_p are the carrier mobilities, and C_{ox} is the oxide capacitance per unit area. Ignoring the effect of channel length modulation and body bias, the gain of the circuit is _____ (rounded off to 1 decimal place).

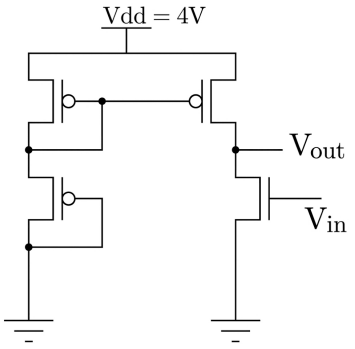


Fig. Q2.19.2

Q2.19.3 NAT 2019 2M

In the circuit shown, $V_1 = 0$ and $V_2 = V_{dd}$. The other relevant parameters are mentioned in the figure. Ignoring the effect of channel length modulation and the body effect, the value of I_{out} is _____ mA (rounded off to 1 decimal place).

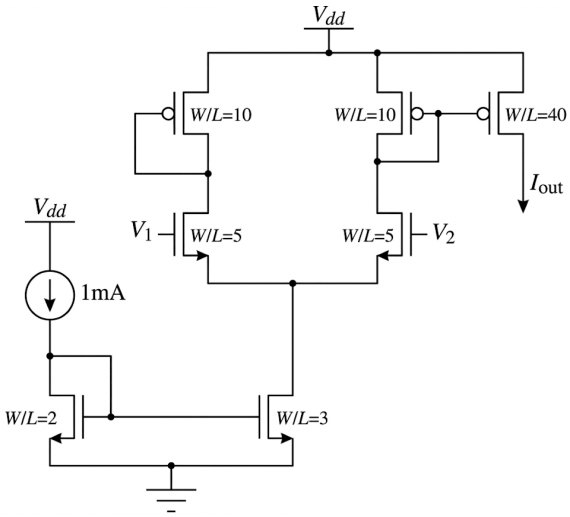


Fig. Q2.19.3

Q2.19.4 NAT 2019 2M

Consider a long-channel MOSFET with a channel length 1 μm and width 10 μm. The device parameters are acceptor concentration $N_A = 5 \times 10^{16}\,\text{cm}^{-3}$, electron mobility $\mu_n = 800\,\text{cm}^2/\text{V-s}$, oxide capacitance/area $C_{ox} = 3.45 \times 10^{-7}\,\text{F/cm}^2$, threshold voltage $V_T = 0.7\,\text{V}$. The drain saturation current (I_{Dsat}) for a gate voltage of 5 V is _____ mA *(rounded off to two decimal places).* $[\varepsilon_0 = 8.854 \times 10^{-14}\,\text{F/cm}, \varepsilon_{Si} = 11.9]$

Q2.18.1 MCQ 2018 1M

Two identical nMOS transistors M_1 and M_2 are connected as shown below. The circuit is used as an amplifier with the input connected between G and S terminals and the output taken between D and S terminals. V_{bias} and V_D are so adjusted that both transistors are in saturation.

The transconductance of this combination is defined as $g_m = \dfrac{\partial i_D}{\partial v_{GS}}$ while the output resistance is $r_o = \dfrac{\partial v_{DS}}{\partial i_D}$, where i_D is the current flowing into the drain of M_2. Let g_{m1}, g_{m2} be the transconductances and r_{o1}, r_{o2} be the output resistances of transistors M_1 and M_2, respectively.

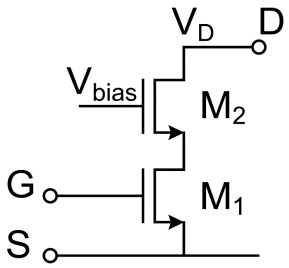


Fig. Q2.18.1

Which of the following statements about estimates for g_m and r_o is correct?

A. $g_m \approx g_{m1} \cdot g_{m2} \cdot r_{o2}$ and $r_o \approx r_{o1} + r_{o2}$.

B. $g_m \approx g_{m1} + g_{m2}$ and $r_o \approx r_{o1} + r_{o2}$.

C. $g_m \approx g_{m1}$ and $r_o \approx r_{o1} \cdot g_{m2} \cdot r_{o2}$.

D. $g_m \approx g_{m1}$ and $r_o \approx r_{o2}$.

Q2.18.2 MCQ 2018 1M

A good transimpedance amplifier has

A. low input impedance and high output impedance.

B. high input impedance and high output impedance.

C. high input impedance and low output impedance.

D. low input impedance and low output impedance.

Q2.18.3 NAT 2018 2M

In the circuit shown below, the (W/L) value for M_2 is twice that for M_1. The two nMOS transistors are otherwise identical. The threshold voltage V_T for both transistors is 1.0 V. Note that V_{GS} for M_2 must be > 1.0 V.

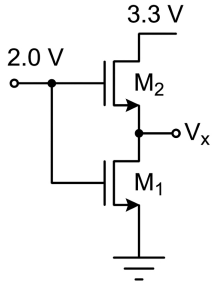


Fig. Q2.18.3

Current through nMOS transistors can be modeled as

$$I_{DS} = \mu C_{ox}\left(\frac{W}{L}\right)\left((V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2\right) \text{ for } V_{DS} \le V_{GS} - V_T$$

$$I_{DS} = \mu C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2/2 \quad \text{for } V_{DS} \ge V_{GS} - V_T$$

The voltage (in volts, accurate to two decimal places) at V_x is _____.

Q2.17.1 NAT 2017 2M

For the circuit shown, assume that the NMOS transistor is in saturation. Its threshold voltage $V_{tn} = 1\,\text{V}$ and its transconductance parameter $\mu_n C_{ox}(W/L) = 1\,\text{mA/V}^2$. Neglect channel length modulation and body bias effects. Under these conditions, the drain current I_D in mA is _____.

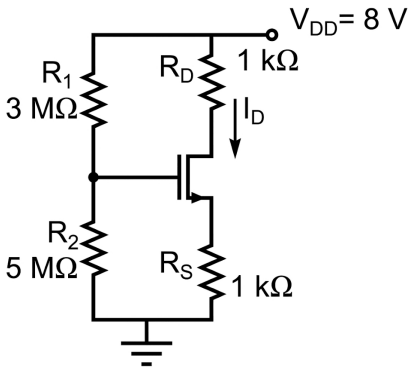


Fig. Q2.17.1

Q2.17.2 MCQ 2017 1M

An n-channel enhancement mode MOSFET is biased at $V_{GS} > V_{TH}$ and $V_{DS} > (V_{GS} - V_{TH})$, where V_{GS} is the gate-to-source voltage, V_{DS} is the drain-to-source voltage and V_{TH} is the threshold voltage. Considering channel length modulation effect to be significant, the MOSFET behaves as a

A. voltage source with zero output impedance

B. voltage source with non-zero output impedance

C. current source with finite output impedance

D. current source with infinite output impedance

Q2.17.3 NAT 2017 1M

Consider an n-channel MOSFET having width W, length L, electron mobility in the channel μ_n and oxide capacitance per unit area C_{ox}. If gate-to-source voltage $V_{GS} = 0.7\,V$, drain-to-source voltage $V_{DS} = 0.1\,V$, $(\mu_n C_{ox}) = 100\,\mu A/V^2$, threshold voltage $V_{TH} = 0.3\,V$ and $(W/L) = 50$, then the transconductance g_m (in mA/V) is _____

Q2.17.4 MCQ 2017 2M

Two n-channel MOSFETs, T1 and T2, are identical in all respects except that the width of T2 is double that of T1. Both the transistors are biased in the saturation region of operation, but the gate overdrive voltage $(V_{GS} - V_{TH})$ of T2 is double that of T1, where V_{GS} and V_{TH} are the gate-to-source voltage and threshold voltage of the transistors, respectively. If the drain current and transconductance of T1 are I_{D1} and g_{m1} respectively, the corresponding values of these two parameters for T2 are

A. $8I_{D1}$ and $2g_{m1}$ **B.** $8I_{D1}$ and $4g_{m1}$
C. $4I_{D1}$ and $4g_{m1}$ **D.** $4I_{D1}$ and $2g_{m1}$

Q2.17.5 MCQ 2017 2M

Assuming that transistors M_1 and M_2 are identical and have a threshold voltage of 1 V, the state of transistors M_1 and M_2 are respectively

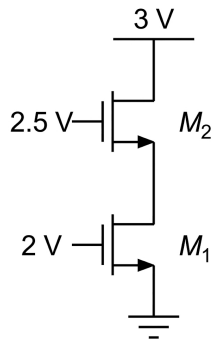


Fig. Q2.17.5

A. Saturation, Saturation **B.** Linear, Linear
C. Linear, Saturation **D.** Saturation, Linear

Q2.16.1 NAT 2016 2M

Consider a long-channel NMOS transistor with source and body connected together. Assume that the electron mobility is independent of V_{GS} and V_{DS}. Given,

$g_m = 0.5\,\mu A/V$ for $V_{DS} = 50\,mV$ and $V_{GS} = 2\,V$

$g_d = 8\,\mu A/V$ for $V_{GS} = 2\,V$ and $V_{DS} = 0\,V$

where $g_m = \dfrac{\partial I_D}{\partial V_{GS}}$ and $g_d = \dfrac{\partial I_D}{\partial V_{DS}}$

The threshold voltage (in volts) of the transistor is _____.

Q2.16.2 MCQ 2016 2M

In the circuit shown in the figure, the channel length modulation of all transistors is non-zero ($\lambda \neq 0$). Also, all transistors operate in saturation and have negligible body effect. The ac small signal voltage gain (V_o/V_{in}) of the circuit is

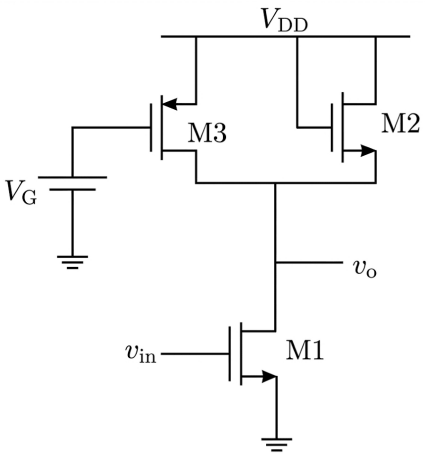


Fig. Q2.16.2

A. $-g_{m1}(r_{o1}||r_{o2}||r_{o3})$

B. $-g_{m1}\left(r_{o1}||\frac{1}{g_{m3}}||r_{o3}\right)$

C. $-g_{m1}\left(r_{o1}||\left(\frac{1}{g_{m2}}||r_{o2}\right)||r_{o3}\right)$

D. $-g_{m1}\left(r_{o1}||\left(\frac{1}{g_{m3}}||r_{o3}\right)||r_{o2}\right)$

Q2.16.3 NAT 2016 2M

In the circuit shown in the figure, transistor M1 is in saturation and has transconductance $g_m = 0.01$ siemens. Ignoring internal parasitic capacitances and assuming the channel length modulation λ to be zero, the small signal input pole frequency (in kHz) is _____

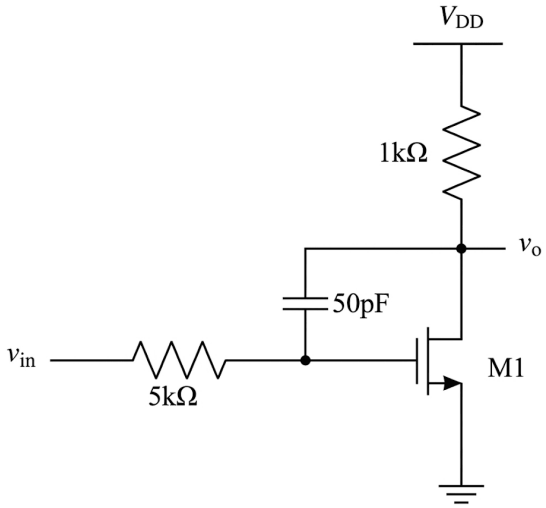


Fig. Q2.16.3

Q2.16.4 NAT 2016 2M

Consider an n-channel metal oxide semiconductor field effect transistor (MOSFET) with a gate-to-source voltage of 1.8 V. Assume that $\frac{W}{L} = 4$, $\mu_N C_{ox} = 70\times10^{-6}\,\text{AV}^{-2}$, the threshold voltage is 0.3 V, and the channel length modulation parameter is $0.09\,\text{V}^{-1}$. In the saturation region, the drain conductance (in micro seimens) is _____

Q2.16.5 MCQ 2016 1M

A long-channel NMOS transistor is biased in the linear region with $V_{DS} = 50\,\text{mV}$ and is used as a resistance. Which one of the following statements is NOT correct?

A. If the device width W is increased, the resistance decreases.

B. If the threshold voltage is reduced, the resistance decreases.

C. If the device length L is increased, the resistance increases.

D. If V_{GS} is increased, the resistance increases.

Q2.15.1 MCQ 2015 2M

For the NMOSFET in the circuit shown, the threshold voltage is V_{th} , where $V_{th} > 0$. The source voltage V_{SS} is varied from 0 to V_{DD}. Neglecting the channel length modulation, the drain current I_D as a function of V_{SS} is represented by

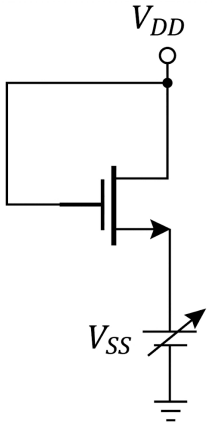
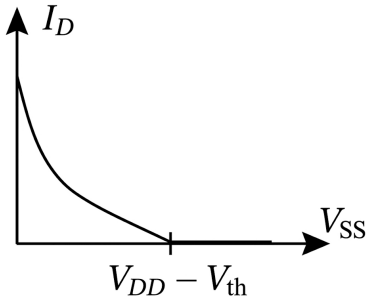
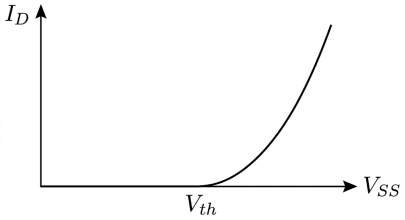


Fig. Q2.15.1

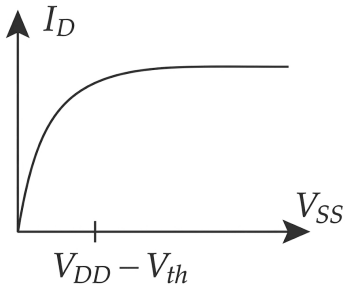
A.



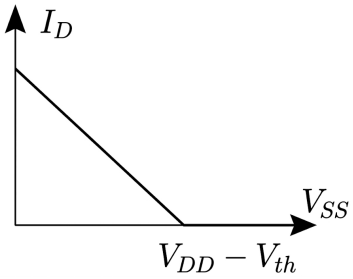
B.



C.



D.



Q2.15.2 NAT 2015 2M

In the circuit shown, both the enhancement mode NMOS transistors have the following characteristics:

$k_n = \mu_n C_{ox}(W/L) = 1\ \text{mA/V}^2$; $V_{TN} = 1\ \text{V}$

Assume that the channel length modulation parameter λ is zero and body is shorted to source. The minimum supply voltage V_{DD} (in volts) needed to ensure that transistor M_1 operates in saturation mode of operation is _____.

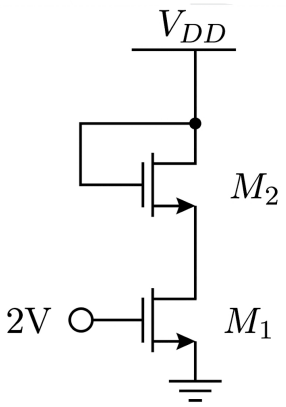


Fig. Q2.15.2

Q2.15.3 NAT 2015 2M

A MOSFET in saturation has a drain current of 1 mA for $V_{DS} = 0.5\ \text{V}$. If the channel length modulation coefficient is $0.05\ \text{V}^{-1}$, the output resistance (in kΩ) of the MOSFET is _____.

Q2.15.4 NAT 2015 2M

The current in an enhancement mode NMOS transistor biased in saturation mode was measured to be 1 mA at a drain-source voltage of 5 V. When the drain-source voltage was increased to 6 V while keeping gate-source voltage same, the drain current increased to 1.02 mA. Assume that drain to source saturation voltage is much smaller than the applied drain-source voltage. The channel length modulation parameter λ (in V^{-1}) is _____.

Report the result rounded off to three decimal places.

Q2.14.1 MCQ 2014 1M

A good current buffer has

A. low input impedance and low output impedance

B. low input impedance and high output impedance

C. high input impedance and low output impedance

D. high input impedance and high output impedance

Q2.14.2 MCQ 2014 1M

In the ac equivalent circuit shown in the figure, if i_{in} is the input current and R_F is very large, the type of feedback is

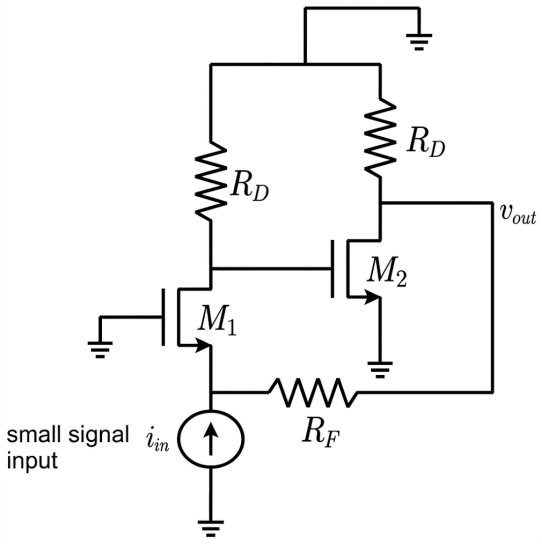


Fig. Q2.14.2

A. voltage-voltage feedback

B. voltage-current feedback

C. current-voltage feedback

D. current-current feedback

Q2.14.3 NAT 2014 2M

A depletion type N-channel MOSFET is biased in its linear region for use as a voltage controlled resistor. Assume threshold voltage $V_{TH} = -0.5\,V$, $V_{GS} = 2.0\,V$, $V_{DS} = 5\,V$, $W/L = 100$, $C_{ox} = 10^{-8}\,F/cm^2$ and $\mu_n = 800\,cm^2/V\text{-}s$. The value of the resistance of the voltage controlled resistor (in Ω) is _____.

Q2.14.4 MCQ 2014 2M

For the n-channel MOS transistor shown in the figure, the threshold voltage V_{Th} is 0.8 V. Neglect channel length modulation effects. When the drain voltage $V_D = 1.6\,V$, the drain current I_D was found to be 0.5 mA. If V_D is adjusted to be 2 V by changing the values of R and V_{DD}, the new value of I_D (in mA) is

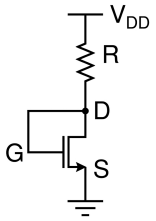


Fig. Q2.14.4

A. 0.625 **B.** 0.75 **C.** 1.125 **D.** 1.5

Q2.14.5 NAT 2014 2M

For the MOSFETs shown in the figure, the threshold voltage $|V_t| = 2\,V$ and $K = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right) = 0.1\,mA/V^2$. The value of I_D (in mA) is _____.

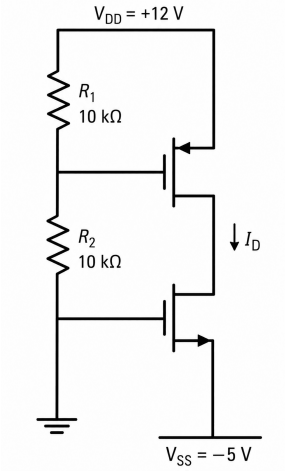


Fig. Q2.14.5

Q2.14.6 MCQ 2014 1M

The desirable characteristics of a transconductance amplifier are

A. high input resistance and high output resistance

B. high input resistance and low output resistance

C. low input resistance and high output resistance

D. low input resistance and low output resistance

Q2.14.7 NAT 2014 2M

The slope of the I_D vs. V_{GS} curve of an n-channel MOSFET in linear regime is $10^{-3}\Omega^{-1}$ at $V_{DS} = 0.1\,V$. For the same device, neglecting channel length modulation, the slope of the $\sqrt{I_D}$ vs. V_{GS} curve (in $\sqrt{A}/V$) under saturation regime is approximately _____.

Q2.14.8 NAT 2014 2M

For the MOSFET M_1 shown in the figure, assume $W/L = 2$, $V_{DD} = 2.0\,V$, $\mu_n C_{ox} = 100\,\mu A/V^2$ and $V_{TH} = 0.5\,V$. The transistor M_1 switches from saturation region to linear region when V_{in} (in Volts) is _____.

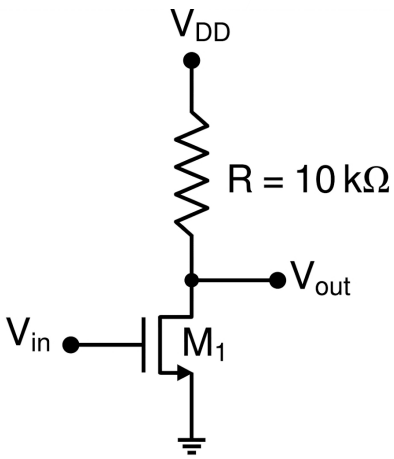


Fig. Q2.14.8

Q2.13.1 MCQ 2013 1M

In a voltage-voltage feedback as shown below, which one of the following statements is TRUE if the gain k is increased?

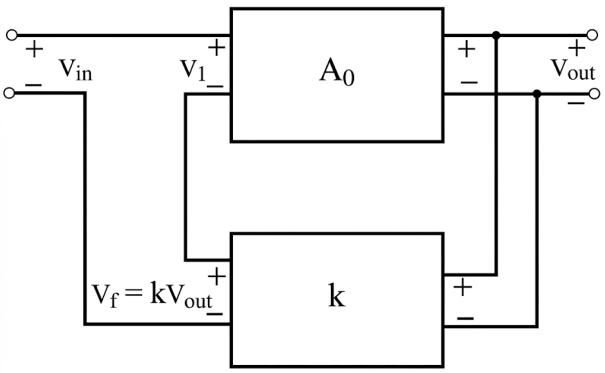


Fig. Q2.13.1

A. The input impedance increases and output impedance decreases.

B. The input impedance increases and output impedance also increases.

C. The input impedance decreases and output impedance also decreases.

D. The input impedance decreases and output impedance increases.

Q2.13.2 MCQ 2013 1M

In a MOSFET operating in the saturation region, the channel length modulation effect causes

A. an increase in the gate-source capacitance

B. a decrease in the transconductance

C. a decrease in the unity-gain cutoff frequency

D. a decrease in the output resistance

Q2.13.3 MCQ 2013 2M

The small-signal resistance (i.e., dV_B/dI_D) in kΩ offered by the n-channel MOSFET M shown in the figure below, at a bias point of $V_B = 2$ V is (device data for M: device transconductance parameter $k_N = \mu_n C'_{ox}(W/L) = 40\,\mu$A/V^2, threshold voltage $V_{TN} = 1$ V, and neglect body effect and channel length modulation effects)

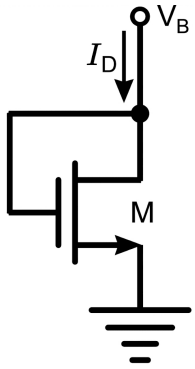


Fig. Q2.13.3

A. 12.5 **B.** 25 **C.** 50 **D.** 100

Q2.13.4 MCQ 2013 2M

The ac schematic of an NMOS common-source stage is shown in the figure below, where part of the biasing circuits has been omitted for simplicity. For the n-channel MOSFET M, the transconductance $g_m = 1$ mA/V, and body effect and channel length modulation effect are to be neglected. The lower cutoff frequency in Hz of the circuit is approximately at

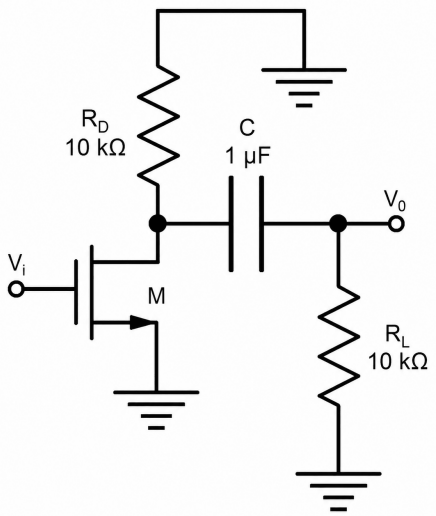


Fig. Q2.13.4

A. 8 **B.** 32 **C.** 50 **D.** 200

Q2.11.1 MCQ 2011 2M

In the circuit shown below, for the MOS transistors, $\mu_n C_{ox} = 100\,\mu A/V^2$ and the threshold voltage $V_T = 1$ V. The voltage V_x at the source of the upper transistor is

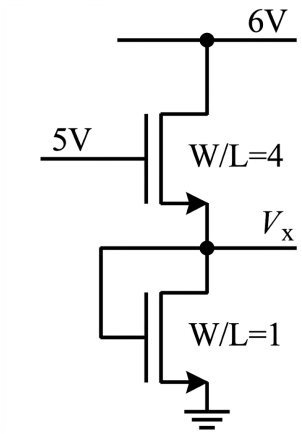


Fig. Q2.11.1

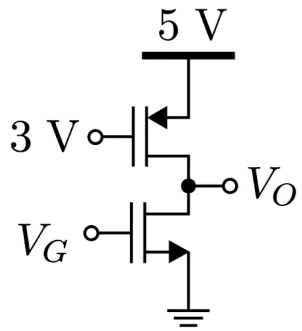
A. 1 V **B.** 2 V **C.** 3 V **D.** 3.67 V

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q2.09.1 and Q2.09.2:

Consider the CMOS circuit shown, where the gate voltage V_G of the n-MOSFET is increased from zero, while the gate voltage of the p-MOSFET is kept constant at 3 V. Assume that, for both transistors, the magnitude of the threshold voltage is 1 V and the product of the transconductance parameter and the (W/L) ratio, i.e. the quantity $\mu C_{ox}(W/L)$, is 1 mA · V^{-2}.



Q2.09.1 MCQ 2009 2M

For small increase in V_G beyond 1 V, which of the following gives the correct description of the region of operation of each MOSFET ?

A. Both the MOSFETs are in saturation region

B. Both the MOSFETs are in triode region

C. n-MOSFET is in triode and p-MOSFET is in saturation region

D. n-MOSFET is in saturation and p-MOSFET is in triode region

Q2.09.2 MCQ 2009 2M

Estimate the output voltage V_0 for $V_G = 1.5$ V. [Hint: Use the appropriate current-voltage equation for each MOSFET, based on the answer to Q2.09.1]

A. $4 - \frac{1}{\sqrt{2}}$ V **B.** $4 + \frac{1}{\sqrt{2}}$ V

C. $4 - \frac{\sqrt{3}}{2}$ V **D.** $4 + \frac{\sqrt{3}}{2}$ V

Q2.08.1 MCQ 2008 1M

The drain current of a MOSFET in saturation is given by $I_D = K(V_{GS} - V_T)^2$ where K is a constant. The magnitude of the transconductance g_m is

A. $\frac{K(V_{GS} - V_T)^2}{V_{DS}}$ **B.** $2K(V_{GS} - V_T)$

C. $\frac{I_d}{V_{GS} - V_{DS}}$ **D.** $\frac{K(V_{GS} - V_T)^2}{V_{GS}}$

Q2.08.2 MCQ 2008 2M

For the circuit shown in the following figure, transistors M1 and M2 are identical NMOS transistors. Assume that M2 is in saturation and the output is unloaded.

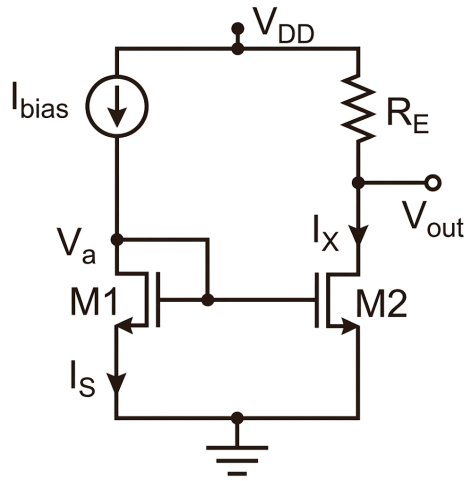


Fig. Q2.08.2

The current I_X is related to I_{bias} as

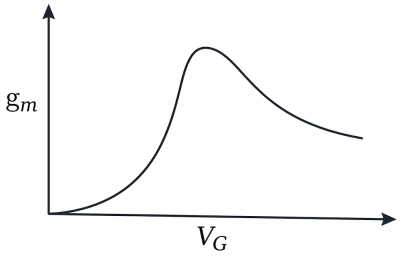
A. $I_X = I_{bias} + I_S$ **B.** $I_X = I_{bias}$

C. $I_X = I_{bias} - I_S$ **D.** $I_X = I_{bias} - \left(V_{DD} - \frac{V_{out}}{R_E}\right)$

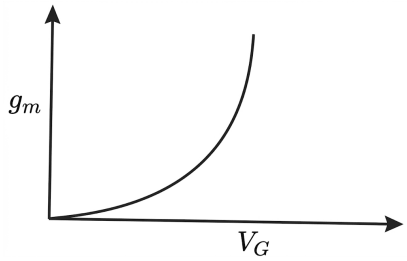
Q2.08.3 MCQ 2008 2M

The measured transconductance g_m of an NMOS transistor operating in the linear region is plotted against the gate voltage V_G at a constant drain voltage V_D. Which of the following figures represents the expected dependence of g_m on V_G?

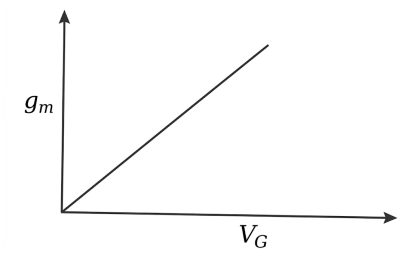
A.



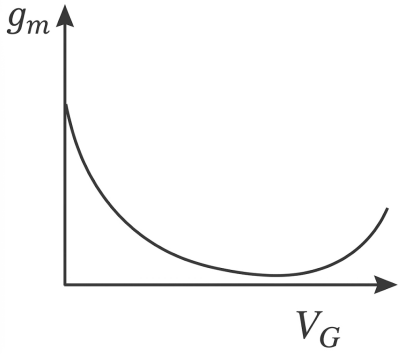
B.



C.



D.



Q2.08.4 MCQ 2008 2M

Two identical NMOS transistors M1 and M2 are connected as shown below. V_{bias} is chosen so that both transistors are in saturation. The equivalent g_m of the pair is defined to be $\frac{\partial I_{out}}{\partial V_i}$ at constant V_{out}.

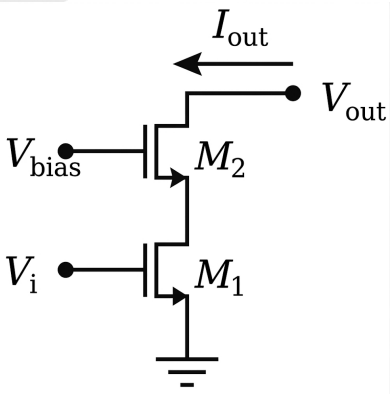


Fig. Q2.08.4

The equivalent g_m of the pair is

A. the sum of individual g_m's of the transistors

B. the product of individual g_m's of the transistors

C. nearly equal to the g_m of M1

D. nearly equal to g_m/g_0 of M2

Q2.06.1 MCQ 2006 1M

An n-channel depletion MOSFET has following two points on its $I_D - V_{GS}$ curve:
(i) $V_{GS} = 0$ at $I_D = 12$ mA and
(ii) $V_{GS} = -6$ Volts at $I_D = 0$
Which of the following Q-points will give the highest transconductance gain for small signals?

A. $V_{GS} = -6$ Volts **B.** $V_{GS} = -3$ Volts

C. $V_{GS} = 0$ Volts **D.** $V_{GS} = 3$ Volts

Q2.05.1 MCQ 2005 2M

For an n-channel MOSFET and its transfer curve shown in figure, the threshold voltage is

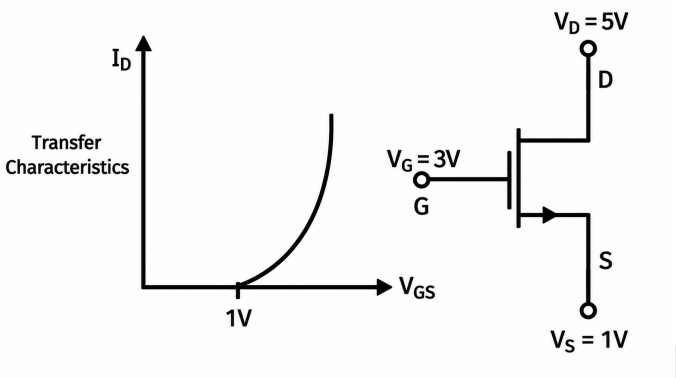


Fig. Q2.05.1

A. 1 V and the device is in active region.

B. −1 V and the device is in saturation region.

C. 1 V and the device is in saturation region.

D. −1 V and the device is in active region.

Q2.05.2 MCQ 2005 2M

Both transistors T_1 and T_2 in figure have a threshold voltage of 1 Volt. The device parameters K_1 and K_2 of T_1 and T_2 are, respectively, 36 μA/V^2 and 9 μA/V^2. The output voltage V_0 is

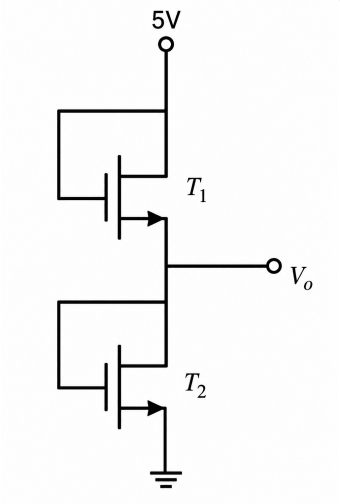


Fig. Q2.05.2

A. 1 V **B.** 2 V **C.** 3 V **D.** 4 V

Q2.05.3 MCQ 2005 1M

The effect of current shunt feedback in an amplifier is to

A. increase the input resistance and decrease the output resistance.

B. increase both input and output resistances.

C. decreases both input and output resistances.

D. decrease the input resistance and increase the output resistance.

Q2.04.1 MCQ 2004 2M

The drain of an n-channel MOSFET is shorted to the gate so that $V_{GS} = V_{DS}$. The threshold voltage (V_T) of MOSFET is 1 V. If the drain current (I_D) is 1 mA for $V_{GS} = 2$ V, then for $V_{GS} = 3$ V, I_D is

A. 2 mA **B.** 3 mA **C.** 9 mA **D.** 4 mA

Q2.03.1 MCQ 2003 2M

An amplifier without feedback has a voltage gain of 50, input resistance of 1 KΩ and output resistance of 2.5 KΩ. The input resistance of the current-shunt negative feedback amplifier using the above amplifier with a feedback factor of 0.2, is

A. $\frac{1}{11}$ KΩ **B.** $\frac{1}{5}$ KΩ **C.** 5 KΩ **D.** 11 KΩ

Q2.03.2 MCQ 2003 2M

Three identical amplifiers with each one having a voltage gain of 50, input resistance of 1 KΩ and output resistance of 250 Ω, are cascaded. The open circuit voltage gain of the combined amplifier is

A. 49 dB **B.** 51 dB **C.** 98 dB **D.** 102 dB

Q2.01.1 MCQ 2001 1M

The effective channel length of a MOSFET in saturation decreases with increase in

A. gate voltage **B.** drain voltage
C. source voltage **D.** body voltage

Q2.99.1 MCQ 1999 1M

Negative feedback in an amplifier

A. reduces gain

B. increases frequency and phase distortions

C. reduces bandwidth

D. increase noise

Q2.99.2 MCQ 1999 1M

An amplifier has an open-loop gain of 100, an input impedance of 1 kΩ, and an output impedance of 100 Ω. A feedback network with a feedback factor of 0.99 is connected to the amplifier in a voltage series feedback mode. The new input and output impedances, respectively, are

A. 10 Ω and 1 Ω
B. 10 Ω and 10 kΩ
C. 100 kΩ and 1 Ω
D. 10 kΩ and 1 kΩ

Q2.99.3 MCQ 1999 1M

An amplifier is assumed to have a single pole high frequency transfer function. The rise time of its output response to a step function input is 35 nsec. The upper −3 dB frequency (in MHz) for the amplifier to a sinusoidal input is approximately at

A. 4.55 **B.** 10 **C.** 20 **D.** 28.6

Q2.98.1 MCQ 1998 1M

Two identical FETs, each characterised by the parameters g_m and r_d are connected in parallel. The composite FET is then characterized by the parameters

A. $\frac{g_m}{2}$ and $2r_d$
B. $\frac{g_m}{2}$ and $\frac{r_d}{2}$
C. $2g_m$ and $\frac{r_d}{2}$
D. $2g_m$ and $2r_d$

Q2.98.2 MCQ 1998 1M

In a shunt-shunt negative feedback amplifier, as compared to the basic amplifier

A. both, input and output impedances, decrease

B. input impedance decreases but output impedance increases

C. input impedance increases but output impedance decreases

D. both, input and output impedances increase

Q2.98.3 MCQ 1998 1M

In the MOSFET amplifier shown in the figure, the signal outputs V_1 and V_2 obey the relationship

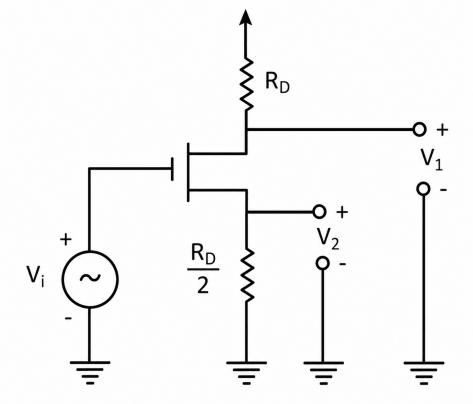


Fig. Q2.98.3

A. $V_1 = \frac{V_2}{2}$
B. $V_1 = -\frac{V_2}{2}$
C. $V_1 = 2V_2$
D. $V_1 = -2V_2$

Q2.97.1 NAT 1997 5M

Given an NMOS circuit as shown in the figure
The specifications of the circuit are
$V_{DD} = 10$ V; $\beta = K = \mu_n C_{ox}(W/L) = 10^{-4}$ Amp/V^2
$V_T = 1$ V and $I_{DS} = 0.5$ mA.
Evaluate V_{DS} and R_D for the circuit. Neglect body - effect for V_T.

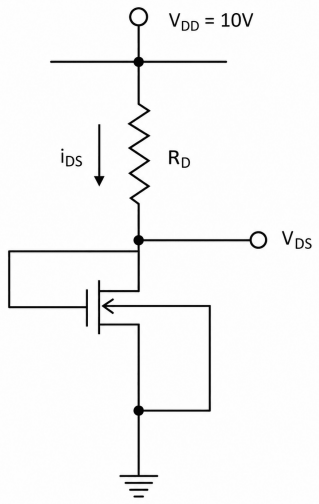


Fig. Q2.97.1

Q2.92.1 NAT 1992 8M

An n-channel MOSFET(T) having a V_T of 2 V [Threshold voltage] is used in the circuit shown in figure. Initially, it is OFF and is in steady state. At time $t = 0$, a step voltage of magnitude 4V is applied at the input so that the MOSFET turns 'ON' instantaneously. Draw the equivalent circuit

and calculate the time taken (ns) for the output V_o to fall to 5V. The device constant of the MOSFET, $K = 5$ mA/V^2, $R_{DS} = \infty, C_{DS} = 0; C_{DG} = 0$.

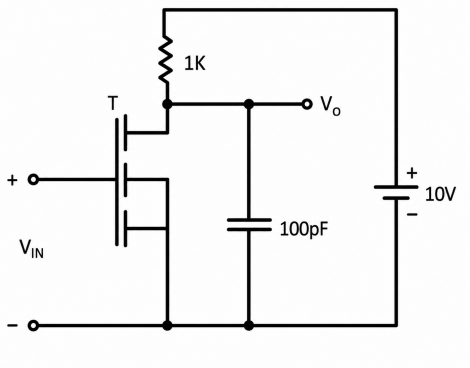


Fig. Q2.92.1

Report the time in ns rounded off to two decimal places.

Q2.91.1 NAT 1991 2M

In figure, the n-channel MOSFETs are identical and their current voltage characteristics are given by the following expressions:

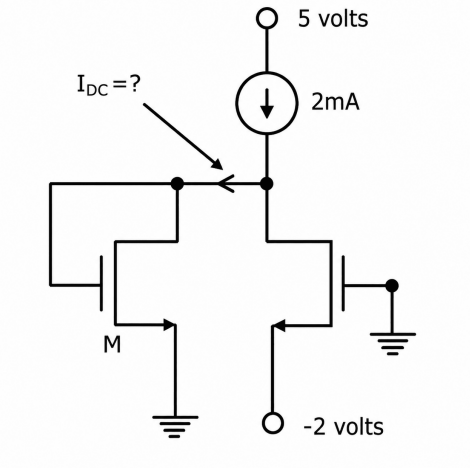


Fig. Q2.91.1

For $V_{DS} < V_{GS} - 1, I_D = \left[(V_{GS} - 1)V_{DS} - \frac{V_{DS}^2}{2}\right] mA$

For $V_{DS} \geq (V_{GS} - 1), I_D = (V_{GS} - 1)^2 mA$

Where V_{GS} and V_{DS} are the gate source and drain source voltages respectively and I_D is the drain current.

The current I_{DC} flowing through the transistor 'M' is equal to _____

Q2.88.1 MSQ 1988 1M

In MOSFET devices, the N-channel type is better than the P-channel type in the following respects

- [ ] **A.** It has better noise immunity
- [ ] **B.** It is faster
- [ ] **C.** It is TTL compatible
- [ ] **D.** It has better drive capability

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q2.26.1 2.56	Q2.25.1 B,D	Q2.25.2 D
Q2.24.1 A	Q2.24.2 A	Q2.24.3 B
Q2.23.1 D	Q2.23.2 D	Q2.23.3 2
Q2.22.1 C	Q2.22.2 -2.0	Q2.22.3 C
Q2.21.1 D	Q2.21.2 0.5	Q2.20.1 B
Q2.20.2 C	Q2.19.1 C	Q2.19.2 -900

Q2.19.3 6.0	Q2.19.4 25.4 to 25.6	Q2.18.1 C
Q2.18.2 D	Q2.18.3 0.42	Q2.17.1 2
Q2.17.2 C	Q2.17.3 0.5	Q2.17.4 B
Q2.17.5 C	Q2.16.1 1.2	Q2.16.2 C
Q2.16.3 57.87	Q2.16.4 28.35	Q2.16.5 D
Q2.15.1 A	Q2.15.2 3	Q2.15.3 20

Q2.15.4 0.022	Q2.14.1 B	Q2.14.2 B
Q2.14.3 499 to 501	Q2.14.4 C	Q2.14.5 0.9
Q2.14.6 A	Q2.14.7 0.0707	Q2.14.8 1.5
Q2.13.1 A	Q2.13.2 D	Q2.13.3 B
Q2.13.4 A	Q2.11.1 C	Q2.09.1 D
Q2.09.2 D	Q2.08.1 B	Q2.08.2 B
Q2.08.3 *	Q2.08.4 C	Q2.06.1 D
Q2.05.1 C	Q2.05.2 C	Q2.05.3 D
Q2.04.1 D	Q2.03.1 A	Q2.03.2 C
Q2.01.1 B	Q2.99.1 A	Q2.99.2 C
Q2.99.3 B	Q2.98.1 C	Q2.98.2 A
Q2.98.3 D	Q2.97.1 *	Q2.92.1 28.77
Q2.91.1 1	Q2.88.1 B,D	

PrepFusion

UNIT

III

Operational Amplifier

Topics

1. Basics of Op-Amp, Comparator and Virtual Short
2. Inverting and Non-Inverting Amplifiers
3. Adder, Differential and Instrumentation Amplifiers
4. Log, Anti-Log and Rectifier Using Op-Amp
5. Filters Using Op-Amp
6. Offsets in Op-Amp
7. CMRR, Slew Rate and Frequency Response of Op-Amp
8. Integrator Using Op-Amp
9. Differentiator Using Op-Amp
10. Active Clipper, Clamper and Peak Detector
11. Multivibrators Using Op-Amp
12. Oscillators Using Op-Amp
13. Schmitt Triggers

Unit Snapshot*

Stream: EC	**Questions:** 97	**MCQ:** 61	**MSQ:** 3	**NAT:** 33
1M: 37	**2M:** 56	**Descriptive:** 4	**Avg Weightage** ~3M	**Range:** 1M(2011)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 1M

The ideal OP-AMP circuit shown in the Figure produces output voltage $V_o = x$ when the Switch, S, is open. Which of the options represents the output voltage when S is closed?

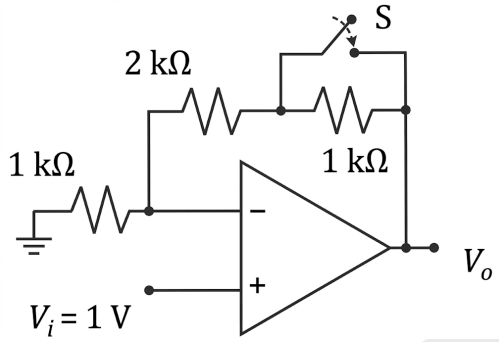


Fig. Q3.26.1

A. x **B.** $\frac{2}{3}x$ **C.** $\frac{3}{4}x$ **D.** $\frac{1}{2}x$

Q3.26.2 MCQ 2026 2M

A circuit using an ideal OP-AMP is shown in the Figure. Which of the following options gives the correct value of the current I_x?

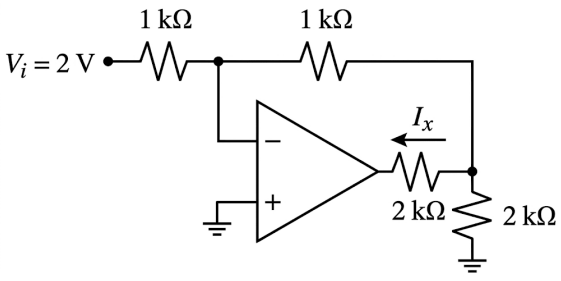


Fig. Q3.26.2

A. 2.0 mA **B.** 1.5 mA **C.** 3.0 mA **D.** 0 mA

Q3.26.3 NAT 2026 2M

Consider an ideal OP-AMP circuit as shown in the Figure. The resistances $R_1 = R_2 = R_3 = R_4 = 50\,\text{k}\Omega$. The magnitude of the closed loop gain is _____. *(rounded off to two decimal places)*

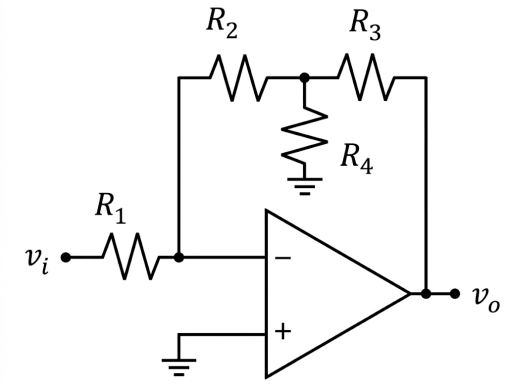


Fig. Q3.26.3

Q3.25.1 MSQ 2025 1M

Which of the following statements is/are TRUE with respect to an ideal opamp?

- **A.** It has an infinite input resistance
- **B.** It has an infinite output resistance
- **C.** It has an infinite open-loop differential gain
- **D.** It has an infinite open-loop common-mode gain

Q3.25.2 NAT 2025 1M

All the components in the bandpass filter given below are ideal. The lower −3 dB frequency of the filter is 1 MHz. The upper −3 dB frequency (in MHz, *rounded off to the nearest integer*) is _____.

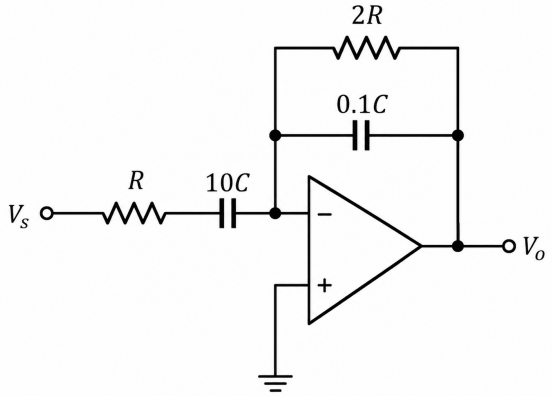


Fig. Q3.25.2

Q3.24.1 MCQ 2024 2M

The opamps in the circuit shown are ideal, but have saturation voltages of ±10 V.

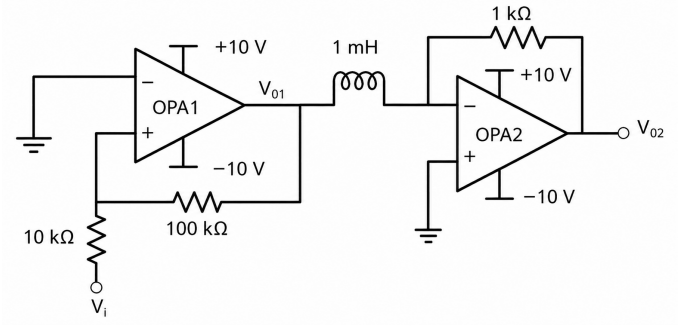


Fig. Q3.24.1

Assume that the initial inductor current is 0 A. The input voltage (V_i) is a triangular signal with peak voltages of ±2 V and time period of 8 μs. Which one of the following statements is true?

A. V_{01} is delayed by 2 μs relative to V_i, and V_{02} is a triangular waveform.

B. V_{01} is not delayed relative to V_i, and V_{02} is a trapezoidal waveform.

C. V_{01} is not delayed relative to V_i, and V_{02} is a triangular waveform.

D. V_{01} is delayed by 1 μs relative to V_i, and V_{02} is a trapezoidal waveform.

Q3.24.2 MCQ 2024 2M

In the circuit below, the opamp is ideal.

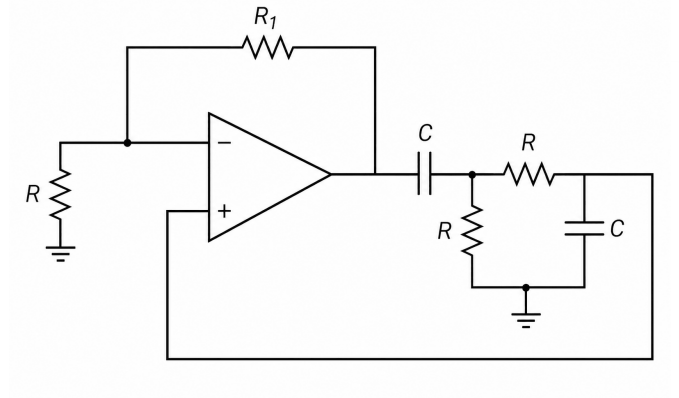


Fig. Q3.24.2

If the circuit is to show sustained oscillations, the respective values of R_1 and the corresponding frequency of oscillation are _____.

A. $29R$ and $1/(2\pi\sqrt{6}RC)$ **B.** $2R$ and $1/(2\pi RC)$
C. $29R$ and $1/(2\pi RC)$ **D.** $2R$ and $1/(2\pi\sqrt{6}RC)$

Q3.23.1 MCQ 2023 2M

The $\frac{V_{OUT}}{V_{IN}}$ of the circuit shown below is

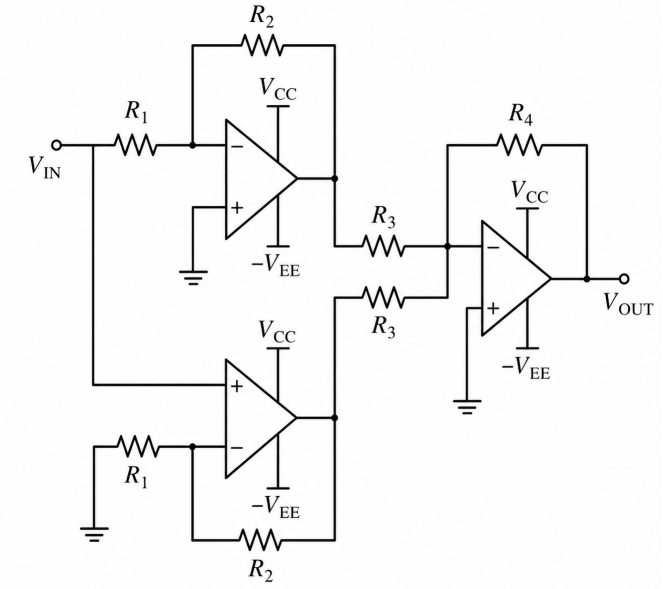


Fig. Q3.23.1

A. $-\frac{R_4}{R_3}$ **B.** $\frac{R_4}{R_3}$ **C.** $1+\frac{R_4}{R_3}$ **D.** $1-\frac{R_4}{R_3}$

Q3.22.1 MSQ 2022 1M

An ideal OPAMP circuit with a sinusoidal input is shown in the figure. The 3 dB frequency is the frequency at which the magnitude of the voltage gain decreases by 3 dB from the maximum value. Which of the options is/are correct?

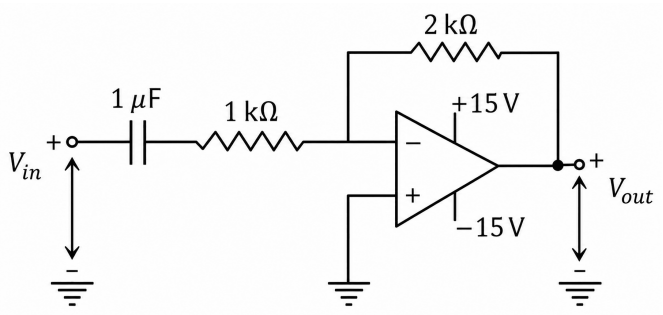


Fig. Q3.22.1

- **A.** The circuit is a low pass filter.
- **B.** The circuit is a high pass filter.
- **C.** The 3 dB frequency is 1000 rad/s.
- **D.** The 3 dB frequency is $\frac{1000}{3}$ rad/s.

Q3.22.2 MCQ 2022 2M

For the following circuit with an ideal OPAMP, the difference between the maximum and the minimum values of the capacitor voltage (V_c) is _____.

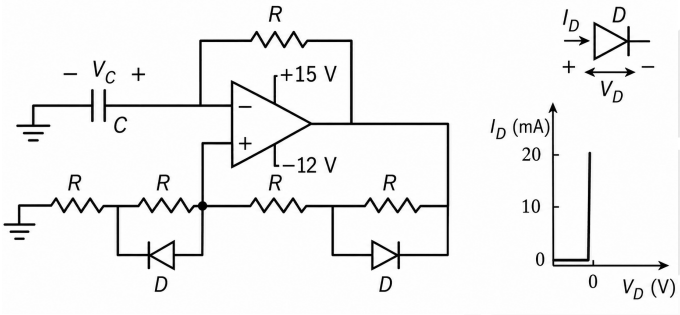


Fig. Q3.22.2

A. 15 V **B.** 27 V **C.** 13 V **D.** 14 V

Q3.22.3 MCQ 2022 2M

A circuit with an ideal OPAMP is shown. The Bode plot for the magnitude (in dB) of the gain transfer function ($A_V(j\omega) = V_{out}(j\omega)/V_{in}(j\omega)$) of the circuit is also provided (here, ω is the angular frequency in rad/s). The values of R and C are _____.

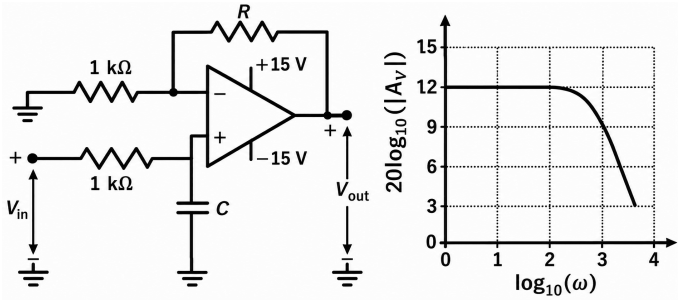


Fig. Q3.22.3

A. $R = 3\,\text{k}\Omega,\ C = 1\,\mu\text{F}$ **B.** $R = 1\,\text{k}\Omega,\ C = 3\,\mu\text{F}$
C. $R = 4\,\text{k}\Omega,\ C = 1\,\mu\text{F}$ **D.** $R = 3\,\text{k}\Omega,\ C = 2\,\mu\text{F}$

Q3.21.1 MCQ 2021 2M

For the circuit with an ideal OPAMP shown in the figure, V_{REF} is fixed.

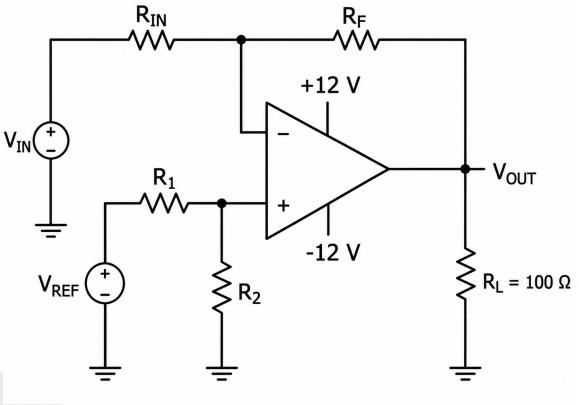


Fig. Q3.21.1

If $V_{OUT} = 1$ volt for $V_{IN} = 0.1$ volt and $V_{OUT} = 6$ volt for $V_{IN} = 1$ volt, where V_{OUT} is measured across R_L connected at the output of this OPAMP, the value of R_F/R_{IN} is

A. 3.285 **B.** 2.860 **C.** 3.825 **D.** 5.555

Q3.21.2 MCQ 2021 1M

Consider the circuit with an ideal OPAMP shown in the figure.

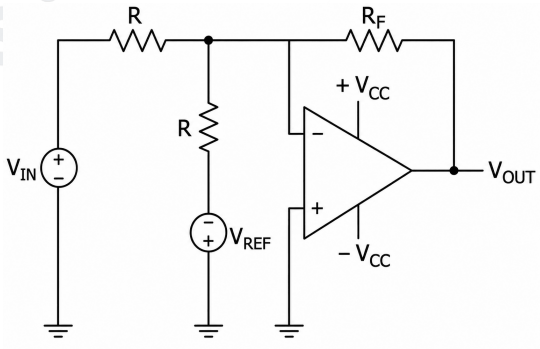


Fig. Q3.21.2

Assuming $|V_{IN}| \ll |V_{CC}|$ and $|V_{REF}| \ll |V_{CC}|$, the condition at which V_{OUT} equals to zero is

A. $V_{IN} = V_{REF}$ **B.** $V_{IN} = 0.5V_{REF}$
C. $V_{IN} = 2V_{REF}$ **D.** $V_{IN} = 2 + V_{REF}$

Q3.21.3 NAT 2021 2M

A circuit with an ideal OPAMP is shown in the figure. A pulse V_{IN} of 20 ms duration is applied to the input. The capacitors are initially uncharged.

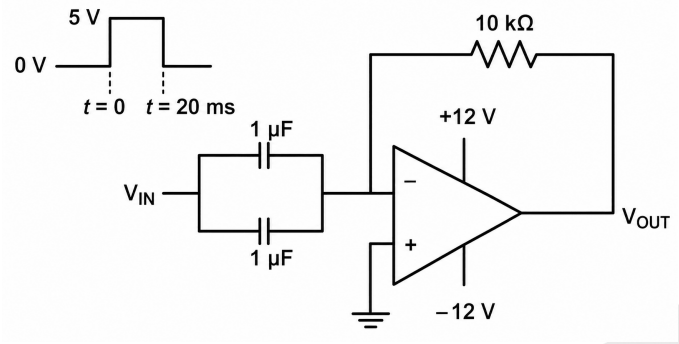


Fig. Q3.21.3

The output voltage V_{OUT} of this circuit at $t = 0^+$ *(in integer)* is _____ V.

Q3.20.1 MCQ 2020 1M

The components in the circuit shown below are ideal. If the op-amp is in positive feedback and the input voltage V_i is a sine wave of amplitude 1 V, the output voltage V_o is

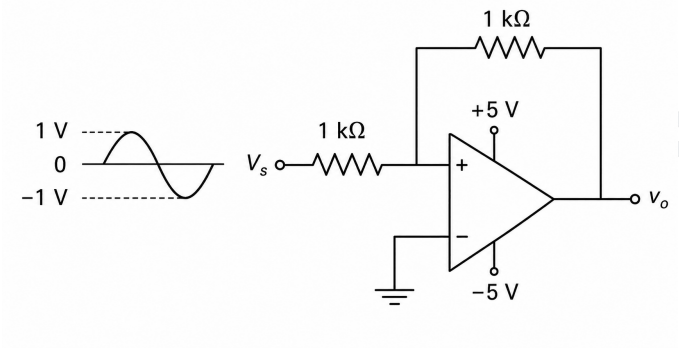


Fig. Q3.20.1

A. a non-inverted sine wave of 2 V amplitude.

B. an inverted sine wave of 1 V amplitude.

C. a square wave of 5 V amplitude.

D. a constant of either +5 V or −5 V.

Q3.20.2 NAT 2020 1M

In the circuit shown below, all the components are ideal. If V_i is +2 V, the current I_o sourced by the op-amp is _____ mA.

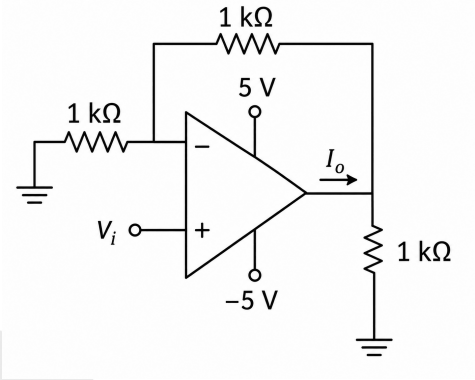


Fig. Q3.20.2

Q3.20.3 MCQ 2020 2M

The components in the circuit given below are ideal. If $R = 2\,\text{k}\Omega$ and $C = 1\,\mu\text{F}$, the −3 dB cut-off frequency of the circuit in Hz is

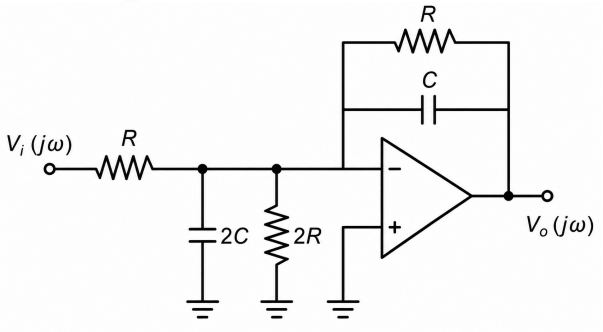


Fig. Q3.20.3

A. 14.92 **B.** 34.46 **C.** 59.68 **D.** 79.58

Q3.18.1 MCQ 2018 1M

In the circuit shown below, the op-amp is ideal and Zener voltage of the diode is 2.5 volts. At the input, unit step voltage is applied, i.e. $v_{IN}(t) = u(t)$ volts. Also, at $t = 0$, the voltage across each of the capacitors is zero.

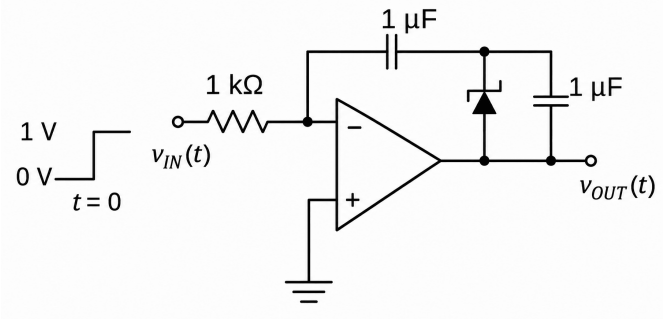


Fig. Q3.18.1

The time t, in milliseconds, at which the output voltage V_{OUT} crosses -10 V is

A. 2.5 **B.** 5 **C.** 7.5 **D.** 10

Q3.18.2 NAT 2018 2M

An op-amp based circuit is implemented as shown below.

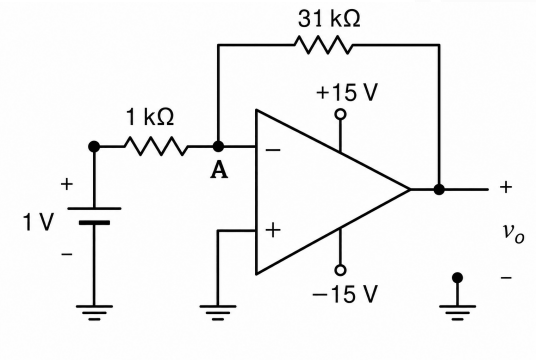


Fig. Q3.18.2

In the above circuit, assume the op-amp to be ideal. The voltage (in volts, correct to one decimal place) at node **A**, connected to the negative input of the op-amp as indicated in the figure is _____.

Q3.17.1 MCQ 2017 1M

For the operational amplifier circuit shown, the output saturation voltages are ±15 V. The upper and lower threshold voltages for the circuit are, respectively,

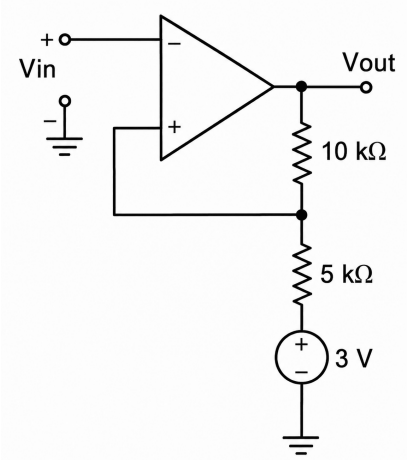


Fig. Q3.17.1

A. +5 V and −5 V **B.** +7 V and −3 V
C. +3 V and −7 V **D.** +3 V and −3 V

Q3.17.2 NAT 2017 2M

The amplifier circuit shown in the figure is implemented using a compensated operational amplifier (op-amp), and has an open-loop voltage gain, $A_0 = 10^5$ V/V and an open-loop cut-off frequency, $f_c = 8$ Hz. The voltage gain of the amplifier at 15 kHz, in V/V, is _____.

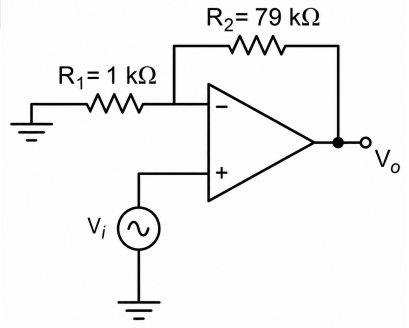


Fig. Q3.17.2

Q3.16.1 MCQ 2016 1M

Consider the oscillator circuit shown in the figure. The function of the network (shown in dotted lines) consisting of the 100 kΩ resistor in series with the two diodes connected back-to-back is to:

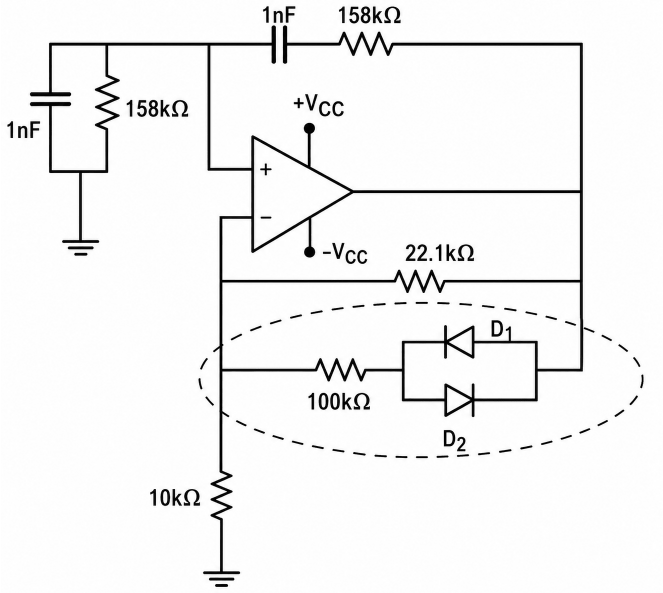


Fig. Q3.16.1

A. introduce amplitude stabilization by preventing the op amp from saturating and thus producing sinusoidal oscillations of fixed amplitude

B. introduce amplitude stabilization by forcing the opamp to swing between positive and negative saturation and thus producing square wave oscillations of fixed amplitude

C. introduce frequency stabilization by forcing the circuit to oscillate at a single frequency

D. enable the loop gain to take on a value that produces square wave oscillations

Q3.16.2 NAT 2016 2M

An ideal opamp has voltage sources $V_1, V_3, V_5, \ldots, V_{N-1}$ connected to the non-inverting input and $V_2, V_4, V_6, \ldots, V_N$ connected to the inverting input as shown in the figure below ($+V_{CC} = 15$ volt, $-V_{CC} = -15$ volt). The voltages $V_1, V_2, V_3, V_4, V_5, V_6, \ldots$ are $1, -1/2, 1/3, -1/4, 1/5, -1/6, \ldots$ volt, respectively. As N approaches infinity, the output voltage (in volt) is _____

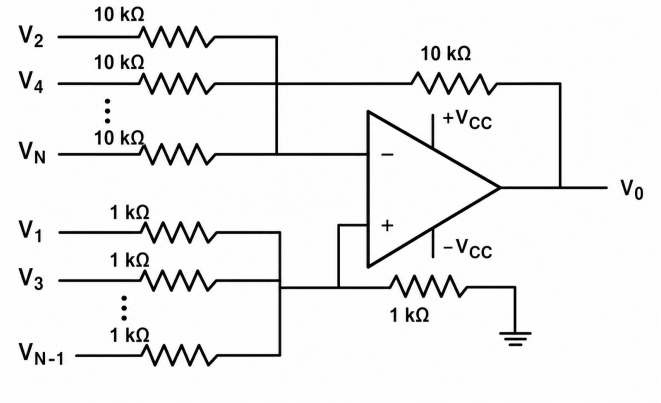


Fig. Q3.16.2

Q3.16.3 NAT 2016 2M

In the opamp circuit shown, the Zener diodes Z1 and Z2 clamp the output voltage V_O to +5 V or −5 V. The switch S is initially closed and is opened at time $t = 0$.

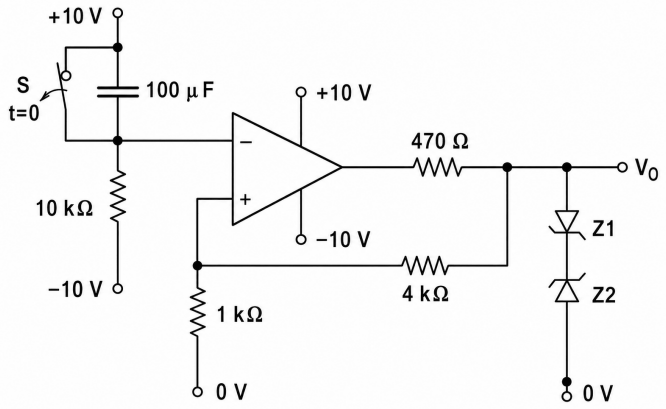


Fig. Q3.16.3

The time $t = t_1$ (in seconds) at which V_O changes state is _____

Q3.16.4 NAT 2016 2M

An opamp has a finite open loop voltage gain of 100. Its input offset voltage V_{ios} (= +5mV) is modeled as shown in the circuit below. The amplifier is ideal in all other respects. V_{input} is 25 mV.

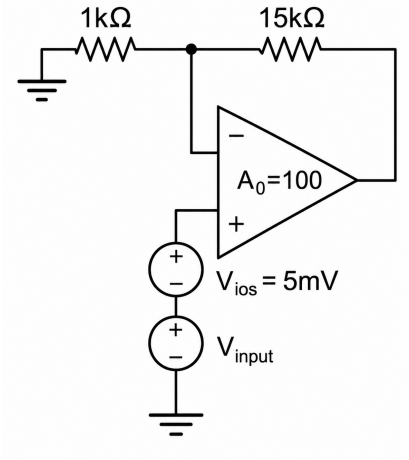


Fig. Q3.16.4

The output voltage (in millivolts) is _____

Q3.16.5 NAT | 2016 | 2M

For the circuit shown in the figure, $R_1 = R_2 = R_3 = 1\,\Omega$, $L = 1\,\mu$H and $C = 1\,\mu$F. If the input $V_{in} = \cos(10^6 t)$, then the overall voltage gain (V_{out}/V_{in}) of the circuit is _____

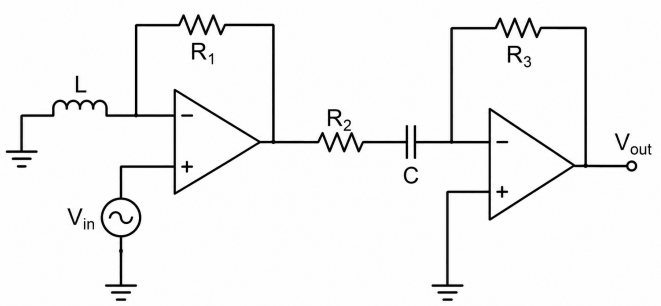


Fig. Q3.16.5

Q3.15.1 NAT | 2015 | 2M

In the circuit shown, assume that the opamp is ideal. The bridge output voltage V_0 (in mV) for $\delta = 0.05$ is _____.

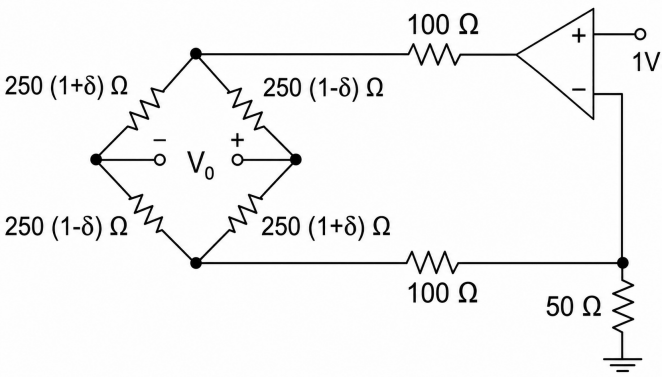


Fig. Q3.15.1

Q3.15.2 MCQ | 2015 | 2M

The circuit shown in the figure has an ideal opamp. The oscillation frequency and the condition to sustain the oscillations, respectively, are

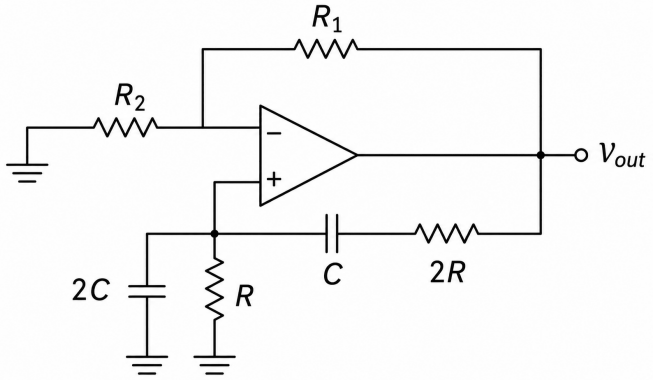


Fig. Q3.15.2

A. $\frac{1}{CR}$ and $R_1 = R_2$

B. $\frac{1}{CR}$ and $R_1 = 4R_2$

C. $\frac{1}{2CR}$ and $R_1 = R_2$

D. $\frac{1}{2CR}$ and $R_1 = 4R_2$

Q3.15.3 NAT | 2015 | 1M

In the circuit shown, $V_0 = V_{0A}$ for switch SW in position A and $V_0 = V_{0B}$ for SW in position B. Assume that the opamp is ideal. The value of $\frac{V_{0B}}{V_{0A}}$ is _____.

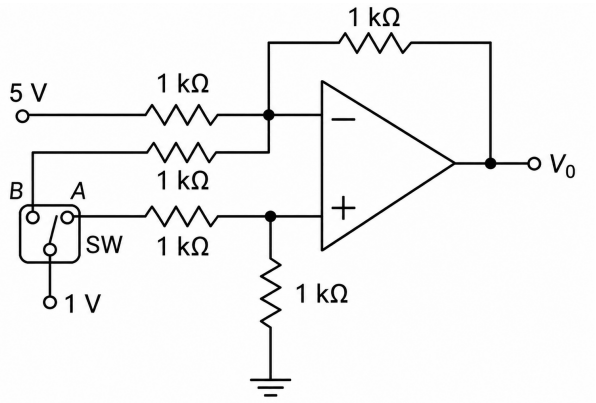


Fig. Q3.15.3

Q3.15.4 NAT 2015 1M

In the bistable circuit shown, the ideal opamp has saturation levels of ±5 V. The value of R_1 (in kΩ) that gives a hysteresis width of 500 mV is _____.

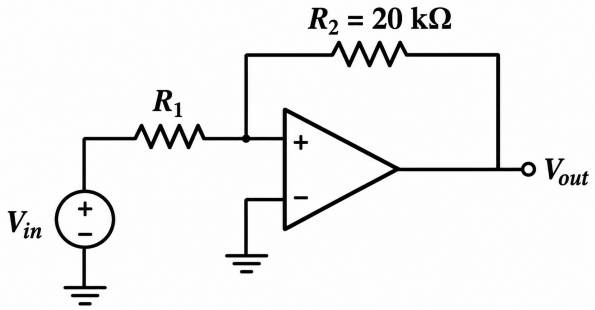


Fig. Q3.15.4

Q3.15.5 NAT 2015 2M

Assuming that the opamp in the circuit shown below is ideal, the output voltage V_o (in volts) is _____.

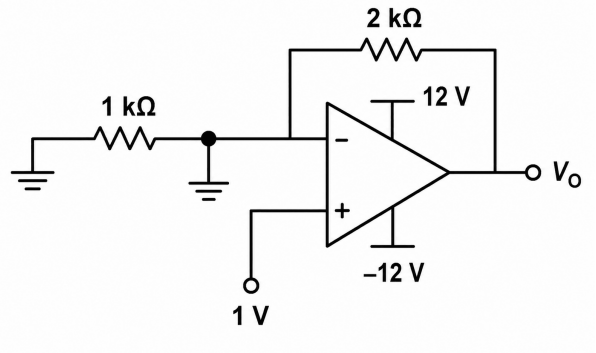


Fig. Q3.15.5

Q3.15.6 NAT 2015 1M

In the circuit shown using an ideal opamp, the 3-dB cut-off frequency (in Hz) is _____.

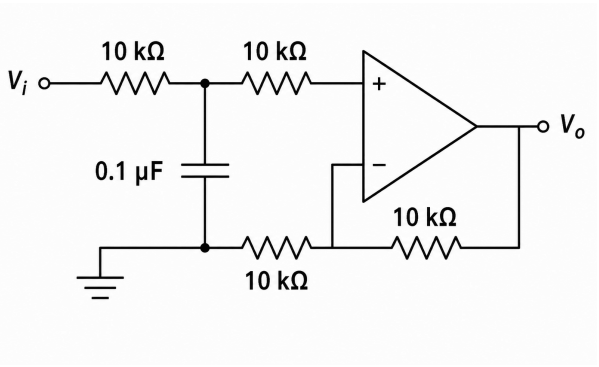


Fig. Q3.15.6

Q3.15.7 NAT 2015 2M

In the circuit shown, assume that the opamp is ideal. If the gain (v_o/v_{in}) is −12, the value of R (in kΩ) is _____.

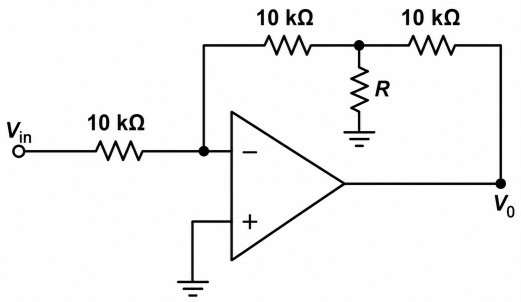


Fig. Q3.15.7

Q3.14.1 NAT 2014 1M

In the low-pass filter shown in the figure, for a cut-off frequency of 5 kHz , the value of R_2 (in kΩ) is _____.

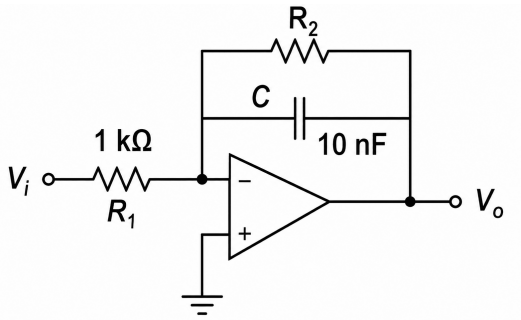


Fig. Q3.14.1

Q3.14.2 MCQ 2014 2M

In the circuit shown, the op-amp has finite input impedance, infinite voltage gain and zero input offset voltage. The output voltage V_{out} is

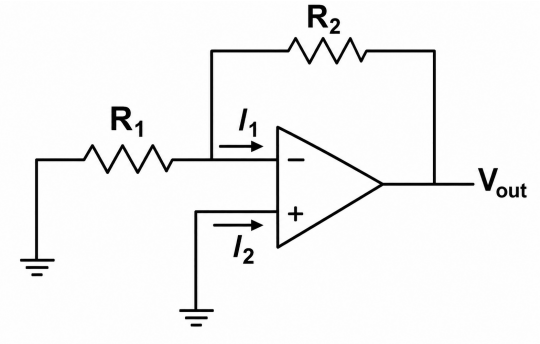


Fig. Q3.14.2

A. $-I_2(R_1 + R_2)$ **B.** I_2R_2
C. I_1R_2 **D.** $-I_1(R_1 + R_2)$

Q3.14.3 MCQ 2014 2M

Assuming that the Op-amp in the circuit shown is ideal, V_o is given by

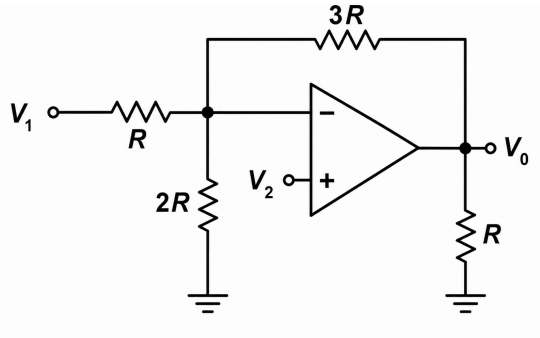


Fig. Q3.14.3

A. $\frac{5}{2}V_1 - 3V_2$ **B.** $2V_1 - \frac{5}{2}V_2$
C. $-\frac{3}{2}V_1 + \frac{7}{2}V_2$ **D.** $-3V_1 + \frac{11}{2}V_2$

Q3.14.4 MCQ 2014 1M

The circuit shown represents

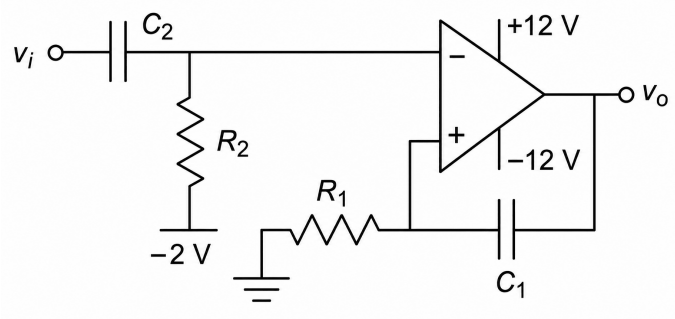


Fig. Q3.14.4

A. a bandpass filter

B. a voltage controlled oscillator

C. an amplitude modulator

D. a monostable multivibrator

Q3.13.1 MCQ 2013 2M

In the circuit shown below the op-amps are ideal. Then V_{out} in Volts is

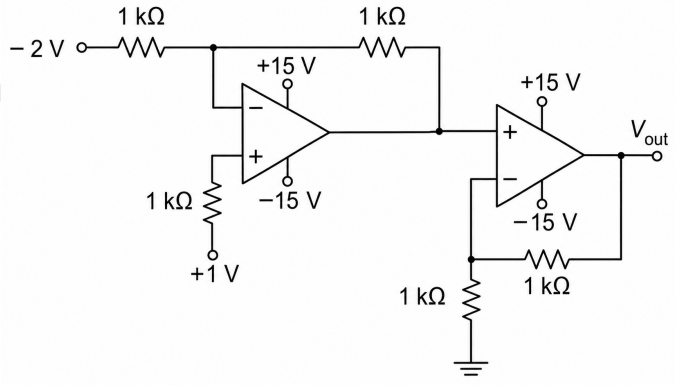


Fig. Q3.13.1

A. 4 **B.** 6 **C.** 8 **D.** 10

Q3.12.1 MCQ 2012 2M

The circuit shown is a

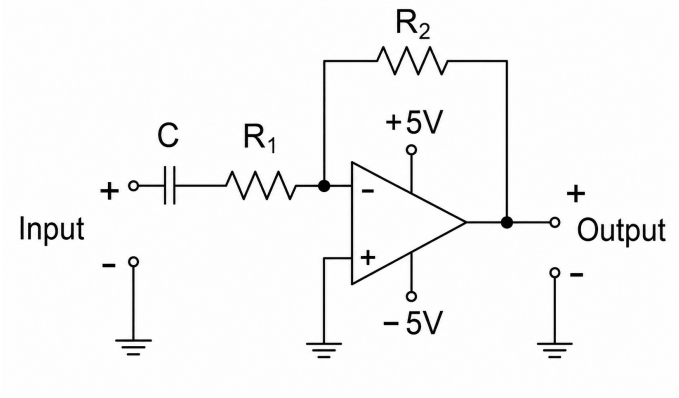


Fig. Q3.12.1

A. low pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

B. high pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

C. low pass filter with $f_{3dB} = \dfrac{1}{R_1 C}$ rad/s

D. high pass filter with $f_{3dB} = \dfrac{1}{(R_1 + R_2)C}$ rad/s

Q3.11.1 MCQ 2011 1M

The circuit below implements a filter between the input current i_i and the output voltage v_o. Assume that the opamp is ideal. The filter implemented is a

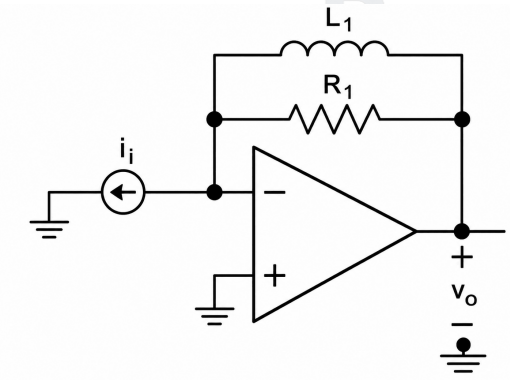


Fig. Q3.11.1

A. low pass filter
B. band pass filter
C. band stop filter
D. high pass filter

Q3.10.1 MCQ 2010 1M

Assuming the OP-AMP to be ideal, the voltage gain of the amplifier shown below is

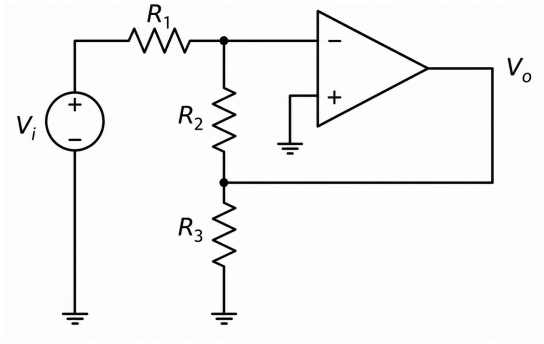


Fig. Q3.10.1

A. $-\dfrac{R_2}{R_1}$ **B.** $-\dfrac{R_3}{R_1}$ **C.** $-\left(\dfrac{R_2 \parallel R_3}{R_1}\right)$ **D.** $-\left(\dfrac{R_2 + R_3}{R_1}\right)$

Q3.10.2 MCQ 2010 2M

The transfer characteristic for the precision rectifier circuit shown below is (assume ideal OP-AMP and practical diodes)

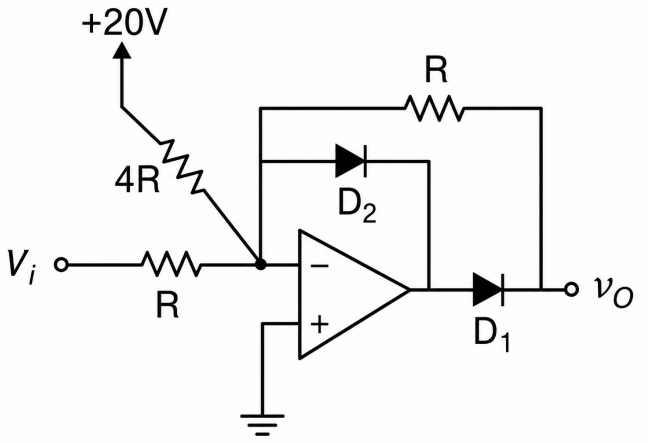
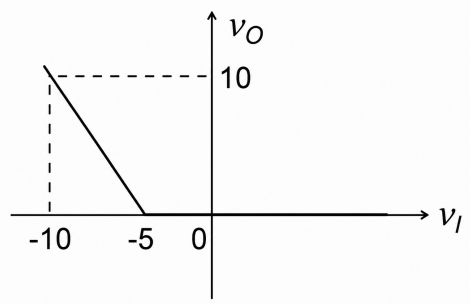
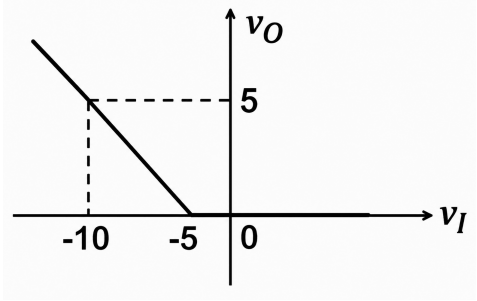


Fig. Q3.10.2

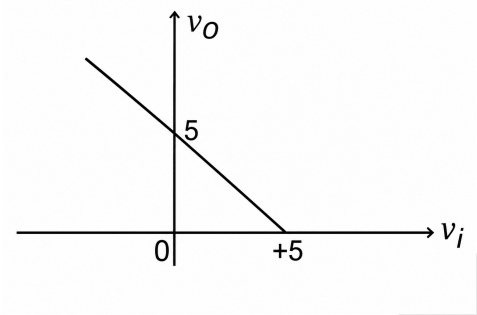
A.



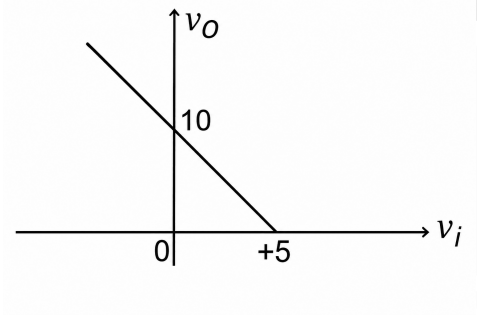
B.



C.



D.



Q3.09.1 MCQ 2009 2M

In the following astable multivibrator circuit, which properties of $v_o(t)$ depend on R_2 ?

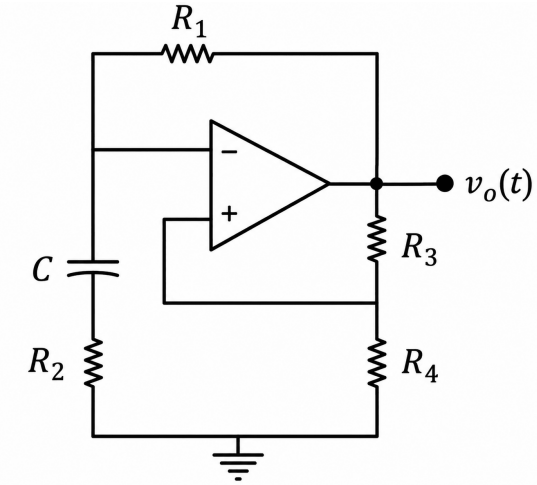


Fig. Q3.09.1

A. Only the frequency

B. Only the amplitude

C. Both the amplitude and the frequency

D. Neither the amplitude nor the frequency

Q3.08.1 MCQ 2008 2M

Consider the following circuit using an ideal OPAMP. The I-V characteristics of the diode is described by the relation $I = I_0\left(e^{\frac{V}{V_T}} - 1\right)$ where $V_T = 25$ mV, $I_0 = 1\mu$A and V is the voltage across the diode (taken as positive for forward bias).

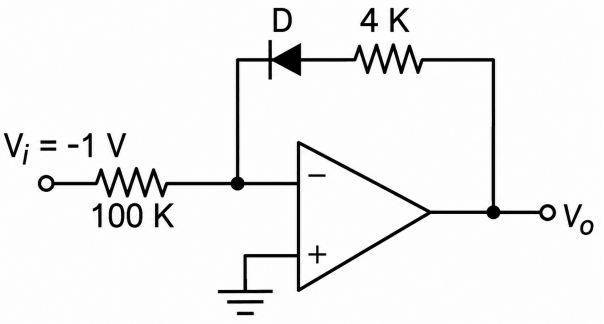


Fig. Q3.08.1

For an input voltage $V_i = -1$ V, the output voltage V_o is

A. 0 V **B.** 0.1 V **C.** 0.7 V **D.** 1.1 V

Q3.08.2 MCQ 2008 2M

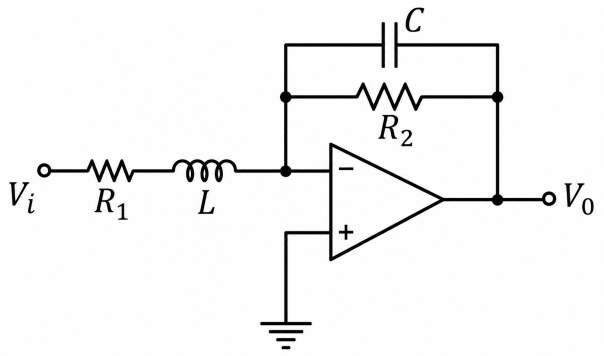


Fig. Q3.08.2

The OPAMP circuit shown above represents a

A. high pass filter **B.** low pass filter
C. band pass filter **D.** band reject filter

Q3.08.3 MCQ 2008 2M

Consider the Schmidt trigger circuit shown below.

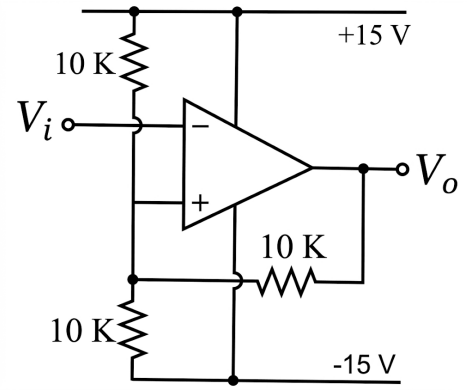


Fig. Q3.08.3

A triangular wave which goes from −12 V to 12 V is applied to the inverting input of the OPAMP. Assume that the output of the OPAMP swings from +15V to −15V. The voltage at the non-inverting input switches between

A. −12V and +12 V
B. −7.5V and +7.5V
C. −5V and +5 V
D. 0V and 5V

Q3.07.1 MCQ 2007 2M

For the Op-Amp circuit shown in the figure, V_o is

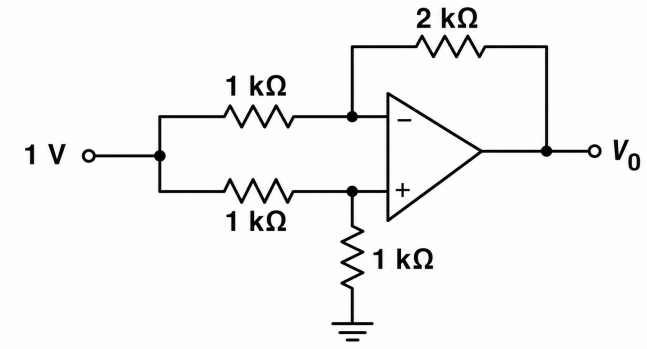


Fig. Q3.07.1

A. −2 V **B.** −1 V **C.** −0.5 V **D.** 0.5 V

Q3.07.2 MCQ 2007 2M

In the Op-Amp circuit shown, assume that the diode current follows the equation $I = I_s \exp(V/V_T)$. For $V_i = 2$ V, $V_o = V_{o1}$, and for $V_i = 4$ V, $V_o = V_{o2}$. The relationship between V_{o1} and V_{o2} is

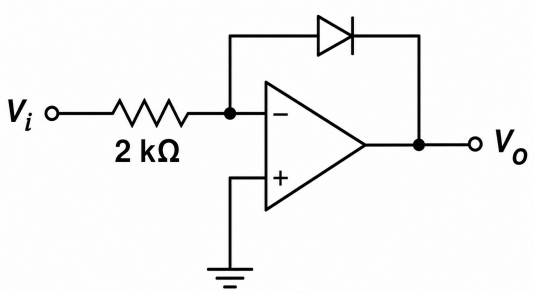


Fig. Q3.07.2

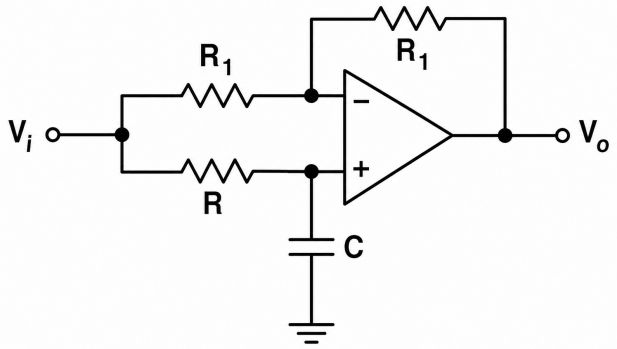
A. $V_{o2} = \sqrt{2}V_{o1}$
B. $V_{o2} = e^2 V_{o1}$
C. $V_{o2} = V_{o1} \ln 2$
D. $V_{o1} - V_{o2} = V_T \ln 2$

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q3.07.3 & Q3.07.4:

Consider the Op-Amp circuit shown in the figure.



Q3.07.3 MCQ 2007 2M

The transfer function $V_o(s)/V_i(s)$ is

A. $\dfrac{1 - sRC}{1 + sRC}$
B. $\dfrac{1 + sRC}{1 - sRC}$
C. $\dfrac{1}{1 - sRC}$
D. $\dfrac{1}{1 + sRC}$

Q3.07.4 MCQ 2007 2M

If $V_i = V_1 \sin(\omega t)$ and $V_o = V_2 \sin(\omega t + \phi)$, then the minimum and maximum values of ϕ (in radians) are respectively

A. $-\pi/2$ and $\pi/2$
B. 0 and $\pi/2$
C. $-\pi$ and 0
D. $-\pi/2$ and 0

Q3.05.1 MCQ 2005 1M

The input resistance R_i of the amplifier shown in figure is

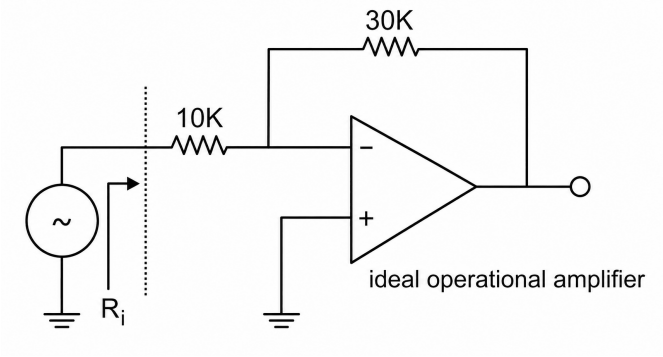


Fig. Q3.05.1

A. $\frac{30}{4}$ kΩ **B.** 10 kΩ **C.** 40 kΩ **D.** infinite

Q3.05.2 MCQ 2005 2M

The voltage e_o indicated in figure has been measured by an ideal voltmeter. Which of the following can be calculated?

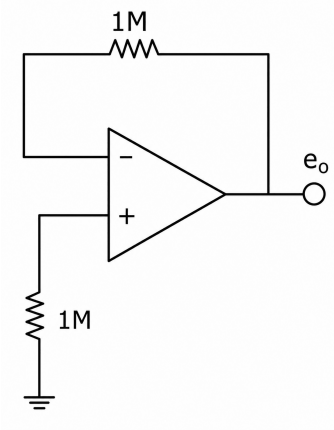


Fig. Q3.05.2

A. Bias current of the inverting input only.

B. Bias current of the inverting and non-inverting inputs only.

C. Input offset current only.

D. Both the bias currents and the input offset current.

Q3.05.3 MCQ 2005 2M

The Op-amp circuit shown in figure is a filter. The type of filter and its cut-off frequency are respectively.

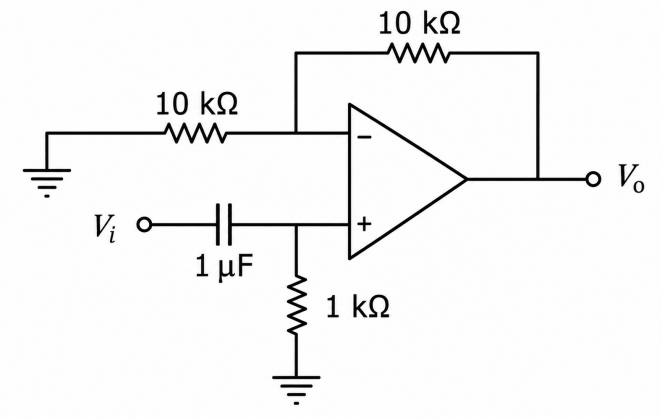


Fig. Q3.05.3

A. high pass, 1000 rad/sec. **B.** low pass, 1000 rad/sec.
C. high pass, 10000 rad/sec. **D.** low pass, 10000 rad/sec.

Q3.05.4 MCQ 2005 2M

Given the ideal operational amplifier circuit shown in figure indicate the correct transfer characteristics assuming ideal diodes with zero cut-in voltage.

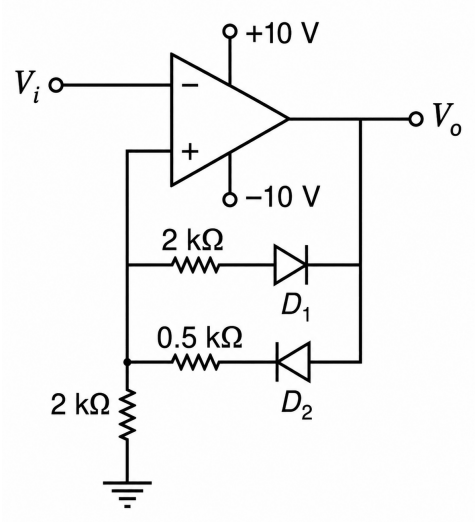
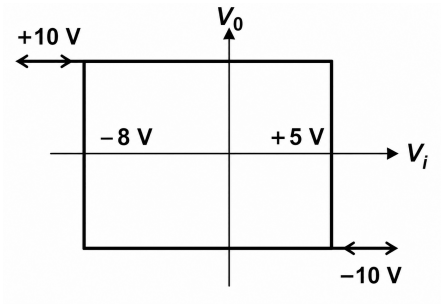
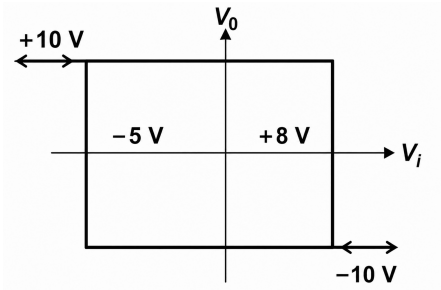


Fig. Q3.05.4

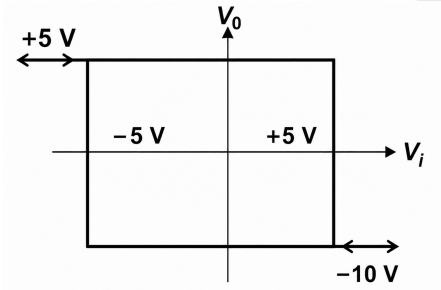
A.



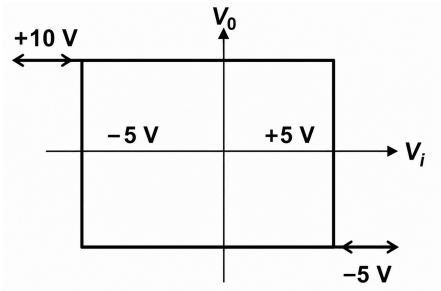
B.



C.



D.



Q3.04.1 MCQ 2004 1M

An ideal op-amp is an ideal

A. voltage controlled current source

B. voltage controlled voltage source

C. current controlled current source

D. current controlled voltage source

Q3.04.2 MCQ 2004 1M

The circuit in Figure is a

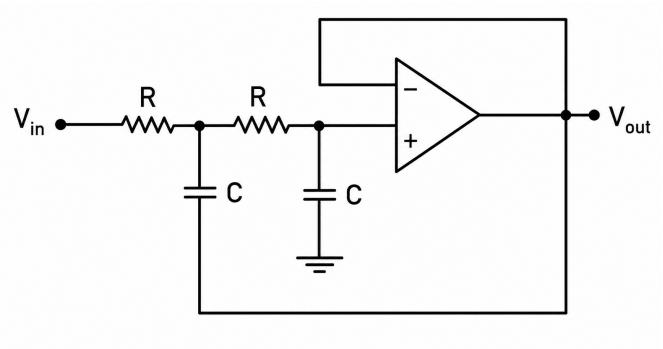


Fig. Q3.04.2

A. low-pass filter **B.** high-pass filter
C. band-pass filter **D.** band-reject filter

Q3.04.3 MCQ 2004 2M

In the op-amp circuit given in figure, the load current I_L is

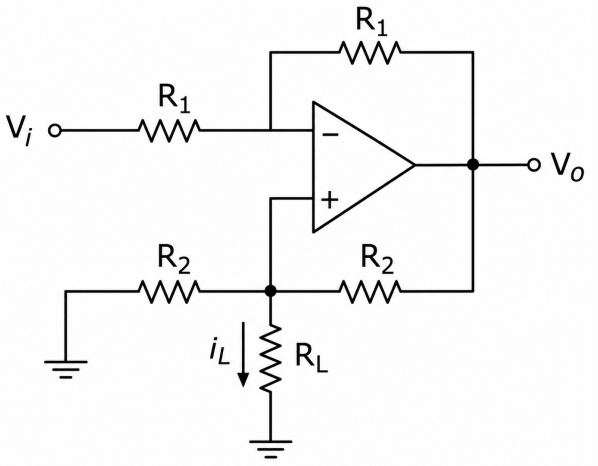


Fig. Q3.04.3

A. $-\frac{V_i}{R_2}$ **B.** $\frac{V_i}{R_2}$ **C.** $-\frac{V_i}{R_L}$ **D.** $\frac{V_i}{R_L}$

Q3.03.1 MCQ 2003 1M

If the input to the ideal comparator shown in figure is a sinusoidal signal of 8 V (peak to peak) without any DC component, then the output of the comparator has a duty cycle of

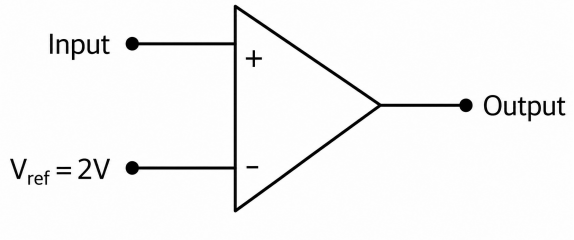


Fig. Q3.03.1

A. $\frac{1}{2}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{6}$ **D.** $\frac{9}{12}$

Q3.03.2 MCQ 2003 1M

If the differential voltage gain and the common mode voltage gain of a differential amplifier are 48 dB and 2 dB respectively, then its common mode rejection ratio is

A. 23 dB **B.** 25 dB **C.** 46 dB **D.** 50 dB

Q3.03.3 MCQ 2003 2M

The oscillator circuit shown in figure has an ideal inverting amplifier. Its frequency of oscillation (in Hz) is

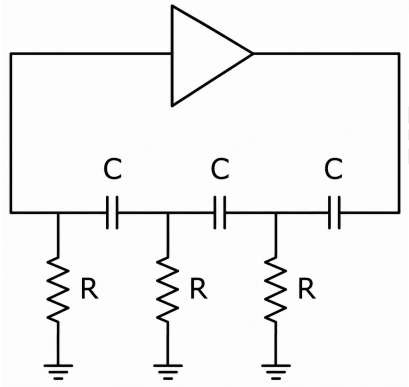


Fig. Q3.03.3

A. $\frac{1}{(2\pi\sqrt{6}RC)}$ **B.** $\frac{1}{(2\pi RC)}$

C. $\frac{1}{(\sqrt{6}RC)}$ **D.** $\frac{\sqrt{6}}{(2\pi RC)}$

Q3.03.4 MCQ 2003 2M

If the op-amp in figure is ideal, the output voltage V_{out} will be equal to

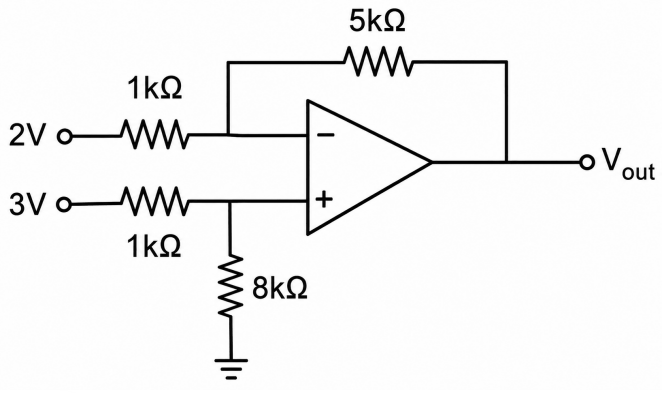


Fig. Q3.03.4

A. 1 V **B.** 6 V **C.** 14 V **D.** 17 V

Q3.02.1 MCQ 2002 1M

An op-amp has a gain-bandwidth product of 1 MHz. A non-inverting amplifier using this op-amp and having a voltage gain of 20 dB will exhibit a 3-dB bandwidth of

A. 50 KHz **B.** 100 KHz **C.** $\frac{1000}{17}$ KHz **D.** $\frac{1000}{7.07}$

Q3.02.2 MCQ 2002 2M

An amplifier using an opamp with a slew-rate SR = $1V/\mu sec$ has a gain of 40 dB. If this amplifier has to faithfully amplify sinusoidal signals from dc to 20 KHz without introducing any slew-rate induced distortion, then the input signal level must not exceed.

A. 795 mV **B.** 395 mV **C.** 79.5 mV **D.** 39.5 mV

Q3.02.3 MCQ 2002 2M

The circuit in the figure employs positive feedback and is intended to generate sinusoidal oscillation. If at a frequency f_0, $B(f) = \Delta\frac{V_f(f)}{V_0(f)} = \frac{1}{6}\angle 0°$, then to sustain oscillation at this frequency

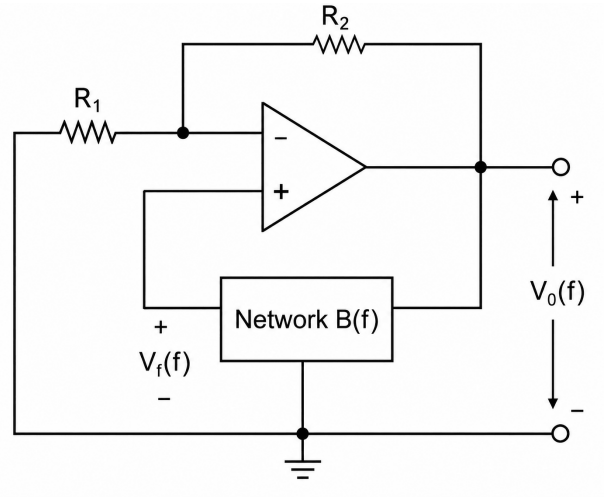


Fig. Q3.02.3

A. $R_2 = 5R_1$ **B.** $R_2 = 6R_1$

C. $R_2 = \frac{R_1}{6}$ **D.** $R_2 = \frac{R_1}{5}$

Q3.01.1 MCQ 2001 1M

The ideal OP-AMP has the following characteristics.

A. $R_i = \infty, A = \infty, R_0 = 0$

B. $R_i = 0, A = \infty, R_0 = 0$

C. $R_i = \infty, A = \infty, R_0 = \infty$

D. $R_i = 0, A = \infty, R_0 = \infty$

Q3.01.2 MCQ 2001 2M

The inverting OP-AMP shown in the figure has an open-loop gain of 100. The closed loop gain V_0/V_s is

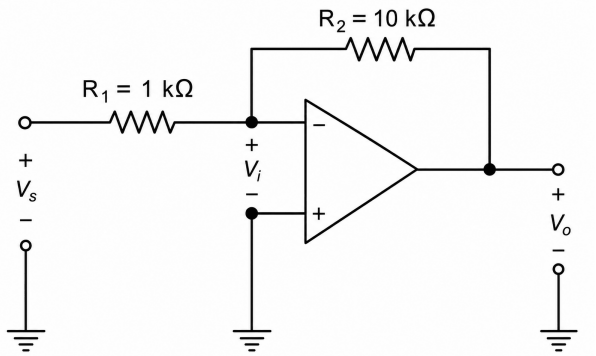


Fig. Q3.01.2

A. -8 **B.** -9 **C.** -10 **D.** -11

Q3.01.3 MCQ 2001 2M

In the given figure, assume the OP-AMPs to be ideal. The output v_0 of the circuit is:

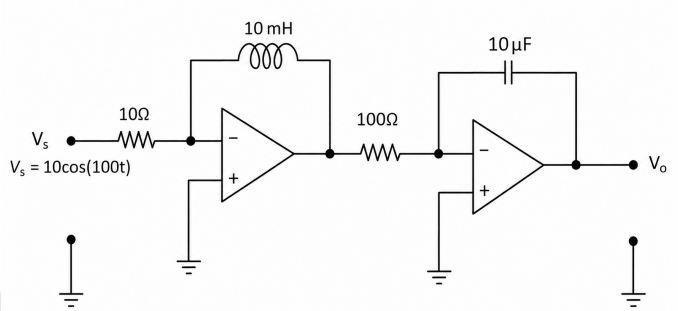


Fig. Q3.01.3

A. $10\cos(100t)$ **B.** $10\int_0^t \cos(100\tau)d\tau$

C. $10^{-4}\int_0^t \cos(100\tau)d\tau$ **D.** $10^{-4}\frac{d}{dt}\cos(100t)$

Q3.00.1 MCQ 2000 1M

In the circuit of below figure, V_o is

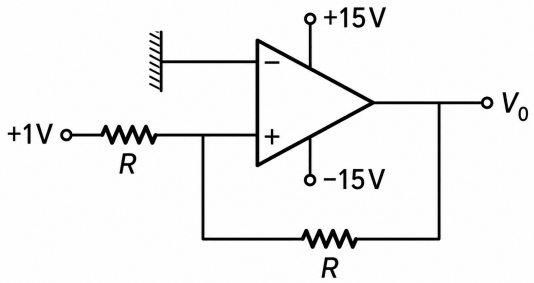


Fig. Q3.00.1

A. -1 V **B.** 2 V **C.** $+1$ V **D.** $+15$ V

Q3.00.2 MCQ 2000 1M

If the op-amp in below figure, is ideal, then v_o, is

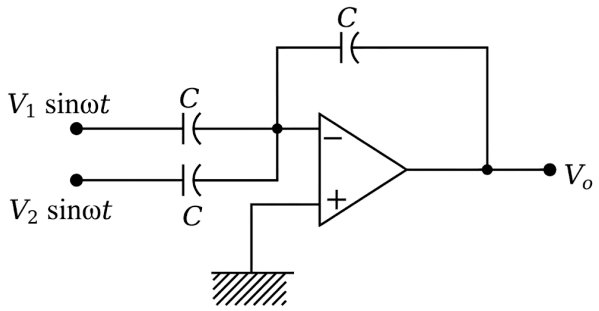


Fig. Q3.00.2

A. zero
B. $(V_1 - V_2)\sin\omega t$
C. $-(V_1 + V_2)\sin\omega t$
D. $(V_1 + V_2)\sin\omega t$

Q3.00.3 MCQ 2000 1M

Assume that the op-amp of figure is ideal. If V_i is a triangular wave, then v_o will be

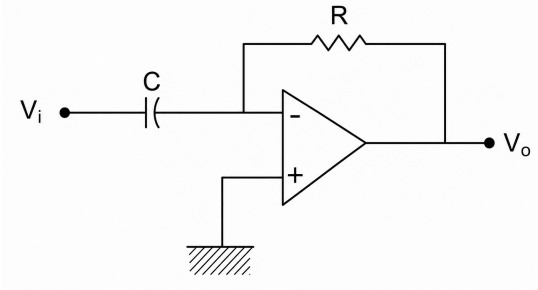


Fig. Q3.00.3

A. square wave
B. triangular wave
C. parabolic wave
D. sine wave

Q3.00.4 MCQ 2000 2M

If the op-amp in figure, has an input offset voltage of 5 mV and an open-loop voltage gain of 10, 000, then V_o will be

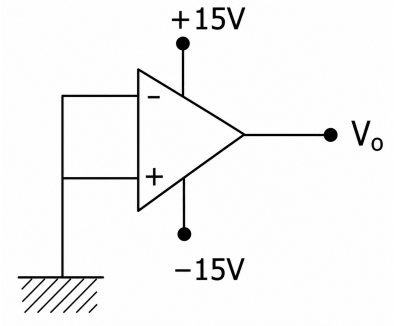


Fig. Q3.00.4

A. 0 V
B. 5 mV
C. +15 V or −15 V
D. +50 V or −50 V

Q3.99.1 NAT 1999 5M

Neatly sketch and label the dc transfer characteristic (*i.e.*, V_o vs. V_{in}) of the circuit shown in figure, as V_{in} varies from −2 V to +2 V. Assume ideal op-amp, and the diodes have a forward voltage of 0.6 V and zero incremental resistance.

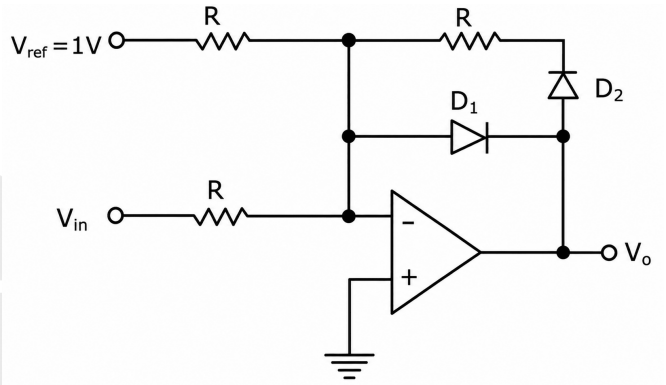


Fig. Q3.99.1

Q3.98.1 MCQ 1998 1M

One input terminal of high gain comparator circuit is connected to ground and a sinusoidal voltage is applied to the other input. The output of comparator will be

A. a sinusoid
B. a full rectified sinusoid
C. a half rectified sinusoid
D. a square wave

Q3.98.2 NAT 1998 5M

Determine the input impedance of the circuit of figure and investigate if it can be inductive.

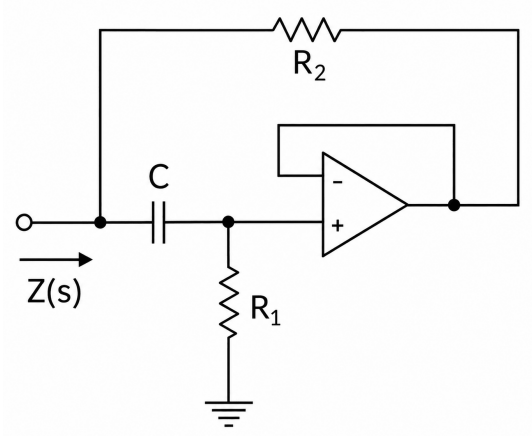


Fig. Q3.98.2

Q3.97.1 MCQ 1997 1M

The output voltage V_o of the circuit shown in the figure is

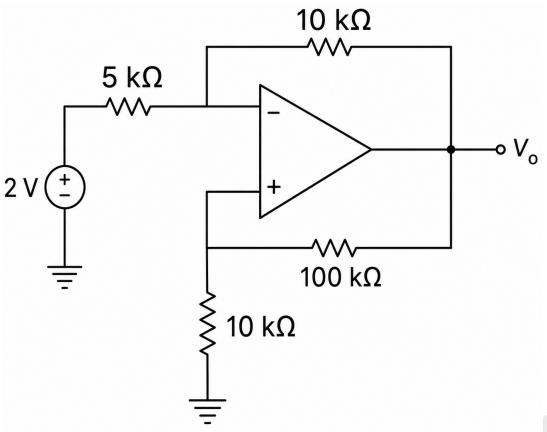


Fig. Q3.97.1

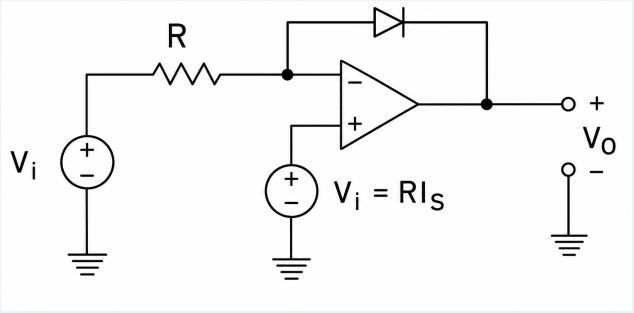
A. −4 V **B.** 6 V **C.** 5 V **D.** −5.5 V

Common Data Questions

The next 2 questions share the data below.

Common data for the next two questions:

Consider the circuit given in the figure using an ideal operational amplifier.



The characteristics of the diode are given by the relation

$$I = I_s\left(e^{\frac{V}{V_T}} - 1\right)$$

where V is the forward voltage across the diode.

Q3.97.2 NAT 1997 3M

Express V_o as function of V_i assuming $V_i > 0$

Q3.97.3 NAT 1997 2M

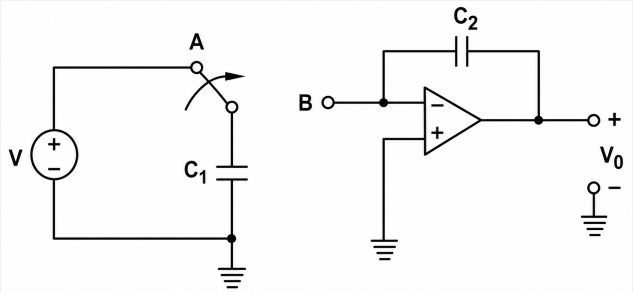
If $R = 100\,\text{k}\Omega$, $I_s = 1\,\mu\text{A}$ and (kT/q)= 25 mV, find the input voltage V_i for which $V_o = 0$.

Common Data Questions

The next 4 questions share the data below.

Common data for the next four questions:

In the circuit shown in the figure, the operational amplifier is ideal and that $V_o = 0$V initially. The switch is connected first to 'A' charging C_1 to the voltage V. It is then connected to the point 'B'. This process is repeated f times per second.



Q3.97.4 NAT 1997 1M

Calculate the charge transferred per second from node A to node B.

Q3.97.5 NAT 1997 2M

Derive the average rate of change of the output voltage V_o.

Q3.97.6 NAT 1997 1M

If the capacitor and the switch are removed and a resistor is connected between points A and B, find the value of the resistor to get the same average rate of change of the output voltage ?

Q3.97.7 NAT 1997 1M

If the repetition rate of the switching action is 10^4 times per second, $C_1 = 100$ pF, $C_2 = 10$ pF and $V = 10$ mV, what is the average rate of change of the output voltage ?

Q3.96.1 MCQ 1996 2M

Value of R in the oscillator circuit shown in the given figure, so chosen that it just oscillates at an angular frequency of ω. The value of ω and the required value of R will respectively be

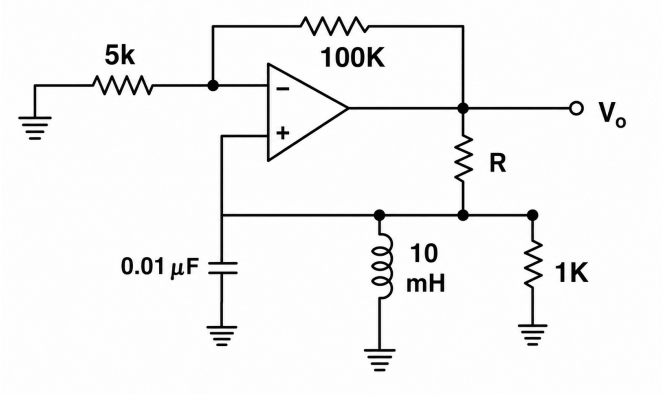


Fig. Q3.96.1

A. 10^5 rad/sec, 2×10^4 Ω

B. 2×10^4 rad/sec, 2×10^4 Ω

C. 2×10^4 rad/sec, 10^5 Ω

D. 10^5 rad/sec, 10^5 Ω

Q3.95.1 NAT 1995 1M

An OP-AMP is used as a zero-crossing detector. If the maximum output available from the OP-AMP is ±12 V $p-p$, and the slew rate of the OP-AMP is 12 V/μsec, then the maximum frequency(in KHz) of the input signal that can be applied without causing a reduction in the $p-p$ output is _____.

Q3.95.2 NAT 1995 1M

In the given circuit the given figure, if the voltage inputs $V-$ and $V+$ are to be amplified by the same amplification factor, the value of R(in kilo ohms) should be

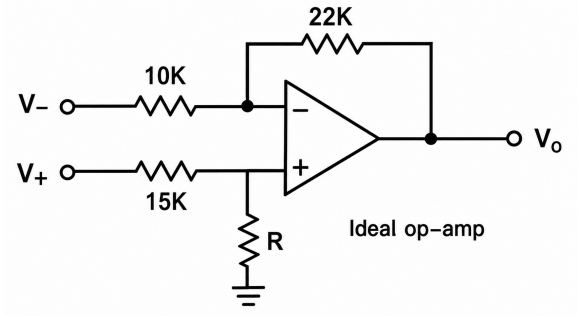


Fig. Q3.95.2

Q3.93.1 NAT 1993 1M

For the ideal op-amp circuit of figure shown, determine the output voltage V_o.

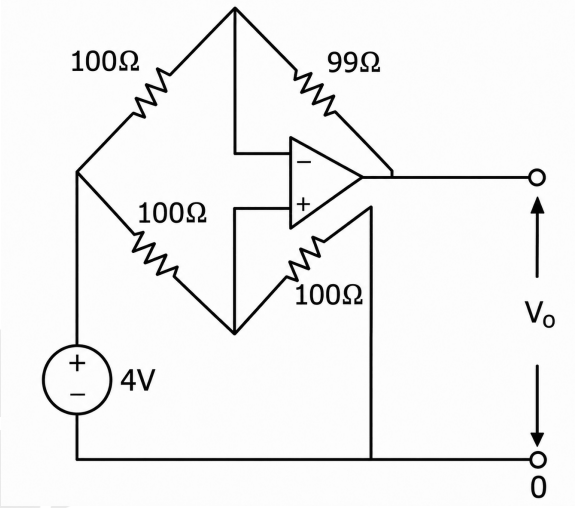


Fig. Q3.93.1

Q3.93.2 NAT 1993 2M

Find the output voltage, V_0 in the following circuit (figure), assuming that the op-amps are ideal.

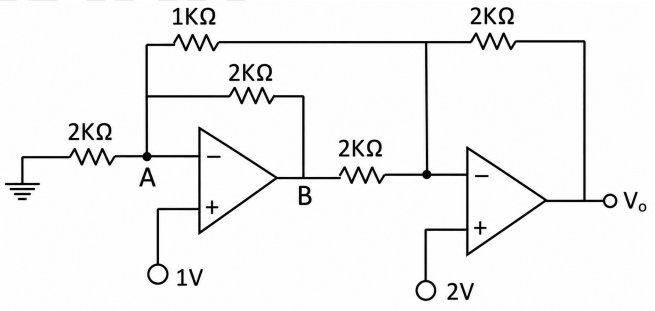


Fig. Q3.93.2

Q3.92.1 MCQ 1992 2M

An op. amp. has an offset voltage of 1 mV and is ideal in all other respects. If this op. amp. is used in the circuit shown in the given figure, the output voltage will be (select the nearest value)

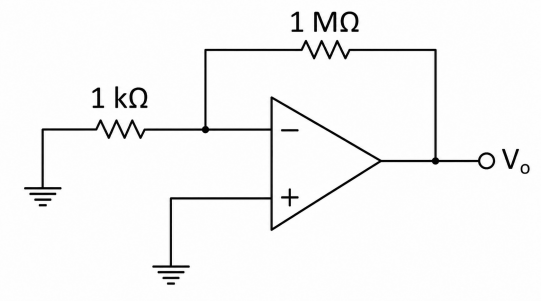


Fig. Q3.92.1

A. 1 mV **B.** 1 V **C.** ±1 V **D.** 0 V.

Q3.92.2 MCQ 1992 2M

The circuit of given figure, uses an ideal op amp. For small positive values of V_{in}, the circuit works as

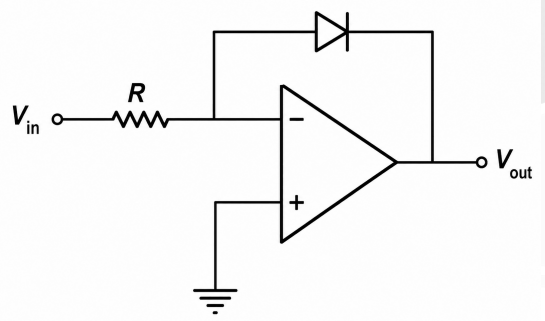


Fig. Q3.92.2

A. a halfwave rectifier **B.** a differentiator
C. a logarithmic amplifier **D.** an exponential amplifier

Q3.92.3 NAT 1992 2M

Assume that the operational amplifier in given figure is ideal. The current, I, through the 1 KΩ resistor is _____

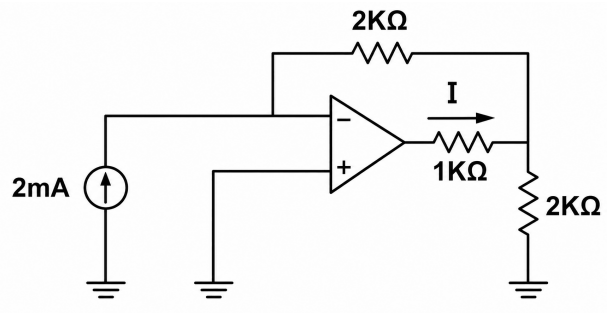


Fig. Q3.92.3

Q3.91.1 MCQ 1991 1M

Two non-inverting amplifiers, one having a unity gain and the other having a gain of twenty are made using identical operational amplifiers. As a compared to the unity gain amplifier, the amplifier with gain twenty has

A. less negative feedback **B.** greater input impedance
C. less bandwidth **D.** none of the above

Q3.91.2 NAT 1991 1M

In order that the circuit of Figure works properly as differentiator, it should be modified to _____ (draw the modified circuit).

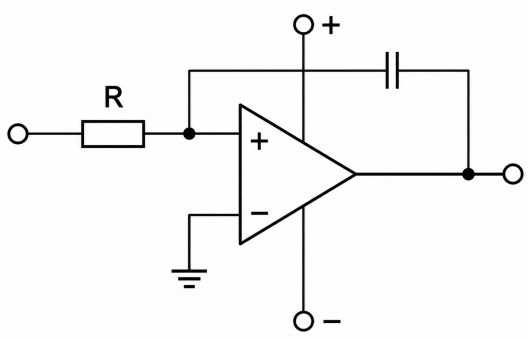


Fig. Q3.91.2

Q3.91.3 NAT 1991 5M

In figure, the operational amplifiers are ideal and their outputs can swing between −15 and +15 volts. Sketch on same diagram, the waveform of voltages V_1 and V_2 as a function of time. You must give the values of important parameters of this sketch.

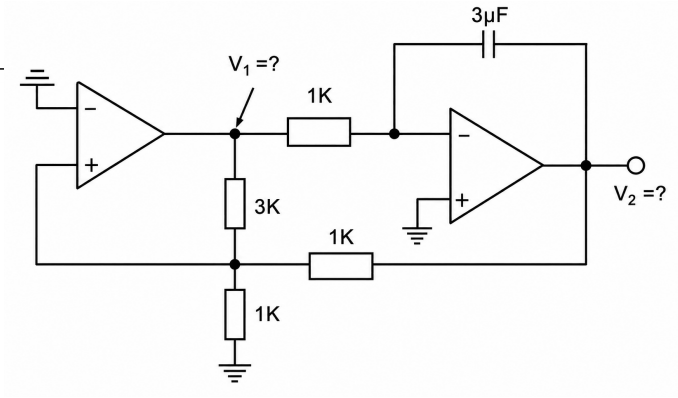


Fig. Q3.91.3

Q3.90.1 MCQ 1990 2M

The CMRR of the differential Amplifier of the figure is equal to

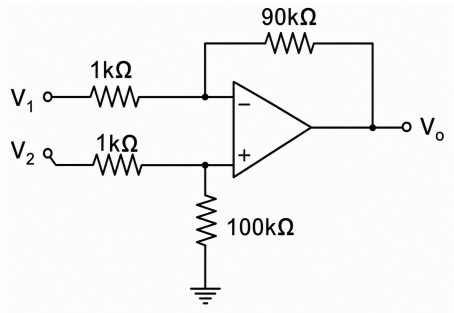


Fig. Q3.90.1

A. 1000 **B.** 0 **C.** 900 **D.** 1800

Q3.90.2 MCQ 1990 2M

The Op-Amp of figure shown below has a very poor open loop voltage gain of 45 but is otherwise ideal. The gain of the amplifier equals

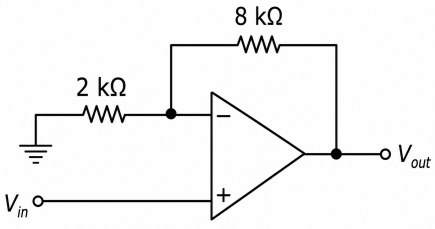


Fig. Q3.90.2

A. 5 **B.** 20 **C.** 4 **D.** 4.5

Q3.89.1 MSQ 1989 2M

Refer to figure shown below

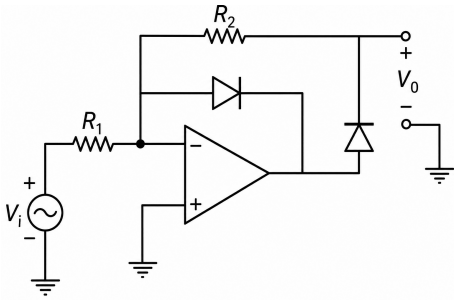


Fig. Q3.89.1

A. For $V_i > 0$, $V_0 = -\frac{R_2}{R_1}V_i$ **B.** For $V_i > 0$, $V_0 = 0$

C. For $V_i < 0$, $V_0 = -\frac{R_2}{R_1}V_i$ **D.** For $V_i < 0$, $V_0 = 0$

Q3.88.1 MCQ 1988 2M

The Op-Amp shown in figure is ideal, $R = \sqrt{L/C}$. The phase angle between V_0 and V_i at $\omega = \frac{1}{\sqrt{LC}}$ is

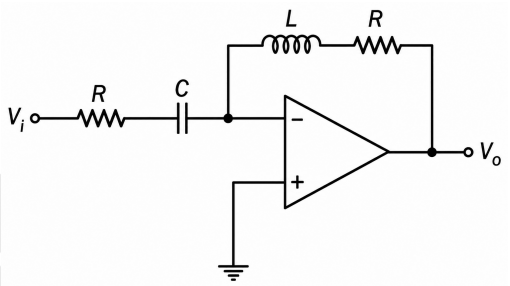


Fig. Q3.88.1

A. $\frac{\pi}{2}$ **B.** π **C.** $\frac{3\pi}{2}$ **D.** 2π

Q3.87.1 NAT 1987 2M

In figure if the CMRR of the operational amplifier is 60 dB, then the magnitude of the output voltage is _____ mV.

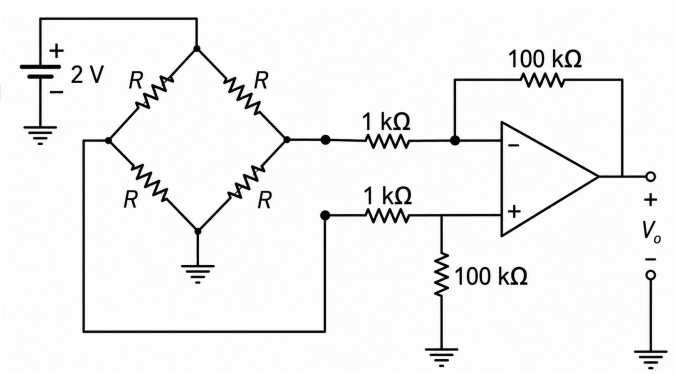


Fig. Q3.87.1

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q3.26.1 C	Q3.26.2 C	Q3.26.3 3
Q3.25.1 A,C	Q3.25.2 50	Q3.24.1 D
Q3.24.2 B	Q3.23.1 A	Q3.22.1 B,C
Q3.22.2 C	Q3.22.3 A	Q3.21.1 MTA
Q3.21.2 A	Q3.21.3 -12	Q3.20.1 D
Q3.20.2 6	Q3.20.3 D	Q3.18.1 C

Q3.18.2 0.5	Q3.17.1 B	Q3.17.2 44.37
Q3.16.1 A	Q3.16.2 15	Q3.16.3 0.798
Q3.16.4 400 to 425	Q3.16.5 -1	Q3.15.1 250
Q3.15.2 D	Q3.15.3 1.5	Q3.15.4 1
Q3.15.5 12	Q3.15.6 159.15	Q3.15.7 1
Q3.14.1 3.18	Q3.14.2 C	Q3.14.3 D

Q3.14.4 D	Q3.13.1 C	Q3.12.1 B
Q3.11.1 D	Q3.10.1 A	Q3.10.2 B
Q3.09.1 A	Q3.08.1 B	Q3.08.2 B
Q3.08.3 C	Q3.07.1 C	Q3.07.2 D
Q3.07.3 A	Q3.07.4 C	Q3.05.1 B
Q3.05.2 C	Q3.05.3 A	Q3.05.4 B

Q3.04.1 B	Q3.04.2 A	Q3.04.3 A
Q3.03.1 B	Q3.03.2 C	Q3.03.3 A
Q3.03.4 B	Q3.02.1 B	Q3.02.2 C
Q3.02.3 A	Q3.01.1 A	Q3.01.2 B
Q3.01.3 A	Q3.00.1 D	Q3.00.2 C
Q3.00.3 A	Q3.00.4 C	Q3.99.1 *

Q3.98.1 D	Q3.98.2 *	Q3.97.1 D
Q3.97.2 *	Q3.97.3 5.45	Q3.97.4 *
Q3.97.5 *	Q3.97.6 *	Q3.97.7 -1000
Q3.96.1 A	Q3.95.1 250	Q3.95.2 33
Q3.93.1 0.02	Q3.93.2 6	Q3.92.1 C
Q3.92.2 C	Q3.92.3 -4	Q3.91.1 C

Q3.91.2 *	Q3.91.3 *	Q3.90.1 C
Q3.90.2 D	Q3.89.1 B,C	Q3.88.1 C
Q3.87.1 100		

UNIT

IV

BJTs

Topics

1. BJT Physics and Biasing
2. Current Mirror Circuit Using BJT
3. Small Signal Analysis of BJT Amplifiers
4. Differential Amplifiers and Feedback Topologies Using BJTs

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 127	**MCQ:** 79	**MSQ:** 3	**NAT:** 42
1M: 49	**2M:** 70	**Descriptive:** 8	**Avg Weightage** ~4M	**Range:** 1M(2016)–6M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M

Consider the circuit shown in the Figure with $V_i = 3$ V and $V_{CC} = 12$ V. Assume $V_{BE} = 0.7$ V and $\beta_{dc} = 99$ for the BJT. Which of the following options is the correct value of the current I_o?

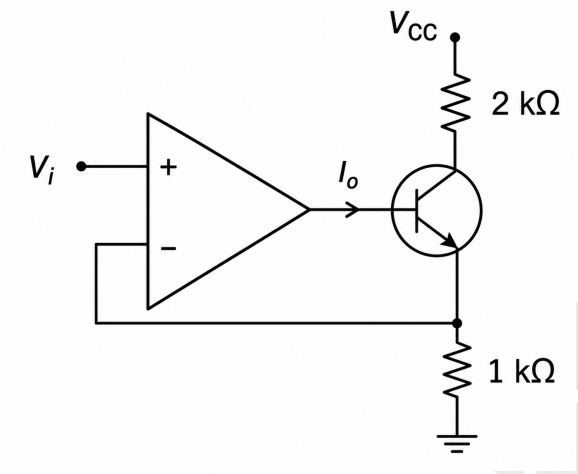


Fig. Q4.26.1

A. 60 μA **B.** 6 μA **C.** 3 μA **D.** 30 μA

Q4.26.2 MCQ 2026 2M

A small signal source, $V_i(t) = A\cos(10^5 t) + B\sin(10^7 t)$ is applied to a BJT circuit as shown in the Figure. Assume zero source resistance, $V_{BE} = 0.7$ V, $\beta_{dc} = 99$, Early voltage = 100 V and Thermal voltage = 25 mV. Effect of internal parasitic capacitances of the BJT may be neglected. Which expression is the best approximation of the output voltage $V_o(t)$?

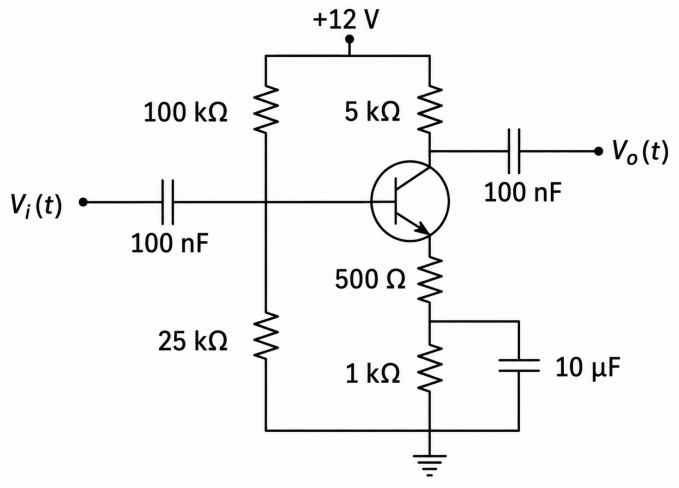


Fig. Q4.26.2

A. $-9.1[A\cos(10^5 t) + B\sin(10^7 t)]$

B. $9.1[A\cos(10^5 t) - B\sin(10^7 t)]$

C. $-190.4[A\cos(10^5 t) + B\sin(10^7 t)]$

D. $190.4[A\cos(10^5 t) - B\sin(10^7 t)]$

Q4.25.1 MCQ 2025 1M

A simplified small-signal equivalent circuit of a BJT-based amplifier is given below. The small-signal voltage gain v_o/v_s (in V/V) is _____.

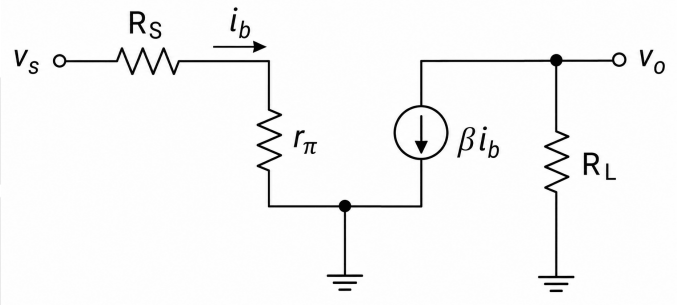


Fig. Q4.25.1

A. $\frac{-\beta R_L}{R_S + r_\pi}$ **B.** $\frac{+\beta R_L}{R_S}$ **C.** $\frac{-\beta R_L}{R_S}$ **D.** $\frac{+\beta R_L}{R_S + r_\pi}$

Q4.25.2 MCQ 2025 1M

The ideal BJT in the circuit given below is biased in the active region with a β of 100. If I_B is 10 μA, then V_{CE} (in Volts, *rounded off to two decimal places*) is _____.

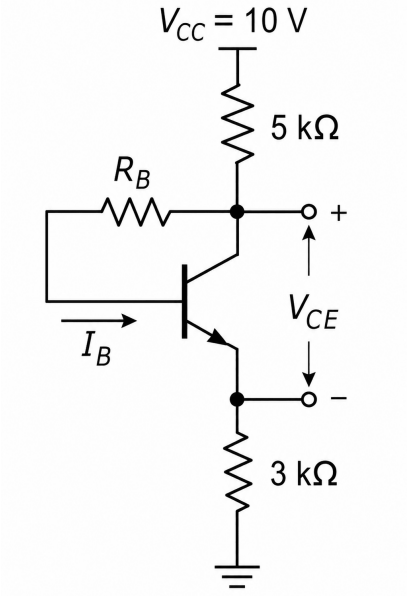


Fig. Q4.25.2

A. 4.95 **B.** 3.03 **C.** 1.92 **D.** 3.73

Q4.25.3 MCQ 2025 2M

In the circuit shown, the identical transistors Q1 and Q2 are biased in the active region with $\beta = 120$. The Zener diode is in the breakdown region with $V_Z = 5$ V and $I_Z = 25$ mA. If $I_L = 12$ mA and $V_{EB1} = V_{EB2} = 0.7$ V, then the values of R_1 and R_2 (in kΩ, *rounded off to one decimal place*) are _____, respectively.

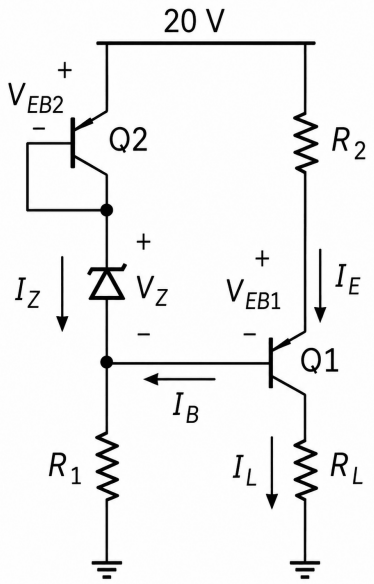


Fig. Q4.25.3

A. 0.6 and 0.4 **B.** 1.4 and 2.5
C. 14.0 and 25.0 **D.** 6.0 and 4.0

Q4.20.1 NAT 2020 2M

In the voltage regulator shown below, V_I is the unregulated input at 15 V. Assume $V_{BE} = 0.7$ V and the base current is negligible for both the BJTs. If the regulated output V_O is 9 V, the value of R_2 is _____ Ω.

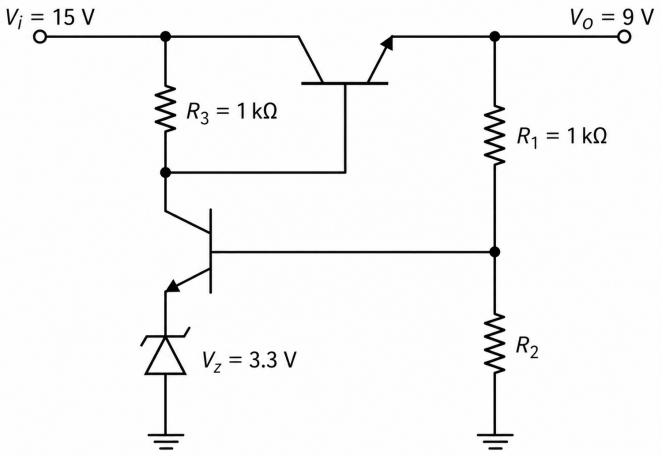


Fig. Q4.20.1

Q4.20.2 MCQ 2020 2M

For the BJT in the amplifier shown below, $V_{BE} = 0.7$ V, $kT/q = 26$ mV. Assume that BJT output resistance (r_o) is very high and the base current is negligible. The capacitors are also assumed to be short circuited at signal frequencies. The input v_i is direct coupled. The low frequency voltage gain v_o/v_i of the amplifier is

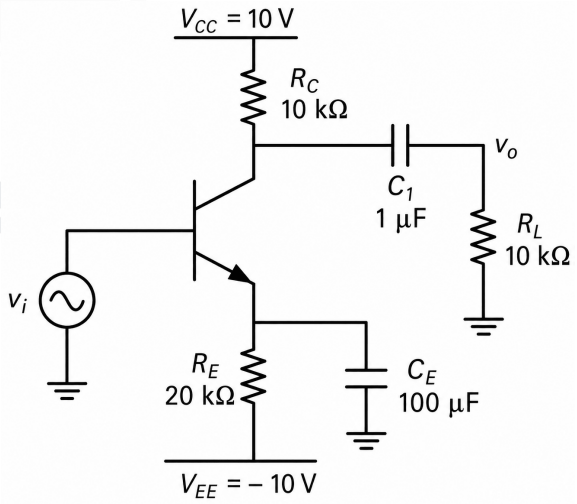


Fig. Q4.20.2

A. –89.42 **B.** –128.21 **C.** –178.85 **D.** –256.42

Q4.17.1 MCQ 2017 1M

The Miller effect in the context of a Common Emitter amplifier explains

A. an increase in the low-frequency cutoff frequency

B. an increase in the high-frequency cutoff frequency

C. a decrease in the low-frequency cutoff frequency

D. a decrease in the high-frequency cutoff frequency

Q4.17.2 NAT 2017 2M

In the figure shown, the npn transistor acts as a switch.

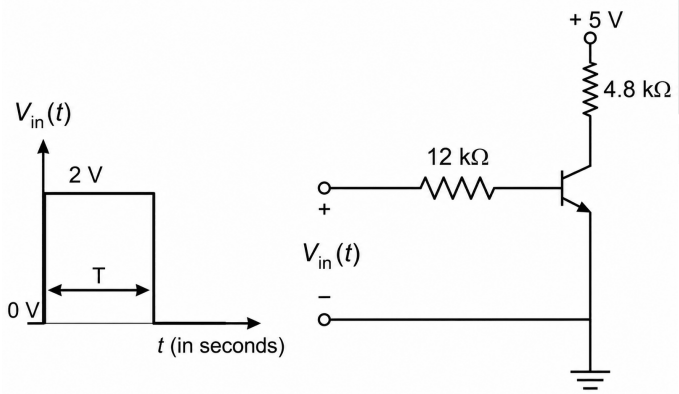


Fig. Q4.17.2

For the input $V_{in}(t)$ as shown in the figure, the transistor switches between the cut-off and saturation regions of operation, when T is large. Assume collector-to-emitter voltage at saturation $V_{CE(\text{sat})} = 0.2$ V and base-to-emitter voltage $V_{BE} = 0.7$ V. The minimum value of the common-base current gain (α) of the transistor for the switching should be _____.

Give the gain rounded off to two decimal places.

Q4.17.3 NAT 2017 2M

For the DC analysis of the Common-Emitter amplifier shown, neglect the base current and assume that the emitter and collector currents are equal. Given that $V_T = 25$ mV, $V_{BE} = 0.7$ V, and the BJT output resistance r_o is practically infinite. Under these conditions, the midband voltage gain magnitude, $A_v = |v_o/v_i|$ V/V, is _____.

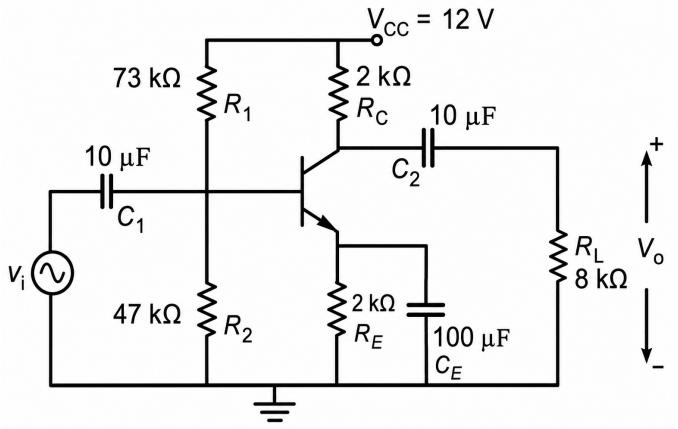


Fig. Q4.17.3

Q4.17.4 NAT 2017 1M

Consider the circuit shown in the figure. Assume base-to-emitter voltage $V_{BE} = 0.8$ V and common base current gain(α) of the transistor is unity.

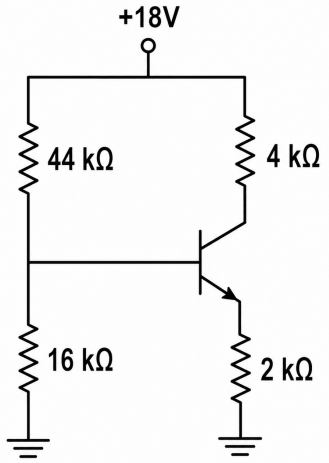


Fig. Q4.17.4

The value of the collector- to - emitter voltage V_{CE} (in volt) is _____.

Q4.17.5 NAT 2017 2M

In the voltage reference circuit shown in the figure, the op-amp is ideal and the transistors $Q_1, Q_2 \ldots, Q_{32}$ are identical in all respects and have infinitely large values of common emitter current gain. The relation $I_C = I_S \exp(V_{BE}/V_T)$, where I_S is the saturation current. Assume that the voltage V_P shown in the figure is 0.7 V and the thermal voltage $V_T = 26$mV.

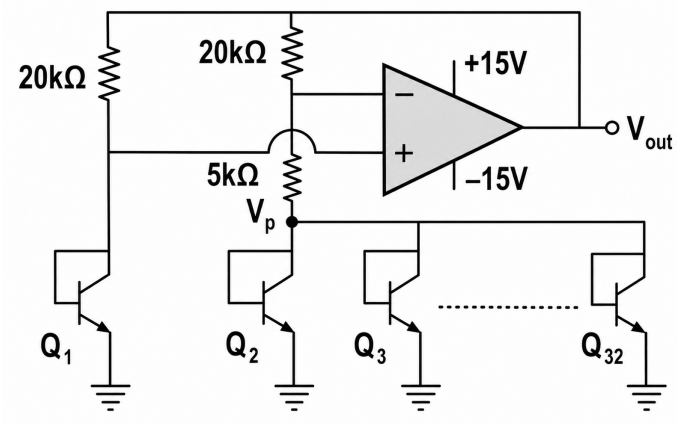


Fig. Q4.17.5

The output voltage V_{out} (in volts) is _____.

Q4.17.6 NAT | 2017 | 2M

In the circuit shown, transistors Q_1 and Q_2 are biased at a collector current of 2.6 mA.

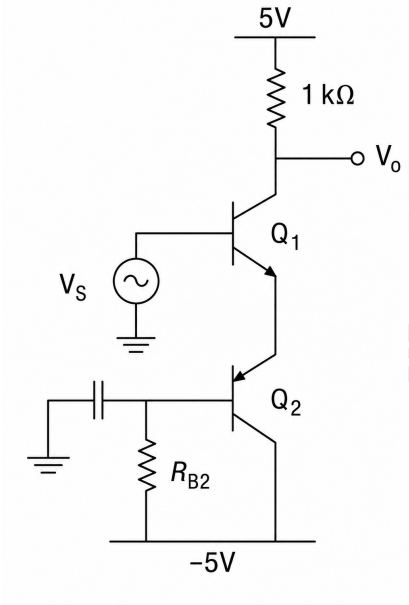


Fig. Q4.17.6

Assuming that transistor current gains are sufficiently large to assume collector current equal to emitter current and thermal voltage of 26 mV, the magnitude of voltage gain V_0/V_s in the mid-band frequency range is _____ (up to second decimal place).

Q4.16.1 NAT | 2016 | 1M

Resistor R_1 in the circuit below has been adjusted so that $I_1 = 1$ mA. The bipolar transistors Q1 and Q2 are perfectly matched and have very high current gain, so their base currents are negligible. The supply voltage V_{cc} is 6 V. The thermal voltage kT/q is 26 mV.

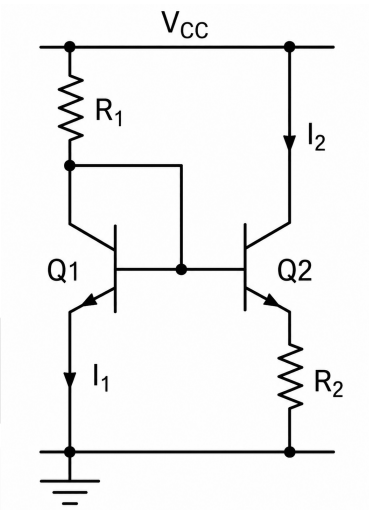


Fig. Q4.16.1

The value of R_2 (in Ω) for which $I_2 = 100\,\mu$A is _____

Q4.16.2 MCQ | 2016 | 1M

Which one of the following statements is correct about an ac-coupled common-emitter amplifier operating in the mid-band region?

A. The device parasitic capacitances behave like open circuits, whereas coupling and bypass capacitances behave like short circuits.

B. The device parasitic capacitances, coupling capacitances and bypass capacitances behave like open circuits.

C. The device parasitic capacitances, coupling capacitances and bypass capacitances behave like short circuits.

D. The device parasitic capacitances behave like short circuits, whereas coupling and bypass capacitances behave like open circuits.

Q4.16.3 NAT 2016 1M

Consider the circuit shown in the figure. Assuming $V_{BE1} = V_{EB2} = 0.7$ volt, the value of the dc voltage V_{C2} (in volt) is _____.

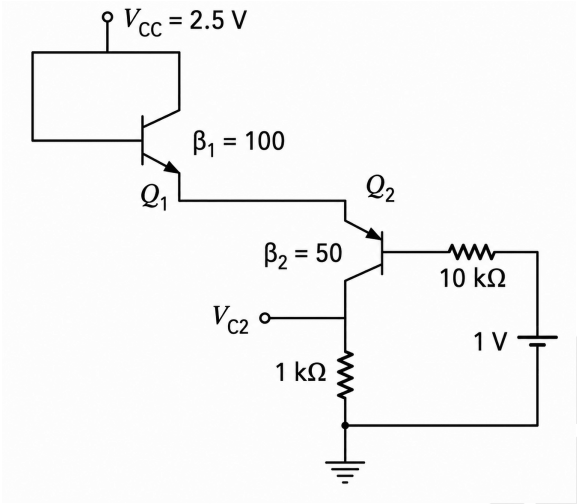


Fig. Q4.16.3

Q4.15.1 NAT 2015 2M

In the circuit shown, $I_1 = 80$ mA and $I_2 = 4$ mA. Transistors T_1 and T_2 are identical. Assume that the thermal voltage V_T is 26 mV at 27°C. At 50°C, the value of the voltage $V_{12} = V_1 - V_2$ (in mV) is _____.

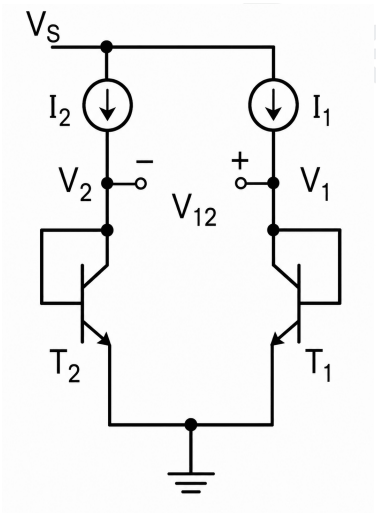


Fig. Q4.15.1

Q4.15.2 NAT 2015 2M

For the voltage regulator circuit shown, the input voltage (V_{in}) is 20 V ± 20% and the regulated output voltage (V_{out}) is 10 V. Assume the opamp to be ideal. For a load R_L drawing 200 mA, the maximum power dissipation in Q_1 (in Watts) is _____.

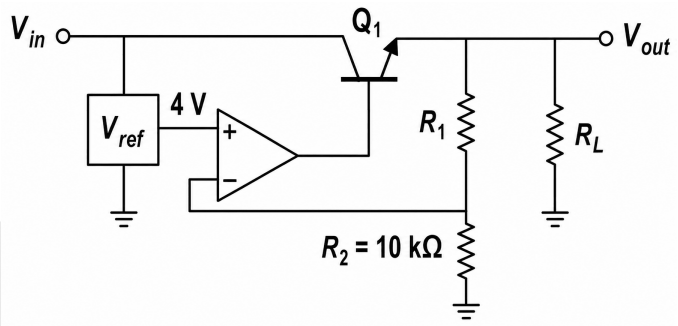


Fig. Q4.15.2

Q4.15.3 NAT 2015 2M

In the AC equivalent circuit shown, the two BJTs are biased in active region and have identical parameters with $\beta \gg 1$. The open circuit small signal voltage gain is approximately _____.

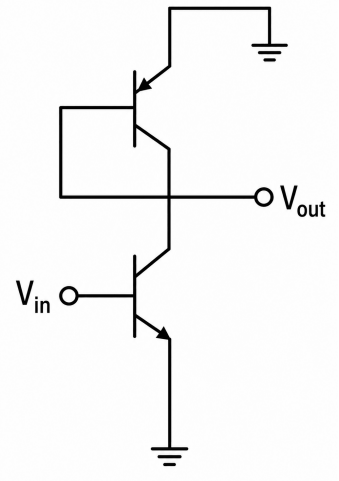


Fig. Q4.15.3

Q4.15.4 NAT 2015 1M

In the circuit shown in the figure, the BJT has a current gain (β) of 50. For an emitter-base voltage $V_{EB} = 600$ mV, the emitter-collector voltage V_{EC} (in Volts) is _____.

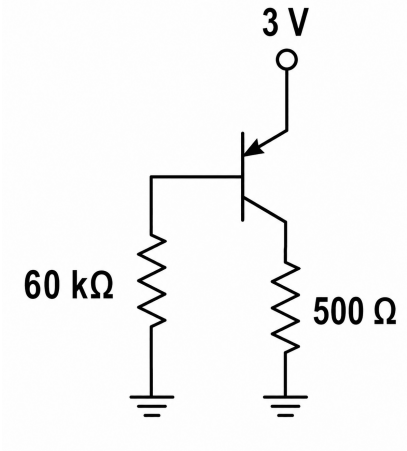


Fig. Q4.15.4

Q4.15.5 NAT 2015 2M

An npn BJT having reverse saturation current $I_S = 10^{-15}$ A is biased in the forward active region with $V_{BE} = 700$ mV. The thermal voltage (V_T) is 25 mV and the current gain (β) may vary from 50 to 150 due to manufacturing variations. The maximum emitter current (in μA) is _____.

Q4.14.1 NAT 2014 2M

A BJT is biased in forward active mode. Assume $V_{BE} = 0.7$ V, $kT/q = 25$ mV and reverse saturation current $I_S = 10^{-13}$A. The transconductance of the BJT (in mA/V) is _____.

Q4.14.2 NAT 2014 2M

In the voltage regulator circuit shown in the figure, the op-amp is ideal. The BJT has $V_{BE} = 0.7$ V and $\beta = 100$, and the zener voltage is 4.7 V. For a regulated output of 9 V, the value of R (in Ω) is _____.

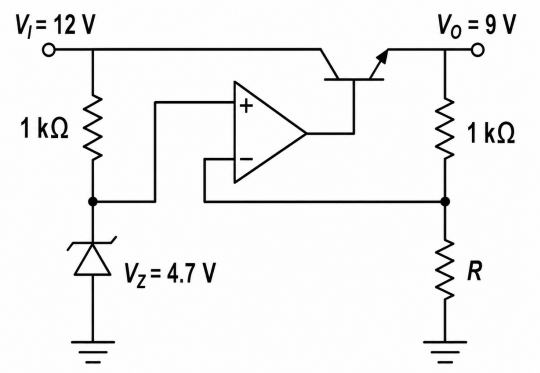


Fig. Q4.14.2

Q4.14.3 NAT 2014 2M

For the amplifier shown in the figure, the BJT parameters are $V_{BE} = 0.7$ V, $\beta = 200$, and thermal voltage $V_T = 25$ mV. The voltage gain (v_o/v_i) of the amplifier is _____.

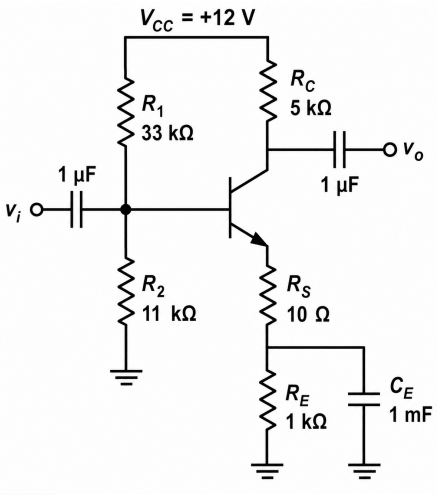


Fig. Q4.14.3

Q4.14.4 MCQ 2014 1M

The feedback topology in the amplifier circuit (the base bias circuit is not shown for simplicity) in the figure is

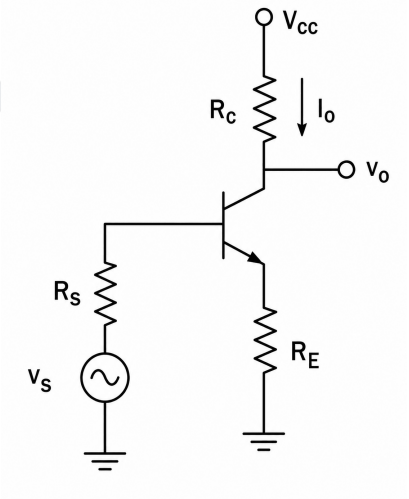


Fig. Q4.14.4

A. Voltage shunt feedback **B.** Current series feedback
C. Current shunt feedback **D.** Voltage series feedback

Q4.14.5 MCQ 2014 1M

In the differential amplifier shown in the figure, the magnitudes of the common-mode and differential-mode gains are A_{cm} and A_d, respectively. If the resistance R_E is increased, then

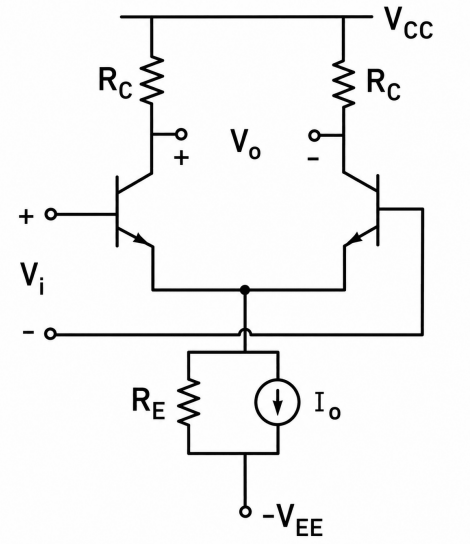


Fig. Q4.14.5

A. A_{cm} increases

B. CMRR increases

C. A_d increases

D. CMRR decreases

Q4.14.6 NAT 2014 1M

In the circuit shown, the PNP transistor has $|V_{BE}| = 0.7$ V and $\beta = 50$. Assume that $R_B = 100$ kΩ. For V_0 to be 5 V, the value of R_C (in kΩ) is _____.

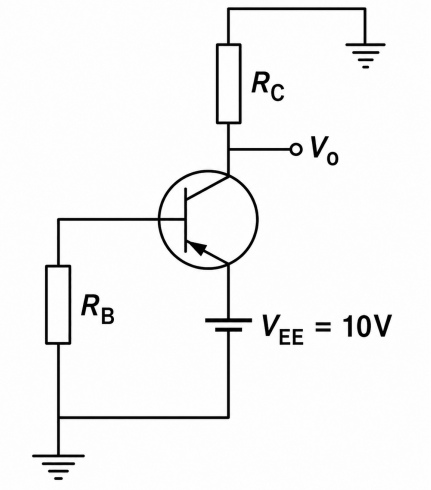


Fig. Q4.14.6

Q4.14.7 MCQ 2014 2M

In the circuit shown, the silicon BJT has $\beta = 50$. Assume $V_{BE} = 0.7$ V and $V_{CE(sat)} = 0.2$ V. Which one of the following statements is correct?

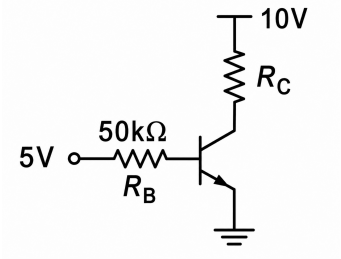


Fig. Q4.14.7

A. For $R_C = 1$ kΩ, the BJT operates in the saturation region

B. For $R_C = 3$ kΩ, the BJT operates in the saturation region

C. For $R_C = 20$ kΩ, the BJT operates in the cut-off region

D. For $R_C = 20$ kΩ, the BJT operates in the linear region

Q4.14.8 MCQ 2014 2M

Consider the common-collector amplifier in the figure (bias circuitry ensures that the transistor operates in forward active region, but has been omitted for simplicity). Let I_C be the collector current, V_{BE} be the base-emitter voltage and V_T be the thermal voltage. Also, g_m and r_o are the small-signal transconductance and output resistance of the transistor, respectively. Which one of the following conditions ensures a nearly constant small signal voltage gain for a wide range of values of R_E?

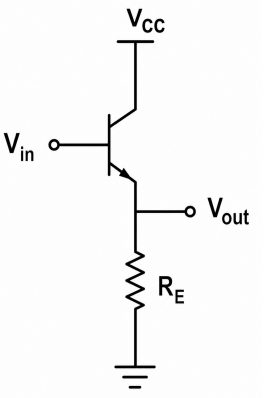


Fig. Q4.14.8

A. $g_m R_E \ll 1$ **B.** $I_C R_E \gg V_T$
C. $g_m r_o \gg 1$ **D.** $V_{BE} \gg V_T$

Q4.14.9 NAT 2014 2M

For the common collector amplifier shown in the figure, the BJT has high β, negligible $V_{CE(sat)}$, and $V_{BE} = 0.7$ V. The maximum undistorted peak-to-peak output voltage v_o (in Volts) is _____.

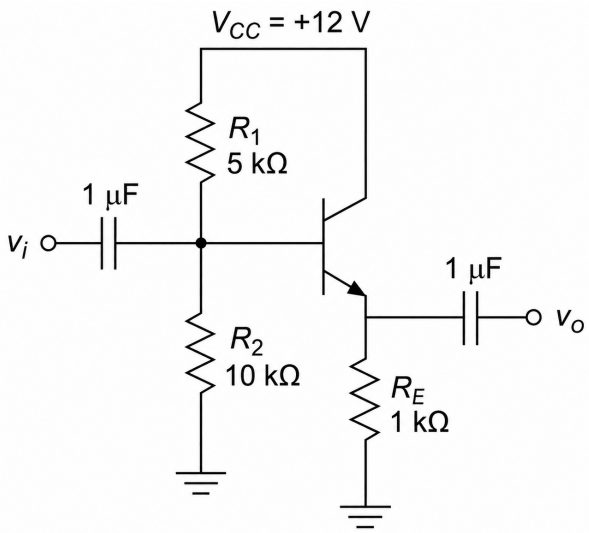


Fig. Q4.14.9

Q4.13.1 MCQ 2013 2M

In the circuit shown below, the silicon npn transistor Q has a very high value of β. The required value of R_2 in kΩ to produce $I_C = 1$ mA is

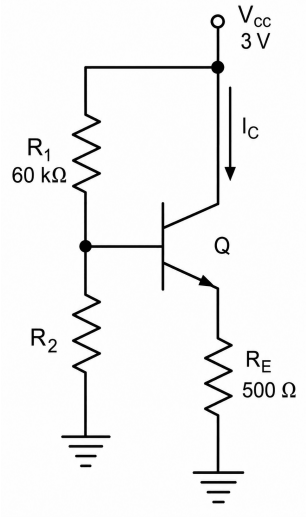


Fig. Q4.13.1

A. 20 **B.** 30 **C.** 40 **D.** 50

Q4.13.2 MCQ 2013 1M

In the circuit shown below what is the output voltage (V_{out}) if a silicon transistor Q and an ideal op-amp are used?

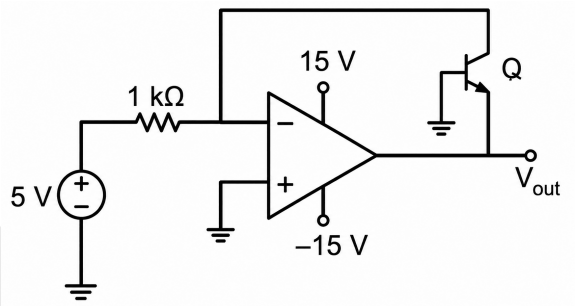


Fig. Q4.13.2

A. −15 V **B.** −0.7 V **C.** +0.7 V **D.** +15 V

Q4.12.1 MCQ 2012 1M

The current i_b through the base of a silicon *npn* transistor is $1 + 0.1\cos(10000\pi t)$ mA. At 300 K, the r_π in the small signal model of the transistor is

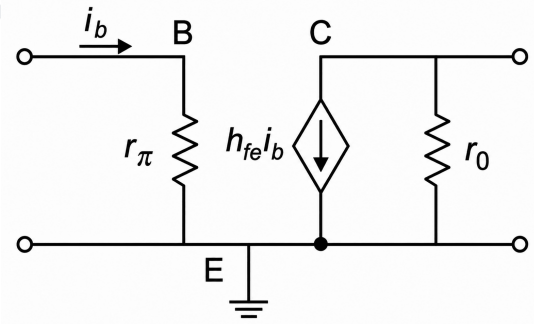


Fig. Q4.12.1

A. 250 Ω **B.** 27.5 Ω **C.** 25 Ω **D.** 22.5 Ω

Q4.12.2 MCQ 2012 2M

The voltage gain A_v of the circuit shown below is

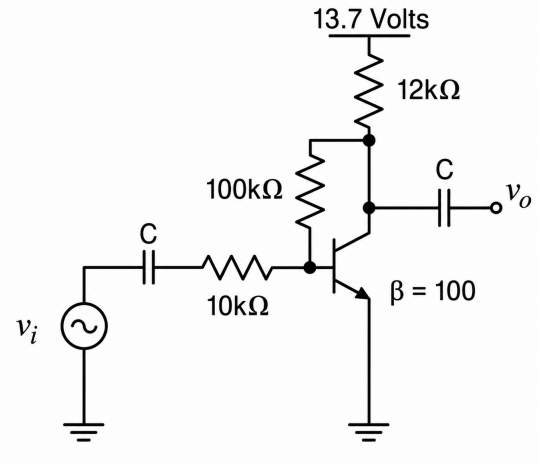


Fig. Q4.12.2

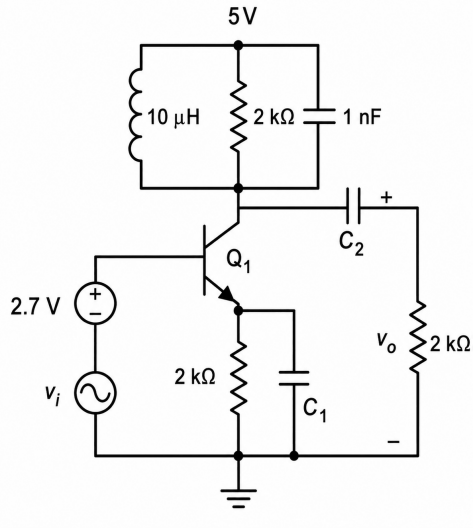


Fig. Q4.11.2

A. maximum **B.** minimum **C.** unity **D.** zero

A. $|A_v| \simeq 200$ **B.** $|A_v| \simeq 100$

C. $|A_v| \simeq 20$ **D.** $|A_v| \simeq 10$

Q4.11.3 MCQ 2011 2M

For the BJT Q_1 in the circuit shown below, $\beta = \infty$, $V_{BEon} = 0.7$ V, $V_{CEsat} = 0.7$ V. The switch is initially closed. At time $t = 0$, the switch is opened. The time t at which Q_1 leaves the active region is

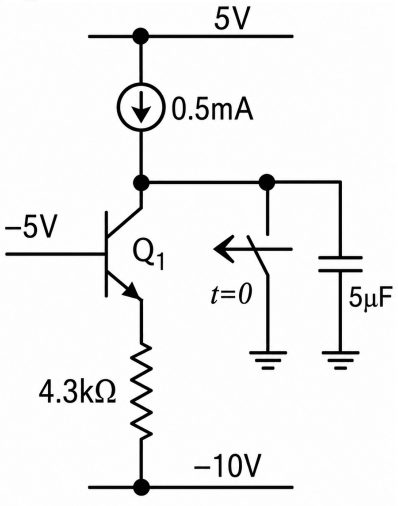


Fig. Q4.11.3

A. 10 ms **B.** 25 ms **C.** 50 ms **D.** 100 ms

Q4.11.1 MCQ 2011 2M

For a BJT, the common-base current gain $\alpha = 0.98$ and the collector base junction reverse bias saturation current $I_{CO} = 0.6\mu$A. This BJT is connected in the common emitter mode and operated in the active region with a base drive current $I_B = 20\mu$A. The collector current I_C for this mode of operation is

A. 0.98 mA **B.** 0.99 mA

C. 1.0 mA **D.** 1.01 mA

Q4.11.2 MCQ 2011 1M

In the circuit shown below, capacitors C_1 and C_2 are very large and are shorts at the input frequency. v_i is a small signal input. The gain magnitude $|v_o/v_i|$ at 10 Mrad/s is

Q4.10.1 MCQ 2010 1M

In the silicon BJT circuit shown below, assume that the emitter area of transistor Q1 is half that of transistor Q2.

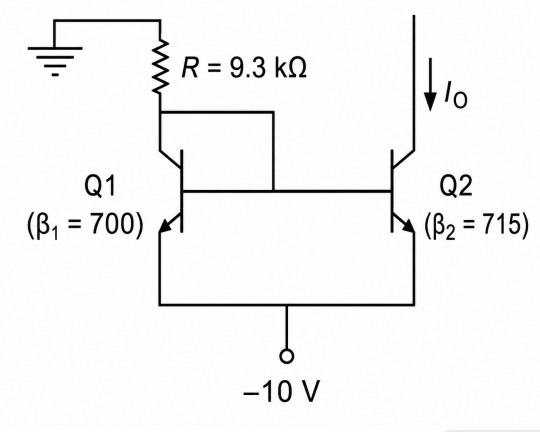


Fig. Q4.10.1

The value of current I_O is approximately

A. 0.5 mA **B.** 2 mA **C.** 9.3 mA **D.** 15 mA

Q4.10.2 MCQ 2010 1M

The amplifier circuit shown below uses a silicon transistor. The capacitors C_C and C_E can be assumed to be short at signal frequency and the effect of output resistance r_o can be ignored. If C_E is disconnected from the circuit, which one of the following statements is TRUE?

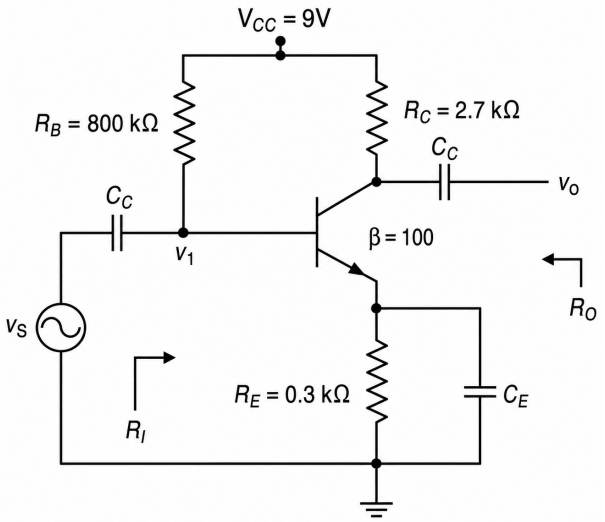


Fig. Q4.10.2

A. The input resistance R_i increases and the magnitude of voltage gain A_v decreases

B. The input resistance R_i decreases and the magnitude of voltage gain A_v increases

C. Both input resistance R_i and the magnitude of voltage gain A_v decrease

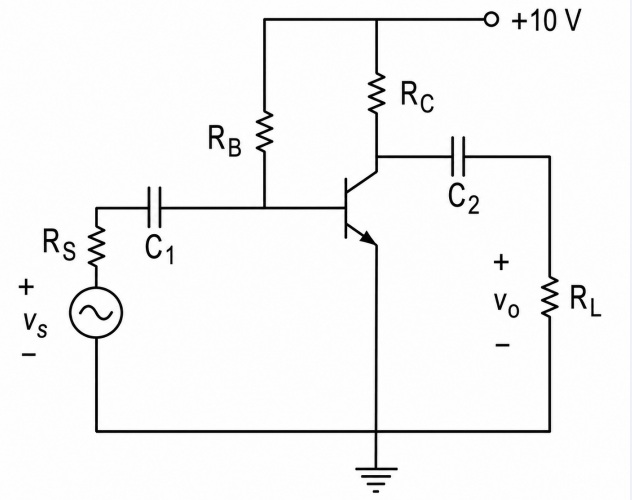
D. Both input resistance R_i and the magnitude of voltage gain A_v increase

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions 4.10.3 and 4.10.4:

Consider the common emitter amplifier shown below with the following circuit parameters: β = 100, g_m = 0.3861 A/V, r_o = ∞, r_π = 259 Ω, R_S = 1 kΩ, R_B = 93 kΩ, R_C = 250 Ω, R_L = 1 kΩ, C_1 = ∞ and C_2 = 4.7 μF.



Q4.10.3 MCQ 2010 2M

The resistance seen by the source v_s is

A. 258 Ω **B.** 1258 Ω **C.** 93 kΩ **D.** ∞

Q4.10.4 MCQ 2010 2M

The lower cut-off frequency due to C_2 is

A. 33.9 Hz **B.** 27.1 Hz **C.** 13.6 Hz **D.** 16.9 Hz

Q4.09.1 MCQ 2009 2M

In the circuit shown below, the op-amp is ideal, the transistor has V_{BE} = 0.6 V and β = 150. Decide whether the feedback in the circuit is positive or negative and determine the voltage V at the output of the op-amp.

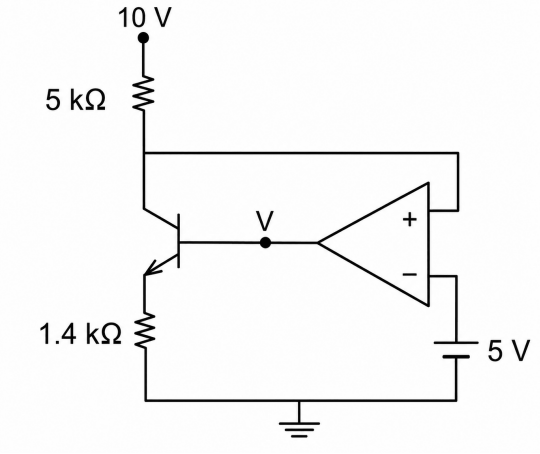


Fig. Q4.09.1

A. Positive feedback, $V = 10$ V

B. Positive feedback, $V = 0$ V

C. Negative feedback, $V = 5$ V

D. Negative feedback, $V = 2$ V

Q4.09.2 MCQ 2009 2M

A small signal source $v_i(t) = A\cos 20t + B\sin 10^6 t$ is applied to a transistor amplifier as shown below. The transistor has $\beta = 150$ and $h_{ie} = 3$ kΩ. Which expression best approximates $v_o(t)$?

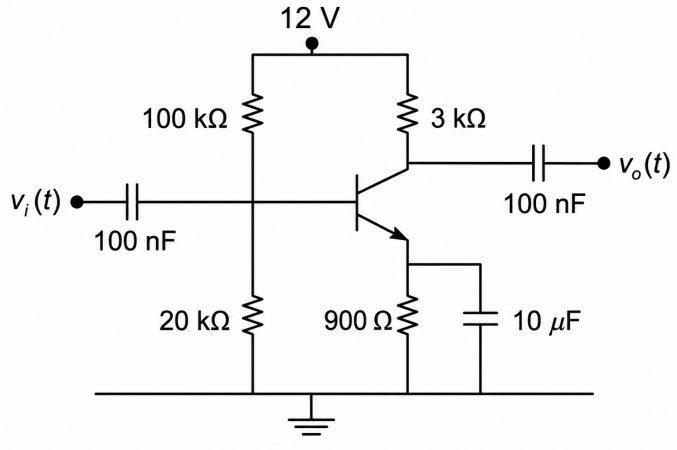


Fig. Q4.09.2

A. $v_o(t) = -1500(A\cos 20t + B\sin 10^6 t)$

B. $v_o(t) = -150(A\cos 20t + B\sin 10^6 t)$

C. $v_o(t) = -1500B\sin 10^6 t$

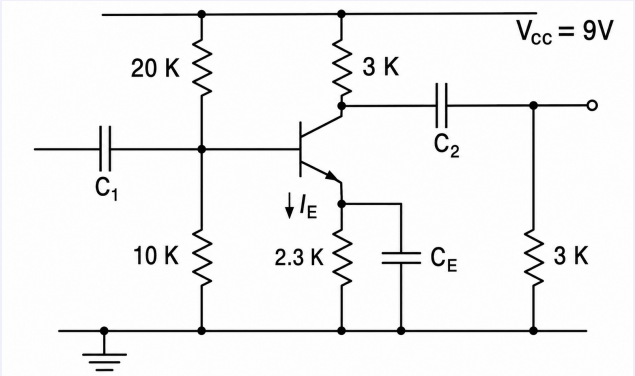
D. $v_o(t) = -150B\sin 10^6 t$

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions 4.08.1 and 4.08.2:

In the following transistor circuit, $V_{BE} = 0.7$ V, $r_e = 25\text{ mV}/I_E$, and β and all the capacitances are very large.



Q4.08.1 MCQ 2008 2M

The value of DC current I_E is

A. 1 mA **B.** 2 mA **C.** 5 mA **D.** 10 mA

Q4.08.2 MCQ 2008 2M

The mid-band voltage gain of the amplifier is approximately

A. -180 **B.** -120 **C.** -90 **D.** -60

Q4.07.1 MCQ 2007 2M

For the BJT circuit shown, assume that the β of the transistor is very large and $V_{BE} = 0.7$ V. The mode of operation of the BJT is

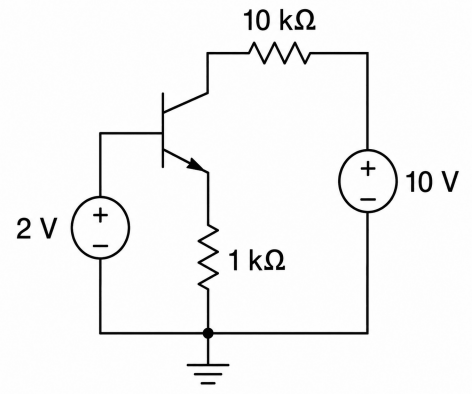


Fig. Q4.07.1

A. cut-off **B.** saturation

C. normal active **D.** reverse active

Common Data Questions

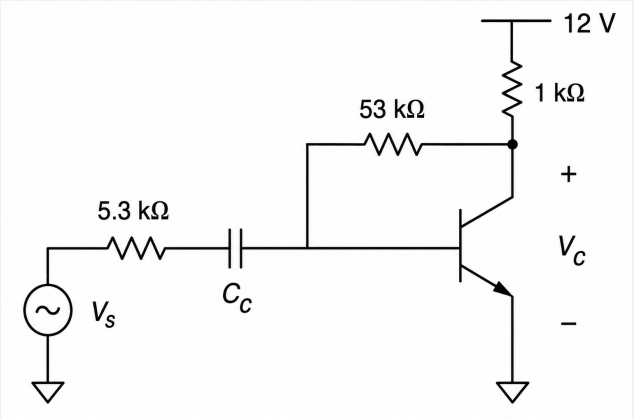
The next 3 questions share the data below.

Common Data for Questions 4.06.1, 4.06.2 and 4.06.3:

In the transistor amplifier circuit shown in the figure below, the transistor has the following parameters:

$$\beta_{DC} = 60, V_{BE} = 0.7 \text{ V}, h_{ie} \to \infty, h_{fe} \to \infty$$

The capacitance C_c can be assumed to be infinite.



In the figure above, the ground has been shown by the symbol ∇

Q4.06.1 MCQ 2006 2M

Under the DC conditions, the collector-to-emitter voltage drop is:

A. 4.8 Volts **B.** 5.3 Volts

C. 6.0 Volts **D.** 6.6 Volts

Q4.06.2 MCQ 2006 2M

If β_{DC} is increased by 10%, the collector-to-emitter voltage drop

A. increases by less than or equal to 10%

B. decreases by less than or equal to 10%

C. increases by more than 10%

D. decreases by more than 10%

Q4.06.3 MCQ 2006 2M

The small-signal gain of the amplifier v_c/v_s is:

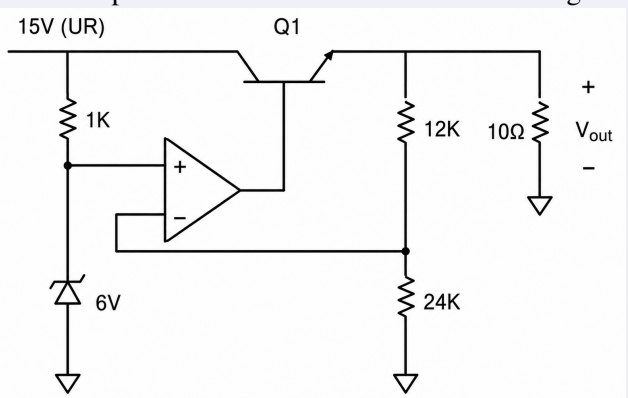
A. -10 **B.** -5.3 **C.** 5.3 **D.** 10

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions 4.06.4 & 4.06.5:

A regulated power supply, shown in figure below, has an unregulated input (UR) of 15 Volts and generates a regulated output V_{out}. Use the component values shown in the figure.



Q4.06.4 MCQ 2006 2M

The power dissipation across the transistor Q1 shown in the figure is:

A. 4.8 Watts **B.** 5.0 Watts

C. 5.4 Watts **D.** 6.0 Watts

Q4.06.5 MCQ 2006 2M

If the unregulated voltage increases by 20%, the power dissipation across the transistor Q1

A. increases by 20% **B.** increases by 50%

C. remains unchanged **D.** decreases by 20%

Q4.05.1 MCQ 2005 1M

The cascode amplifier is a multistage configuration of

A. CC-CB **B.** CE-CB **C.** CB-CC **D.** CE-CC

Q4.05.2 MCQ 2005 2M

For an npn transistor connected as shown in figure, V_{BE} = 0.7 Volts. Given that reverse saturation current of the junction at room temperature 300°K is 10^{-13} A, the emitter current is

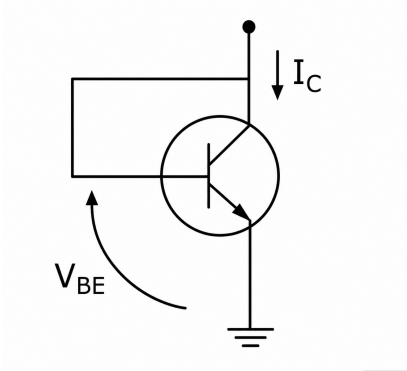


Fig. Q4.05.2

A. 30 mA **B.** 39 mA **C.** 49 mA **D.** 20 mA

Q4.05.3 MCQ 2005 2M

The circuit using a BJT with β = 50 and V_{BE} = 0.7 V is shown in figure. The base current I_B and collector voltage V_C are respectively

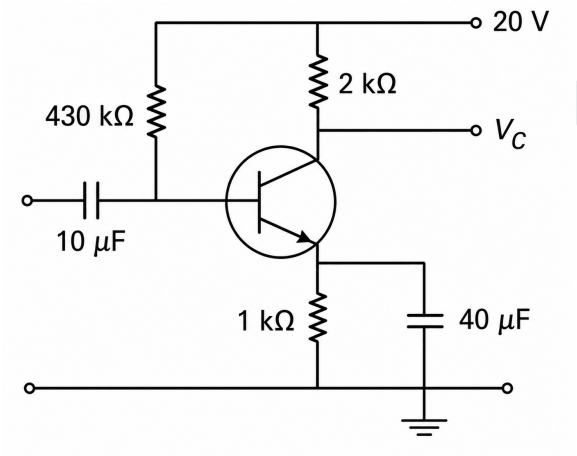


Fig. Q4.05.3

A. 43 μA and 11.4 Volts **B.** 40 μA and 16 Volts

C. 45 μA and 11 Volts **D.** 50 μA and 10 Volts

Q4.04.1 MCQ 2004 1M

Assuming V_{CEsat} = 0.2 V and β = 50, the minimum base current (I_B) required to drive the transistor in the figure to saturation is

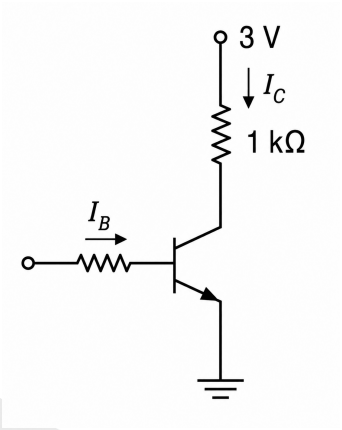


Fig. Q4.04.1

A. 56 μA **B.** 140 mA **C.** 60 μA **D.** 3 mA

Q4.04.2 MCQ 2004 2M

Assuming that the β of the transistor is extremely large and V_{BE} = 0.7 V, I_C and V_{CE} in the circuit shown in Figure, are

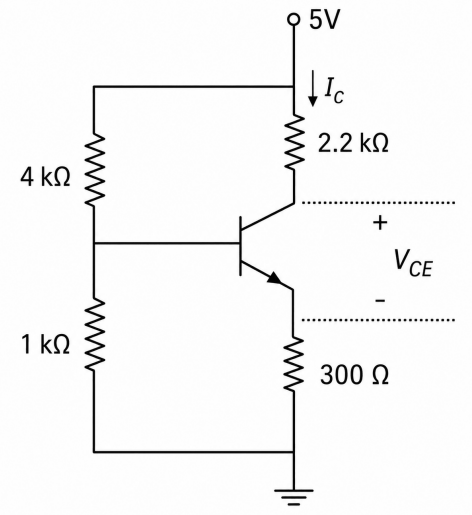


Fig. Q4.04.2

A. I_C = 1 mA, V_{CE} = 4.7 V

B. I_C = 0.5 mA, V_{CE} = 3.75 V

C. I_C = 1 mA, V_{CE} = 2.5 V

D. $I_C = 0.5$ mA, $V_{CE} = 3.9$ V

Q4.04.3 MCQ 2004 2M

A bipolar transistor is operating in the active region with a collector current of 1mA. Assuming that the β of the transistor is 100 and the thermal voltage (V_T) is 25 mV, the transconductance (g_m) and the input resistance (r_π) of the transistor in the common emitter configuration, are

A. $g_m = 25mA/V$ and $r_\pi = 15.625k\Omega$

B. $g_m = 40mA/V$ and $r_\pi = 4.0k\Omega$

C. $g_m = 25mA/V$ and $r_\pi = 2.5k\Omega$

D. $g_m = 40mA/V$ and $r_\pi = 2.5k\Omega$

Q4.03.1 MCQ 2003 2M

In the amplifier circuit shown in figure, the values of R_1 and R_2 are such that the transistor is operating at $V_{CE} = 3$ V and $I_C = 1.5$ mA when its β is 150. For a transistor with β of 200, the operating point (V_{CE}, I_C) is

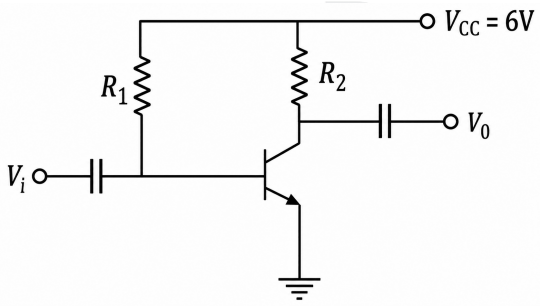


Fig. Q4.03.1

A. (2 V, 2 mA) **B.** (3 V, 2 mA)

C. (4 V, 2 mA) **D.** (4 V, 1 mA)

Q4.03.2 MCQ 2003 2M

The output voltage of the regulated power supply shown in figure is

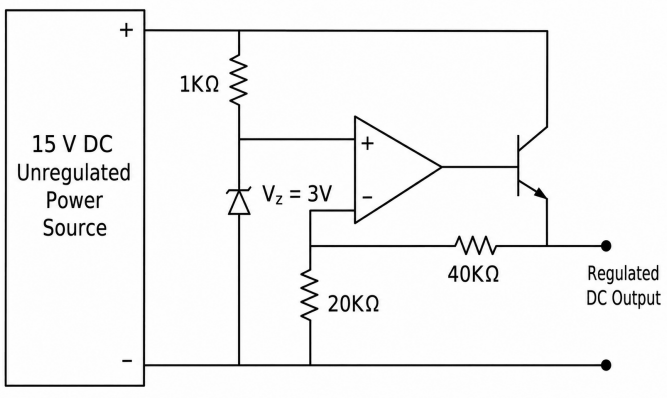


Fig. Q4.03.2

A. 3 V **B.** 6 V **C.** 9 V **D.** 12 V

Q4.02.1 MCQ 2002 1M

Three identical RC-coupled transistor amplifiers are cascaded. If each of the amplifiers has a frequency response as shown in the figure, the overall frequency response is as given in

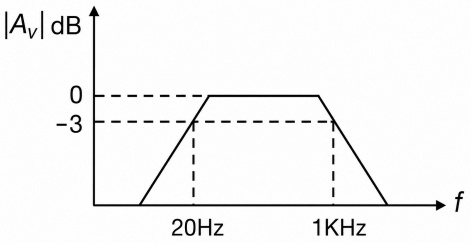
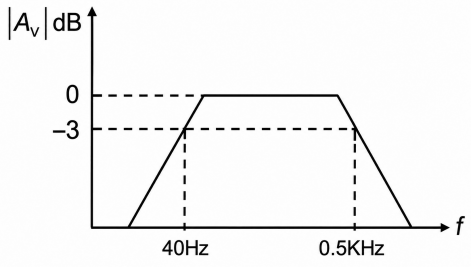
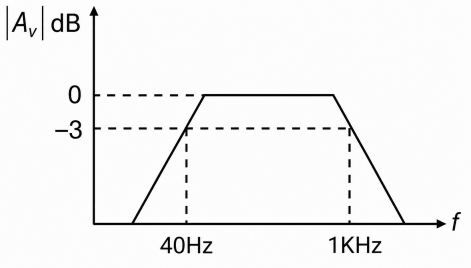


Fig. Q4.02.1

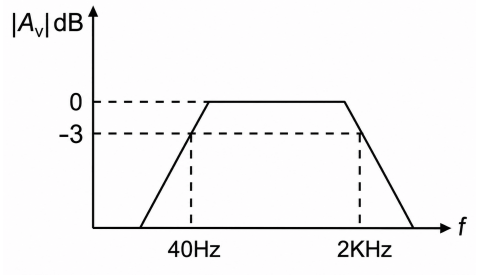
A.



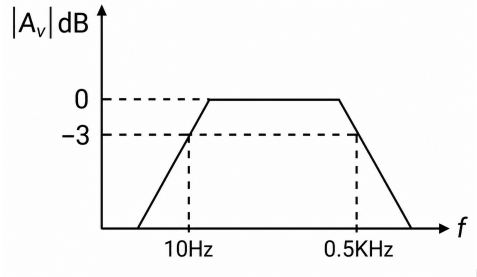
B.



C.



D.



Q4.02.2 MCQ | 2002 | 1M

If the transistor in the figure is in saturation, then

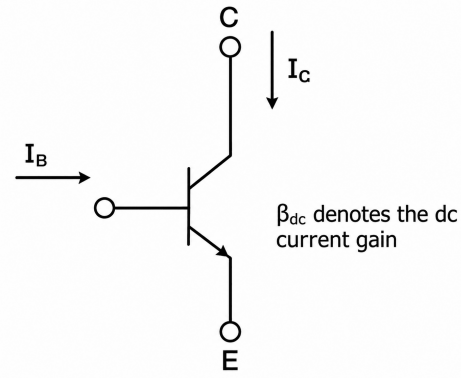


Fig. Q4.02.2

A. I_C is always equal to $\beta_{dc} I_B$

B. I_C is always equal to $-\beta_{dc} I_B$

C. I_C is greater than or equal to $\beta_{dc} I_B$

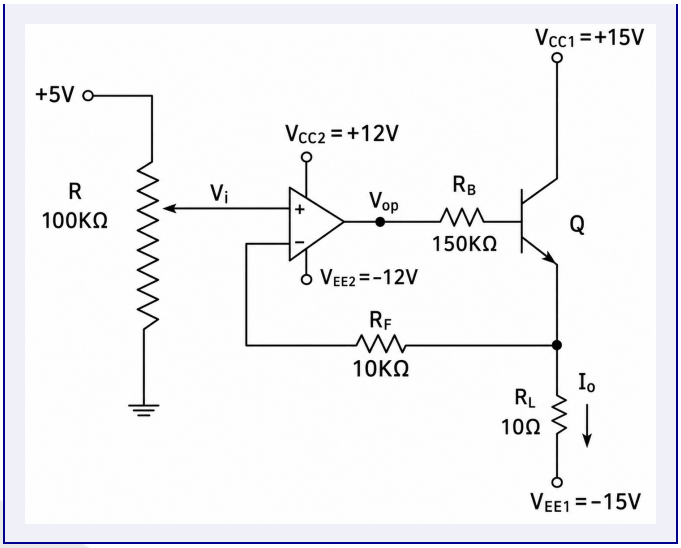
D. I_C is less than or equal to $\beta_{dc} I_B$

Common Data Questions

The next 3 questions share the data below.

Common Data for Questions 4.02.3, 4.02.4 and 4.02.5:

Consider the circuit in below figure. The op-amp used is ideal.



Q4.02.3 NAT | 2002 | 1M

In which mode is the BJT operating (i.e. active, saturation or cutoff)? Justify your answer.

Q4.02.4 NAT | 2002 | 2M

Obtain an expression relating the output current I_o and the input voltage V_i.

Q4.02.5 NAT | 2002 | 2M

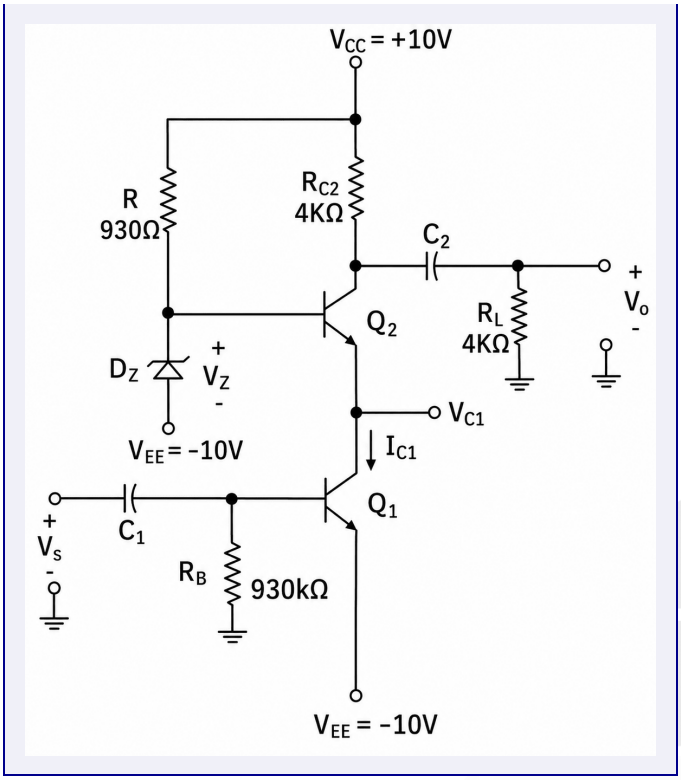
Determine I_o and V_{OP} if $V_i = 2$ Volt (V_{OP}:output of opamp) (Assume $\beta = 99$ and $V_{BE} = 0.7$ V)

Common Data Questions

The next 3 questions share the data below.

Common Data for Questions 4.02.6, 4.02.7 and 4.02.8:

The below figure shows a 2-stage amplifier. The transistors Q1 and Q2 are identical with current gain $\beta = 100$; further $\beta_{dc} = \beta_{ac} = \beta$. The Zener diode D_z has a break down voltage $V_z = 10.7$ volt. Assume that D_z is in breakdown region and its dynamic resistance r_z is zero. The capacitors C1 and C2 are large and provide negligible impedance at signal frequencies.



Q4.02.6 NAT | 2002 | 1M

Identify the configuration in each of the amplifier stages (i.e., whether CE, CC, CB etc.)

Q4.02.7 NAT | 2002 | 2M

Determine the quiescent quantities I_{C1} and V_{C1}.

Q4.02.8 NAT | 2002 | 2M

Derive an expression for the voltage gain $A_V \triangleq \dfrac{V_o}{V_s}$ and determine its value.
(Assume $V_{BE} = 0.7\text{V}, r_0 = \infty$ and Thermal voltage $V_T = 25$ mV)

Q4.01.1 MCQ | 2001 | 1M

The current gain of a BJT is

A. $g_m r_0$ **B.** $\dfrac{g_m}{r_o}$ **C.** $g_m r_\pi$ **D.** $\dfrac{g_m}{r_\pi}$

Q4.01.2 MCQ | 2001 | 2M

An npn BJT has $g_m = 38\text{m A/V}$, $C_\mu = 10^{-14}$ F, $C_\pi = 4\times10^{-13}$F, and DC current gain $\beta_0 = 90$. for this transistor f_T and f_β are

A. $f_T = 1.64 \times 10^8 Hz$ and $f_\beta = 1.47 \times 10^{10} Hz$

B. $f_T = 1.47 \times 10^{10} Hz$ and $f_\beta = 1.64 \times 10^8 Hz$

C. $f_T = 1.33 \times 10^{12} Hz$ and $f_\beta = 1.47 \times 10^{10} Hz$

D. $f_T = 1.47 \times 10^{10} Hz$ and $f_\beta = 1.33 \times 10^{12} Hz$

Q4.01.3 MCQ | 2001 | 2M

The transistor shunt regulator shown in figure has a regulated output voltage of 10 V, when the input varies from 20 V to 30 V. The relevant parameters for the Zener diode and the transistor are: $V_z = 9.5$, $V_{BE} = 0.3\text{V}$, $\beta = 99$. Neglect the current through R_B. Then the maximum power dissipated in the Zener diode (P_Z) and the transistor (P_T) are

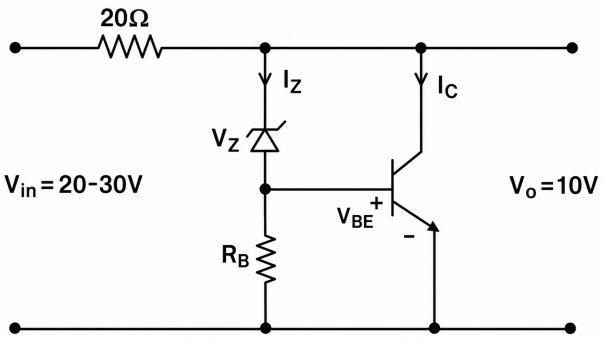


Fig. Q4.01.3

A. $P_Z = 75\text{mW}, P_T = 7.9\text{W}$

B. $P_Z = 85\text{mW}, P_T = 8.9\text{W}$

C. $P_Z = 95\text{mW}, P_T = 9.9\text{W}$

D. $P_Z = 115\text{mW}, P_T = 11.9\text{W}$

Q4.01.4 MCQ | 2001 | 2M

The oscillator circuit shown in figure is

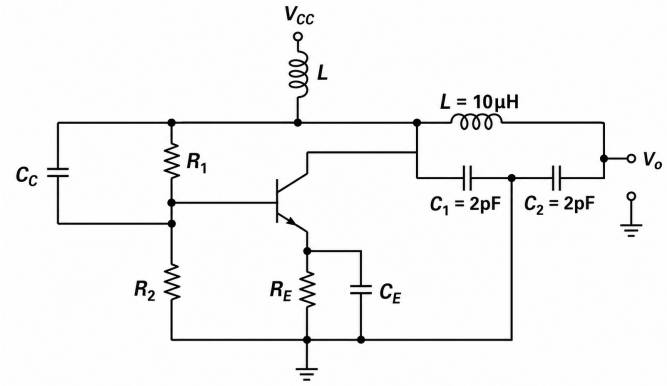


Fig. Q4.01.4

A. Hartley oscillator with $f_{\text{oscillation}} = 79.6MHz$

B. Colpitts oscillator with $f_{oscillation} = 79.6MHz$

C. Hartley oscillator with $f_{oscillation} = 159.2MHz$

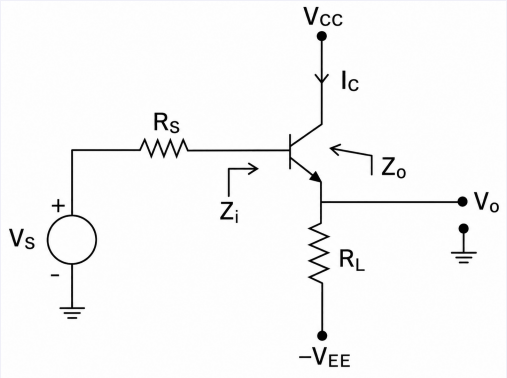
D. Colpitts oscillator with $f_{oscillation} = 159.2MHz$

Common Data Questions

The next 4 questions share the data below.

Common Data for Questions 4.01.5, 4.01.6, 4.01.7 and 4.01.8:

An emitter-follower amplifier is shown in figure. Z_i is the impedance looking into the base of the transistor and Z_0 is the impedance looking into the emitter of the transistor.



Q4.01.5 NAT 2001 1M

Draw the small signal equivalent circuit of the amplifier.

Q4.01.6 NAT 2001 1M

Obtain an expression for Z_i.

Q4.01.7 NAT 2001 1M

Obtain an expression for Z_0.

Q4.01.8 NAT 2001 2M

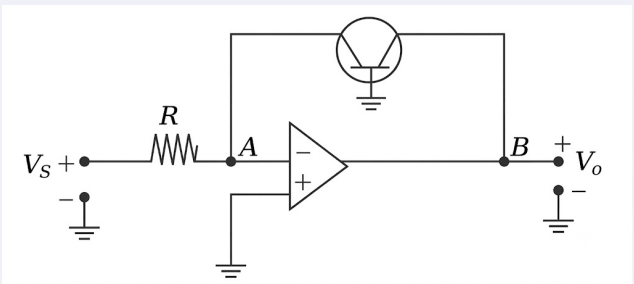
Determine Z_i and Z_0 if a capacitor C_L is connected across R_L.

Common Data Questions

The next 3 questions share the data below.

Common Data for Questions 4.01.9, 4.01.10 and 4.01.11:

Assume that the OP-AMP in figure is ideal. (npn transistor with collector connected to point A and emitter to B)



Q4.01.9 NAT 2001 1M

obtain an expression for v_o in terms of v_s, R, and the reverse saturation current I_s of the transistor.

Q4.01.10 NAT 2001 2M

If R = 1Ω, I_s = 1 pA and the thermal voltage V_T = 25 mV, then what is the value of the output voltage v_o for an input voltage v_s = 1V?

Q4.01.11 NAT 2001 2M

Suppose that the transistor in the feedback path is replaced by a p-n junction diode with a reverse saturation current of I_s. The p-side of the diode is connected to node A and the n-side to node B. Then what is the expression for v_o in terms of v_s, R and I_s?

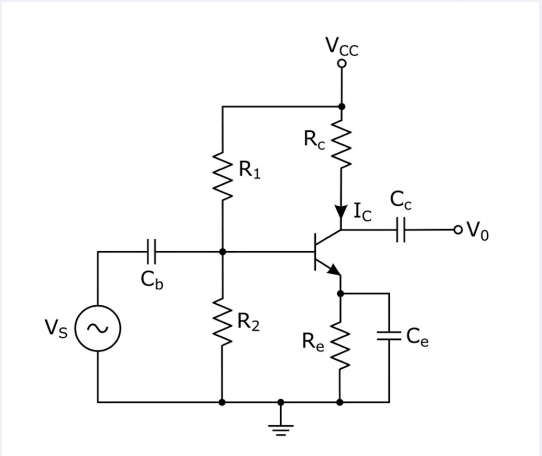
Common Data Questions

The next 3 questions share the data below.

Answer Questions Q4.00.1, Q4.00.2 and Q4.00.3 using the following data. These 3 questions together carry 5 Marks

For the amplifier of given figure,

$I_C = 1.3\,\text{mA}$, $R_C = 2\,\text{k}\Omega$, $R_E = 500\,\Omega$,
$V_T = 26\,\text{mV}$, $\beta = 100$, $V_{CC} = 15\,\text{V}$,
$V_s = 0.01\sin(\omega t)\,\text{V}$ and $C_b = C_c = 10\,\mu\text{F}$.



Q4.00.1 NAT 2000

What is the small-signal voltage gain, $A_v = \dfrac{V_o}{V_s}$?

Q4.00.2 NAT 2000

What is the approximate A_v, if C_e is removed?

Q4.00.3 Descriptive 2000

What will V_o be if C_b is short circuited?

Q4.00.4 MCQ 2000 2M

In the circuit of Figure shown below, assume that the transistor is in active region. It has a large β and its base-emitter voltage is 0.7 V. The value of I_C is

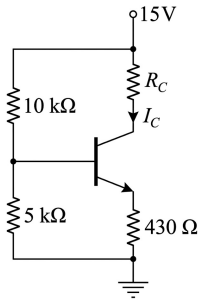


Fig. Q4.00.4

A. Indeterminate since R_C is not given **B.** 1 mA
C. 5 mA **D.** 10 mA

Q4.00.5 MCQ 2000 1M

The current gain of a bipolar transistor drops at high frequencies because of

A. transistor capacitances

B. high current effects in the base

C. parasitic inductive elements

D. the Early effect

Q4.99.1 NAT 1999 5M

A bipolar junction transistor amplifier circuit is shown in the figure. Assume that the current source I_{bias} is ideal, and the transistor has very large β, $r_b = 0$, $r_0 \to \infty$ and $V_T = 26\,\text{mV}$. In the circuit: $I_{\text{bias}} = 0.5\,\text{mA}$, $R_B = 100\,\text{k}\Omega$, $R_E = 50\,\Omega$, $R_L = 1\,\text{k}\Omega$, $C_1 \to \infty$ and $C_2 \to \infty$

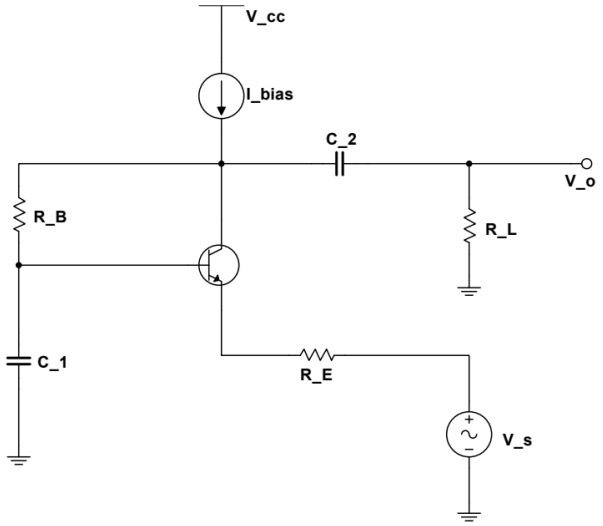


Fig. Q4.99.1

Determine the ac small-signal mid-band voltage gain $\left(\dfrac{V_o}{V_s}\right)$, input resistance (R_i), and output resistance (R_o) of the circuit.

Q4.99.2 MCQ 1999 1M

Negative feedback in an amplifier

A. reduces gain.

B. increases frequency and phase distortions.

C. reduces bandwidth.

D. increases noise.

Q4.99.3 MCQ 1999 1M

In the cascode amplifier shown in Figure below, if the common-emitter stage (Q_1) has a transconductance g_{m1}, and the common-base stage (Q_2) has a transconductance g_{m2}, then the overall transconductance $g = \left(\dfrac{i_o}{v_i}\right)$ of the cascode amplifier is

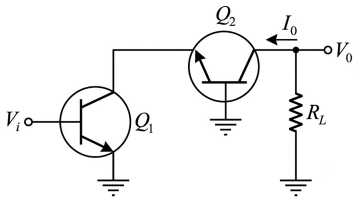


Fig. Q4.99.3

A. g_{m1} **B.** g_{m2} **C.** $\dfrac{g_{m1}}{2}$ **D.** $\dfrac{g_{m2}}{2}$

Q4.99.4 MCQ 1999 2M

An amplifier has an open-loop gain of 100, an input impedance of 1 kΩ and an output impedance of 100 Ω. A feedback network with a feedback factor of 0.99 is connected to the amplifier in a voltage series feedback mode. The new input and output impedance respectively are

A. 10 Ω and 1 Ω
B. 10 Ω and 10 Ω
C. 100 kΩ and 1 Ω
D. 100 kΩ and 1 kΩ

Q4.99.5 MCQ 1999 2M

An npn transistor (with $C_\mu = 0.3$ pF) has a unity gain cut-off frequency f_T of 400 MHz at a DC-bias current I_C = 1 mA. The value of its C_π (in pF) is approximately (V_T = 26 mV).

A. 15 **B.** 30 **C.** 50 **D.** 96

Q4.99.6 MCQ 1999 2M

An amplifier is assumed to have a single-pole high frequency transfer function. The rise time of its output response to a step function input is 35 ns. The upper 3 dB frequency (in MHz) for the amplifier to a sinusoidal input is approximately at

A. 4.55 **B.** 10 **C.** 20 **D.** 28.6

Q4.98.1 NAT 1998 5M

In the circuit of fig., determine the resistance R_o seen by the output terminals. Ignore the effects of R_1 and R_2. For the transistor, $\beta = 99$, $r_\pi = 1$ kΩ.

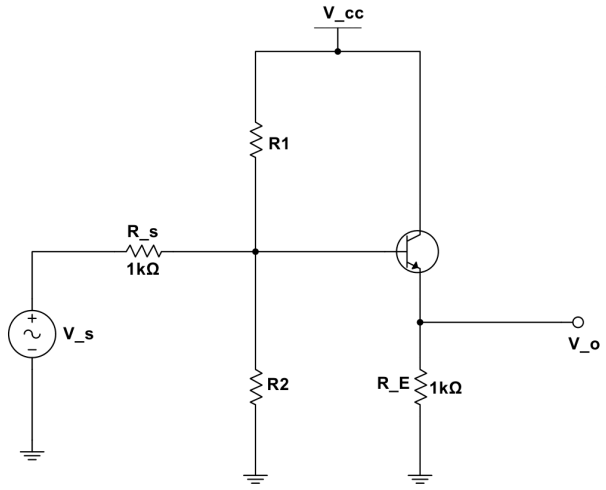


Fig. Q4.98.1

Give the answer in ohms, rounded off to two decimal places.

Q4.98.2 MCQ 1998 1M

A multistage amplifier has a low-pass response with three real poles at $s = -\omega_1$, $-\omega_2$ and $-\omega_3$. The approximate overall bandwidth B of the amplifier will be given by

A. $B = \omega_1 + \omega_2 + \omega_3$
B. $\frac{1}{B} = \frac{1}{\omega_1} + \frac{1}{\omega_2} + \frac{1}{\omega_3}$
C. $B = (\omega_1 + \omega_2 + \omega_3)^{1/3}$
D. $B = \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2}$

Q4.98.3 MCQ 1998 1M

The circuit of the figure is an example of feedback of the following type

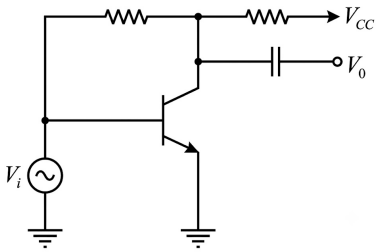


Fig. Q4.98.3

A. current series
B. current shunt
C. voltage series
D. voltage shunt

Q4.98.4 MCQ 1998 1M

In a shunt-shunt negative feedback amplifier, as compared to the basic amplifier

A. both input and output impedance decreases.

B. input impedance decreases but output impedance increases.

C. input impedance increases but output impedance decreases.

D. both input and output impedance increases.

Q4.98.5 MCQ 1998 1M

The f_T of a BJT is related to its g_m, C_π and C_μ as follows

A. $f_T = \frac{C_\pi + C_\mu}{g_m}$
B. $f_T = \frac{2\pi(C_\pi + C_\mu)}{g_m}$
C. $f_T = \frac{g_m}{C_\pi + C_\mu}$
D. $f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$

Q4.98.6 MCQ 1998 1M

From measurement of the rise time of the output pulse of an amplifier whose input is a small amplitude square wave, one can estimate the following parameter of the amplifier

A. gain-bandwidth product **B.** slew rate
C. upper 3-dB frequency **D.** lower 3-dB frequency

Q4.97.1 NAT 1997 5M

In the cascode amplifier circuit shown below, determine the values of R_1, R_2 and R_L, such that the quiescent current through the transistors is 1 mA and the collector voltages $V_{C1} = 3$ V and $V_{C2} = 6$ V. Take $V_{BE} = 0.7$ V. Transistor β to be high and base currents to be negligible.

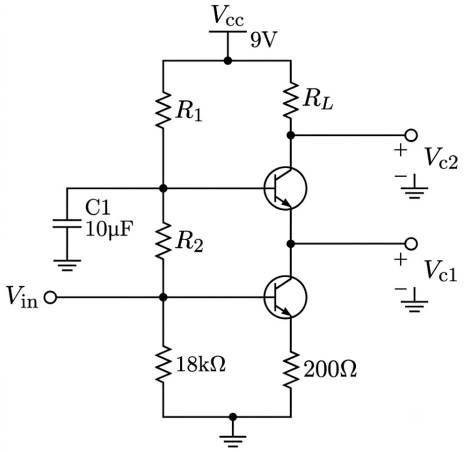


Fig. Q4.97.1

Q4.97.2 MCQ 1997 1M

In the BJT amplifier shown in the figure, the transistor is biased in the forward active region, putting a capacitor across R_E will

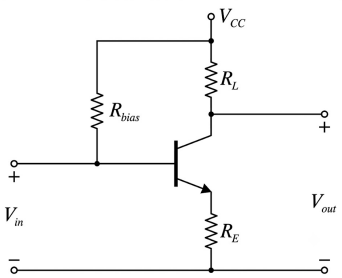


Fig. Q4.97.2

A. decrease the voltage gain and decrease the input impedance.

B. increase the voltage gain and decrease the input impedance.

C. decrease the voltage gain and increase the input impedance.

D. increase the voltage gain and increase the input impedance.

Q4.97.3 MCQ 1997 1M

A cascode amplifier stage is equivalent to

A. a common emitter stage followed by a common base stage.

B. a common base stage followed by an emitter follower.

C. an emitter follower stage followed by a common base stage.

D. a common base stage followed by a common emitter stage.

Q4.97.4 MCQ 1997 2M

Negative feedback in
1. voltage series configuration
2. current shunt configuration
(a) increases input impedance.
(b) decreases input impedance.
(c) increases closed loop gain.
(d) leads to oscillation.

A. 1-a, 2-d **B.** 1-c, 2-b **C.** 1-a, 2-b **D.** 1-b, 2-d

Q4.96.1 NAT 1996 5M

A common-emitter amplifier with an external capacitor C_C connected across the base and the collector of the transistor is shown in Fig. Consider $R_S = 10\,\text{k}\Omega$, $R_C = 10\,\text{k}\Omega$, $C_C = 25\,\text{pF}$, $g_m = 5\,\text{mA/V}$, $r_\pi = 200\,\text{k}\Omega$, $C_\pi = 1.5\,\text{pF}$, $C_\mu = 0.5\,\text{pF}$.

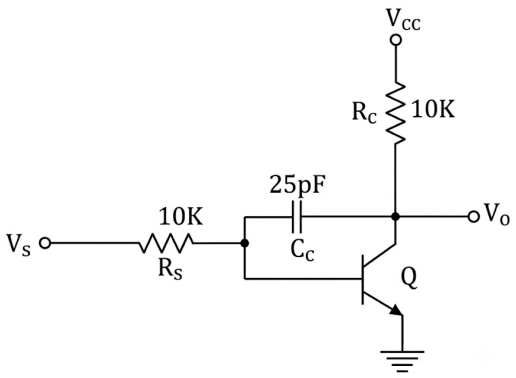


Fig. Q4.96.1

Determine the upper cutoff frequency f_H, in Hz, of the amplifier.

Q4.96.2 MCQ 1996 1M

If a transistor is operating with both of its junctions forward biased, but with the collector base forward bias greater than the emitter-base forward bias, then it is operating in the

A. forward active mode.

B. reverse saturation mode.

C. reverse active mode.

D. forward saturation mode.

Q4.96.3 MCQ 1996 2M

A Darlington stage is shown in the figure, if the transconductance of Q_1 is g_{m1} and Q_2 is g_{m2}, then the overall transconductance $g_m = \left(\frac{I_o}{V_i}\right)$ is given by

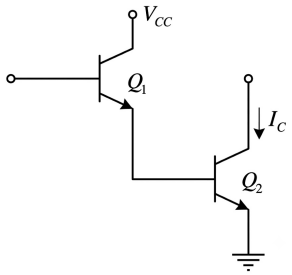


Fig. Q4.96.3

A. g_{m1} **B.** $0.5\,g_{m1}$ **C.** g_{m2} **D.** $0.5g_{m2}$

Q4.96.4 MCQ 1996 3M

Match the following:
(A) Cascode Amplifier
(B) Differential Amplifier
(C) Darlington pair common-collector Amplifier
(1) does not provide current gain.
(2) is a wide band amplifier.
(3) has very low input impedance and very high current gain.
(4) has very high input impedance and very high current gain.
(5) provides high common mode voltage Rejection.

A. A-2, B-5, C-4 **B.** A-4, B-5, C-1
C. A-2, B-1, C-4 **D.** A-3, B-5, C-4

Q4.96.5 MCQ 1996 2M

An npn transistor has a beta cut-off frequency f_β of 1 MHz and Common Emitter short circuit low frequency current gain β_0 of 200. Its unity gain frequency f_T and the alpha cut-off frequency f_α respectively are

A. 200 MHz, 201 MHz **B.** 200 MHz, 199 MHz
C. 199 MHz, 200 MHz **D.** 201 MHz, 200 MHz

Q4.96.6 MCQ 1996 2M

In the circuit shown in figure is a finite gain amplifier with a gain of K, a very large input impedance, and a very low output impedance. The input impedance of the feedback amplifier with the feedback impedance Z connected as shown will be

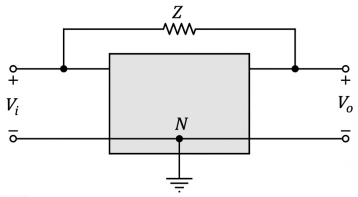


Fig. Q4.96.6

A. $Z\left(1-\frac{1}{K}\right)$ **B.** $Z(1-K)$ **C.** $\frac{Z}{K-1}$ **D.** $\frac{Z}{1-K}$

Q4.95.1 NAT 1995 1M

A transistor having $\alpha = 0.99$ and $V_{BE} = 0.7\,\text{V}$, is used in the circuit of the figure. The value of the collector current (in mA) will be

_____

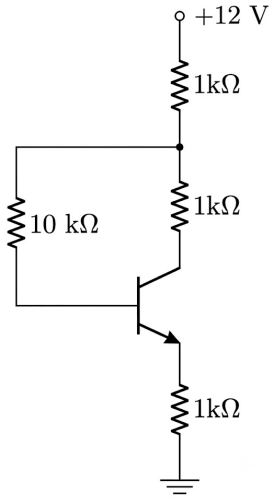


Fig. Q4.95.1

Q4.95.2 MCQ 1995 1M

To obtain very high input and output impedances in a feedback amplifier, the mostly used feedback is

A. voltage-series **B.** current-series
C. voltage-shunt **D.** current-shunt

Q4.95.3 NAT 1995 1M

An RC-Coupled Amplifier is assumed to have a single-pole low frequency transfer function. The maximum lower-cutoff frequency allowed for the amplifier to pass 50 Hz square wave with no more than 10% tilt is

_____ Hz.

Q4.95.4 Descriptive 1995 1M

An amplifier has an open-loop gain of 100 and its lower and upper-cut-off frequencies are 100 Hz and 100 kHz respectively, a feedback network with a feedback factor of 0.99 is connected to the amplifier. The new lower and upper-cut-off frequencies are at

_____ Hz and

_____ MHz.

Q4.94.1 NAT 1994 1M

A Common Emitter transistor Amplifier has a collector current of 1.0 mA when its base current is 25 μA at the room temperature. It's input resistance (in $k\Omega$ is approximately equal to

_____.

Q4.94.2 Descriptive 1994 1M

In order to reduce the harmonic distortion in an Amplifier its dynamic range has to be

_____.

Q4.93.1 MCQ 1993 2M

α cut-off frequency of a bipolar junction transistor

A. increase with the increase in base width

B. increase with the increase in emitter width

C. increases with the increase in collector width

D. increase with decrease in the base width

Q4.93.2 MCQ 1993 1M

For the Amplifier circuit of figure shown below, the transistor has a β of 800. The mid band voltage gain V_o/V_i of the circuit will be

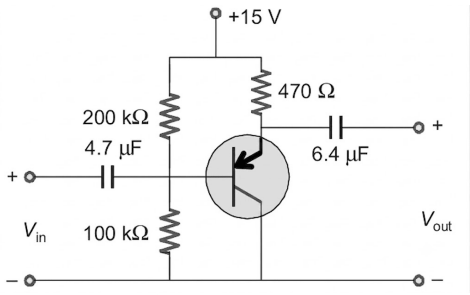


Fig. Q4.93.2

A. 0 **B.** < 1 **C.** ≈ 1 **D.** 800

Q4.93.3 MCQ 1993 1M

In a multi-stage RC-Coupled Amplifier the coupling capacitor.

A. Limits the low frequency response

B. Limits the high frequency response

C. Does not effect the frequency response

D. Blocks the d.c components without effecting the frequency response.

Q4.93.4 MCQ 1993 1M

The Bandwidth of an n-stage tuned Amplifier, with each stage having a bandwidth of B, is given by

A. $\frac{B}{n}$ **B.** $\frac{B}{\sqrt{n}}$ **C.** $B\sqrt{2^{1/n}-1}$ **D.** $\frac{B}{\sqrt{2^{1/n}-1}}$

Q4.93.5 MSQ 1993 1M

Negative feedback in Amplifiers

- ☐ **A.** improves the signal to noise ratio at the input
- ☐ **B.** improves the signal to noise ratio at the output
- ☐ **C.** does not effect the signal to noise ratio at the output
- ☐ **D.** reduces distortion

Q4.92.1 NAT 1992 2M

If the transistor in fig. has high value of β and V_{BE} of 0.65 the current I in mA flowing through the 2 kΩ resistance will be

_____.

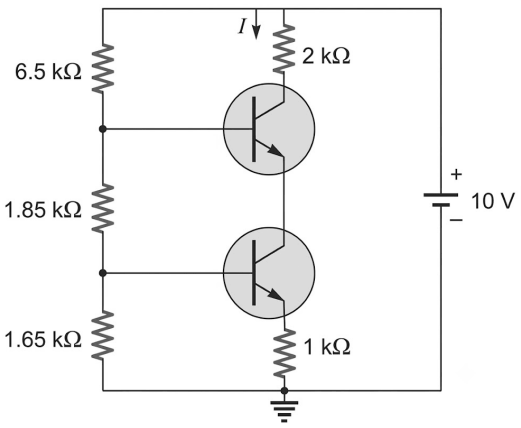


Fig. Q4.92.1

Q4.91.1 MSQ 1991 2M

Two non-inverting amplifiers, one having a unity gain and the other having a gain of twenty, are made using identical operational amplifiers. As compared to the unity gain amplifier, the amplifier with gain twenty has

- **A.** less negative feedback
- **B.** greater input impedance
- **C.** less bandwidth
- **D.** none of the above

Q4.91.2 NAT 1991 2M

In figure all transistors are identical and have a high value of beta. The voltage V_{DC} is equal to

_____.

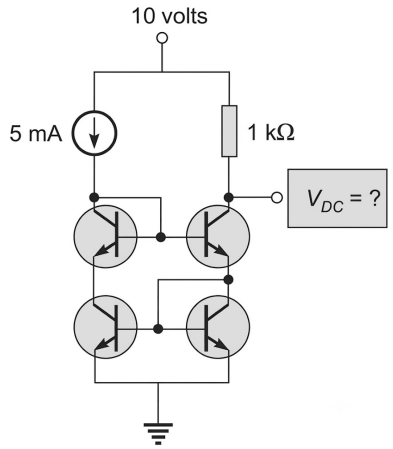


Fig. Q4.91.2

Q4.90.1 MSQ 1990 2M

Which of the following statements are correct for basic transistor Amplifier configurations?

- **A.** CB Amplifiers has low input impedance and low current gain
- **B.** CC Amplifiers has low output impedance and a high current gain.
- **C.** CE Amplifier has very poor voltage gain but very high input impedance
- **D.** The current gain of CB Amplifier is higher than the current gain of CC Amplifiers

Q4.90.2 MCQ 1990 2M

For good stabilized biasing of the transistor of the CE Amplifier of fig, we should have

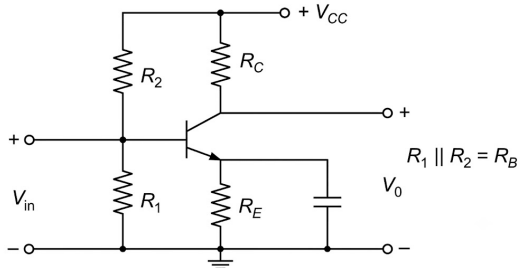
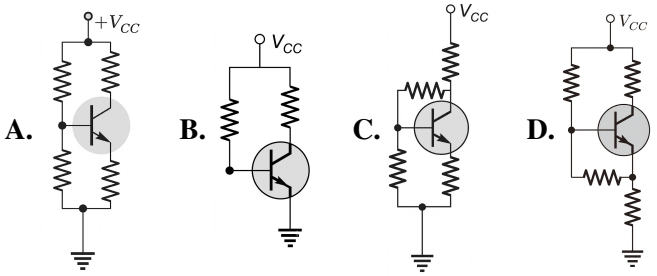


Fig. Q4.90.2

A. $\frac{R_E}{R_B} \ll 1$ **B.** $\frac{R_E}{R_B} \gg 1$ **C.** $\frac{R_E}{R_B} \ll h_{FE}$ **D.** $\frac{R_E}{R_B} \gg h_{FE}$

Q4.89.1 MCQ 1989 2M

Of the four biasing circuits shown in Fig. For a BJT, indicate the one which can have maximum bias stability



Q4.89.2 MCQ 1989 2M

The feedback amplifier shown in Fig. has:

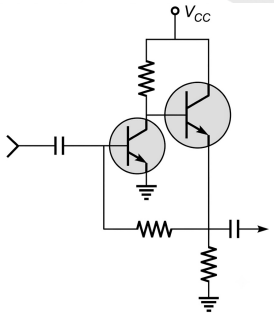


Fig. Q4.89.2

A. current–series feedback with large input impedance and large output impedance.

B. voltage–series feedback with large input impedance and low output impedance.

C. voltage–shunt feedback with low input impedance and low output impedance.

D. current–shunt feedback with low input impedance and output impedance.

Q4.88.1 MCQ 1988 2M

The amplifier circuit shown below uses a composite transistor of a MOSFET and BIPOLAR in cascade. All capacitances are large. g_m of the MOSFET = 2 mA/V, and h_{fe} of the BIPOLAR = 99. The overall transconductance g_m of the composite transistor is (in mA/V):

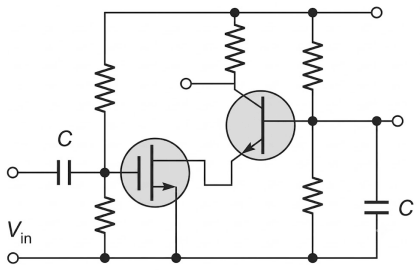


Fig. Q4.88.1

A. 198 **B.** 9.9 **C.** 4.95 **D.** 1.98

Q4.88.2 MCQ 1988 2M

Each transistor in the Darlington pair (see Fig. below) has h_{FE} = 100. The overall h_{FE} of the composite transistor neglecting the leakage currents is

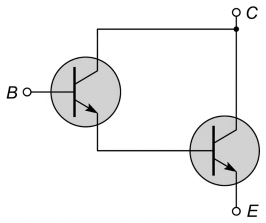


Fig. Q4.88.2

A. 10000 **B.** 10001 **C.** 10100 **D.** 10200

Q4.88.3 MCQ 1988 2M

The quiescent collector current I_C, of a transistor is increased by changing resistances. As a result

A. g_m will not be affected

B. g_m will decrease

C. g_m will increase

D. g_m will increase or decrease depending upon bias stability.

Q4.88.4 MCQ 1988 2M

The transistor in the amplifier shown below has following parameters: h_{fe} = 100, h_{ie} = 2 kΩ, h_{re} = 0, h_{oe} = 0.05 mmho. C is very large. The output impedance is

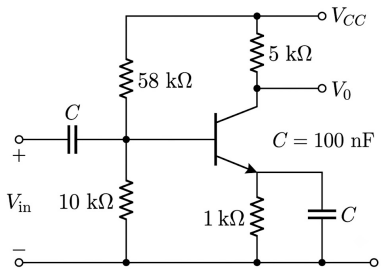


Fig. Q4.88.4

A. 20 kΩ **B.** 16 kΩ **C.** 5 kΩ **D.** 4 kΩ

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q4.26.1 D	**Q4.26.2** A	**Q4.25.1** A
Q4.25.2 C	**Q4.25.3** A	**Q4.20.1** 800
Q4.20.2 A	**Q4.17.1** D	**Q4.17.2** 0.9
Q4.17.3 128	**Q4.17.4** 6	**Q4.17.5** 1.100 to 1.200
Q4.17.6 50	**Q4.16.1** 570 to 610	**Q4.16.2** A
Q4.16.3 0.5	**Q4.15.1** 83.86	**Q4.15.2** 2.8

Q4.15.3 -1	**Q4.15.4** 2	**Q4.15.5** 1475
Q4.14.1 5.7 to 5.9	**Q4.14.2** 1092 to 1094	**Q4.14.3** -240 to -230
Q4.14.4 B	**Q4.14.5** B	**Q4.14.6** 1.07
Q4.14.7 B	**Q4.14.8** B	**Q4.14.9** 9.4
Q4.13.1 C	**Q4.13.2** B	**Q4.12.1** C
Q4.12.2 D	**Q4.11.1** D	**Q4.11.2** A

Q4.11.3 C	**Q4.10.1** B	**Q4.10.2** A
Q4.10.3 B	**Q4.10.4** B	**Q4.09.1** D
Q4.09.2 D	**Q4.08.1** A	**Q4.08.2** D
Q4.07.1 B	**Q4.06.1** C	**Q4.06.2** B
Q4.06.3 A	**Q4.06.4** C	**Q4.06.5** B
Q4.05.1 B	**Q4.05.2** C	**Q4.05.3** B

Q4.04.1 A	**Q4.04.2** C	**Q4.04.3** D
Q4.03.1 A	**Q4.03.2** C	**Q4.02.1** A
Q4.02.2 D	**Q4.02.3** *	**Q4.02.4** *
Q4.02.5 *	**Q4.02.6** *	**Q4.02.7** *
Q4.02.8 -80	**Q4.01.1** C	**Q4.01.2** B
Q4.01.3 C	**Q4.01.4** MTA	**Q4.01.5** *

Q4.01.6 *	**Q4.01.7** *	**Q4.01.8** *
Q4.01.9 *	**Q4.01.10** *	**Q4.01.11** *
Q4.00.1 -100	**Q4.00.2** -3.81	**Q4.00.3** $-\sin(\omega t)\ V$
Q4.00.4 D	**Q4.00.5** A	**Q4.99.1** *
Q4.99.2 A	**Q4.99.3** A	**Q4.99.4** C
Q4.99.5 A	**Q4.99.6** B	**Q4.98.1** 19.61

Q4.98.2 B	**Q4.98.3** D	**Q4.98.4** A
Q4.98.5 D	**Q4.98.6** C	**Q4.97.1** *
Q4.97.2 B	**Q4.97.3** A	**Q4.97.4** C
Q4.96.1 12.85×10^3	**Q4.96.2** B	**Q4.96.3** D
Q4.96.4 A	**Q4.96.5** A	**Q4.96.6** D
Q4.95.1 3.71	**Q4.95.2** B	**Q4.95.3** 1.59

Q4.95.4 **1, 10**	**Q4.94.1** **1**	**Q4.94.2** **large**
Q4.93.1 **D**	**Q4.93.2** **C**	**Q4.93.3** **A**
Q4.93.4 **C**	**Q4.93.5** **B, D**	**Q4.92.1** **1**
Q4.91.1 **A, C**	**Q4.91.2** **5**	**Q4.90.1** **A, B**
Q4.90.2 **B**	**Q4.89.1** **A**	**Q4.89.2** **C**
Q4.88.1 **D**	**Q4.88.2** **D**	**Q4.88.3** **C**
Q4.88.4 **D**		

PrepFusion

UNIT

V

Miscellaneous Topics

Topics

1. CMOS Inverter

Unit Snapshot*

Stream: EC	**Questions:** 1	**MCQ:** 0	**MSQ:** 0	**NAT:** 1
1M: 0	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2019)–2M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q5.19.1 NAT 2019 2M

A CMOS inverter, designed to have a mid-point voltage V_I equal to half of V_{dd}, as shown in the figure, has the following parameters:

$$V_{dd} = 3 \text{ V}$$

$$\mu_n C_{ox} = 100\ \mu\text{A/V}^2 ;\ V_{tn} = 0.7 \text{ V for nMOS}$$

$$\mu_p C_{ox} = 40\ \mu\text{A/V}^2 ;\ |V_{tp}| = 0.9 \text{ V for pMOS}$$

The ratio of $(W/L)_n$ to $(W/L)_p$ is equal to _____ *(rounded off to 3 decimal places).*

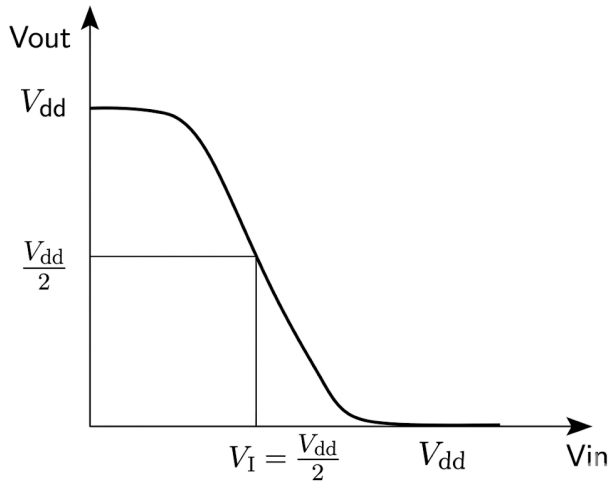


Fig. Q5.19.1

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Answer Key

Q5.19.1 **0.225**		

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PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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